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United States Patent Application Publication

20250281497

Kind Code

A1

Publication Date

September 11, 2025

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FUSED AMINO PYRIMIDINE COMPOUNDS FOR TREATMENT OF SEIZURE DISORDERS

Abstract

Compositions and methods for treating a seizure disorder with a compound according to Formula (I), or pharmaceutically acceptable salts thereof, are provided. The compositions and methods may be used to improve one or more symptoms of a seizure disorder.

Formula (I):

##STR00001##

or pharmaceutically acceptable salts thereof, wherein R.sup.1, R.sup.2, R.sup.7 and ring A have any of the meanings herein defined in the description.

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Family ID: 1000008630714

Appl. No.: 19/071799

Filed: March 06, 2025

Related U.S. Application Data

us-provisional-application US 63561912 20240306

Publication Classification

Int. Cl.: A61K31/5377 (20060101); A61K9/00 (20060101); A61K31/519 (20060101); A61P25/08 (20060101)

U.S. Cl.:

Background/Summary

[0001] This application claims benefit of and priority to U.S. Provisional Application No. 63/561,912, filed on Mar. 6, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] Methods of treating seizure disorders with fused amino pyrimidine compounds are provided.

BACKGROUND

[0003] Seizure disorders typically involve abnormal nerve cell activity in the brain, causing seizures which may be manifested by periods of unusual behavior, sensations, convulsions, diminished consciousness and sometimes loss of consciousness. Seizures can be a symptom of many different disorders that can affect the brain. Epilepsy is a seizure disorder characterized by recurrent seizures. See, e.g., Blume et al., *Epilepsia*. 2001; 42:1212-1218. Epileptic seizures are usually marked by abnormal electrical discharges in the brain and typically manifested by sudden brief episodes of altered or diminished consciousness, involuntary movements, or convulsions. Abnormal electrical activity in the brain may be measured or detected by electroencephalography (EEG). Clinical epileptic seizures can be preceded by abnormal electrical activity detected in EEG recordings as multifocal spikes which provide evidence of epileptic discharges but are not accompanied by seizures. See, e.g., Józwiak et al., *European Journal of Paediatric Neurology*, 2011, 15(5) 424-431 (“Józwiak et al”). Non-epileptic seizures may or may not be accompanied by abnormal electrical activity in the brain and may be caused by psychological issues or stress. Drug or alcohol withdrawal can also cause seizures. Seizure symptoms can vary widely. Some seizures can hardly be noticed, while others are totally disabling. Seizure disorders include epilepsy.

[0004] Hamartomas may be associated with certain seizure disorders. Hamartomas are a mostly benign, focal malformation that resembles a neoplasm in the tissue of its origin. They are composed of tissue elements normally found at that site, but grow in a disorganized manner. Hamartomas can originate in the brain. Tuberous Sclerosis Complex (TSC) is a genetic seizure disorder characterized by hamartomatous growth in various organs. The earliest symptoms of TSC can include heart tumors and cortical tubers, which can be seen even prenatally. Patients who have this disorder can exhibit a high rate of epilepsy and cognitive problems resulting from multiple lesions in the brain. TSC lesions (cortical tubers) typically contain dysmorphic neurons, brightly eosinophilic giant cells and white matter alterations. Seizures associated with TSC can be intractable. Tuber cinereum hamartoma (also known as hypothalamic hamartoma) is a benign tumor in which a disorganized collection of neurons and glia accumulate at the tuber cinereum of the hypothalamus. Symptoms include gelastic seizures, a disorder characterized by spells of involuntary laughter with interval irritability and depressed mood. Antiepileptic treatment before the onset of seizures can reduce epilepsy severity and the risk of mental retardation in infants with TSC. See, e.g., Józwiak et al.

[0005] Infantile spasms (IS) is a specific type of seizure disorder seen in infancy and childhood and is also known as West Syndrome, juvenile spasms or epileptic spasms. West Syndrome is characterized by infantile spasms, developmental regression, and a specific pattern on EEG testing called hypsarrhythmia (chaotic brain waves). The onset of infantile spasms is usually in the first year of life, typically between 3-12 months, typically manifesting around the fifth month. The seizures primarily consist of a sudden bending forward of the body with stiffening of the arms and legs; some children arch their backs as they extend their arms and legs. Spasms tend to occur upon

awakening or after feeding, and often occur in clusters of up to 100 spasms at a time. Infants may have dozens of clusters and several hundred spasms per day. Infantile spasms usually stop by mid-childhood, but may be replaced by other seizure types. The intellectual prognosis for children with infantile spasm is generally poor. Treatment with corticosteroids such as prednisone is standard, although serious side effects can occur. Several antiepileptic medications, such as topiramate may ease some symptoms. Vigabatrin been approved by the U.S. Food and Drug Administration to treat infantile spasms in children ages one month to two years.

[0006] Lennox-Gastaut syndrome (LGS) is a severe seizure disorder. Seizures usually begin before 4 years of age. Seizure types, which vary among patients, include tonic (stiffening of the body, upward deviation of the eyes, dilation of the pupils, and altered respiratory patterns), atonic (brief loss of muscle tone and consciousness, causing abrupt falls), atypical absence (staring spells), and myoclonic (sudden muscle jerks). There may be periods of frequent seizures mixed with brief, relatively seizure-free periods. Many children with Lennox-Gastaut syndrome typically experience some degree of impaired intellectual functioning or information processing, along with developmental delays, and behavioral disturbances. Lennox-Gastaut syndrome can be caused by brain malformations, perinatal asphyxia, severe head injury, central nervous system infection and inherited degenerative or metabolic conditions. In certain cases, no cause can be found. Treatment for Lennox-Gastaut syndrome includes clobazam and anti-epileptic medications such as valproate, lamotrigine, felbamate, or topiramate. There is usually no single antiepileptic medication that will completely control seizures. Children who improve initially may later show tolerance to a drug or have uncontrollable seizures.

[0007] CDKL5 disorder is an X-linked genetic seizure disorder that results in severe neurodevelopmental impairment and early onset, difficult to control seizures. CDKL5 stands for cyclin-dependent kinase-like 5, and is a gene located on the X chromosome. The CDKL5 gene was previously called STK9. Although CDKL5 disorder is primarily associated with females, it has been seen in males as well. A predominant characteristic associated with CDKL5 mutations is the so-called epileptic encephalopathy, the onset of severe seizures in the first six months of life (often within the first 3 months), and poor subsequent neurocognitive development and commonly the presence of repetitive hand movements (stereotypies). Most afflicted children cannot walk, talk or feed themselves, and many are confined wheelchairs, dependent on others for everything. Many also suffer with scoliosis, visual impairment, sensory issues and various gastrointestinal difficulties. Other symptoms of a CDKL5 disorder often include: low or poor muscle tone, hand wringing movements or mouthing of the hands, marked developmental delay, limited or absent speech, lack of eye contact or poor eye contact, gastroesophageal reflux, constipation, small, cold feet, breathing irregularities such as hyperventilation, grinding of the teeth, episodes of laughing or crying for no reason, very limited hand skills, some autistic-like tendencies, scoliosis, cortical visual impairment (CVI), aka “cortical blindness”, apraxia, eating/drinking challenges, sleep difficulties, characteristics such as a sideways glance, and a habit of leg crossing. There are currently no approved drugs to treat CDKL5 disorder. Seizure control is challenging and is often the most difficult health issue to manage. No one antiepileptic drug has been found to be uniformly effective, and often multiple anticonvulsants are needed.

[0008] As described in Kabat J, Król P., Focal cortical dysplasia—review. *Pol J Radiol.* 2012 April; 77(2):35-43, focal cortical dysplasia is a malformation of cortical development, which is the most common cause of medically refractory epilepsy in the pediatric population and the second/third most common etiology of medically intractable seizures in adults. In general, three types of cortical dysplasia are recognized. Type I focal cortical dysplasia with mild symptomatic expression and late onset is more often seen in adults, with changes present in the temporal lobe. Clinical symptoms are more severe in Type II focal cortical dysplasia usually seen in children. In this type, more extensive changes occur outside the temporal lobe with predilection for the frontal lobes. Type III focal cortical dysplasia is one of the above dysplasias with an associated principal lesion in

connection with hippocampal sclerosis, tumor, vascular malformation or acquired pathology during early life. Focal cortical dysplasia is currently recognized as a common cause of neocortical pharmacoresistant epilepsy.

[0009] Medications are used to treat seizure disorders and can be referred to as anti-epileptic drugs (“AED”). The treatment of recurrent seizures predominantly centers on the utilization of at least one AED, with possible adjunctive use of a second or even third agent in the case of monotherapeutic failure. See, Tolman and Faulkner, *Ther Clin Risk Manag.* 2011; 7: 367-375. Benzodiazepines such as diazepam and lorazepam are considered to be the standard first-line treatments for certain seizure disorders such as status epilepticus (SE). See, Deeb et al., *Epilepsia.* 2012 December; 53 Suppl 9(09):79-88. doi: 10.1111/epi.12037. However, approximately 30%-40% of epileptic patients have inadequate seizure control with just one AED and require the use of adjunctive agents. Tolman and Faulkner, *supra*. A subset of this group will have regular and persistent seizure activity despite reasonable doses of multiple AEDs. Accordingly, there remains a need for improved and/or additional therapies for treating seizure disorders.

[0010] Benzodiazepine resistant seizures such as diazepam resistant seizures or lorazepam resistant seizures are considered refractory to treatment. See, Deeb et al., *supra*. According to Deeb et al., *supra*, the reduction of diazepam efficacy suggests that seizures incur changes in the benzodiazepine-sensitive GABA_A receptor system, and it is possible that these alterations could contribute to the maintenance of seizures during status epilepticus (SE) and the recurrence of seizures at later stages. The simplest explanation for reduced benzodiazepine efficacy is a reduction in the number of benzodiazepine receptors. *Id.* Patients exhibiting drug-resistant forms of temporal lobe epilepsies (TLE) have a reduced number of benzodiazepine binding sites in the hippocampus that cannot be accounted for by the loss of neurons in sclerotic areas. *Id.* In certain instances, the affinity of the benzodiazepine binding sites changed in some areas, suggesting a biophysical change in the types of benzodiazepine-sensitive GABA_A receptors. *Id.* Pilocarpine-induced seizures result in a progressive change in EEG patterns that resemble those observed in humans experiencing generalized convulsive SE. *Id.* Studies have shown that SE caused a sharp decline in diazepam efficacy between 10-15 min after the first seizure that worsened over the next 30-180 min. *Id.* These experiments clearly demonstrate that pilocarpine-induced SE in rats causes a progressive reduction in diazepam's therapeutic efficacy. *Id.* There is a need for additional therapeutic options to treat benzodiazepine resistant seizures.

[0011] According to the Department of Environmental Protection. *Recognition and Management of Pesticide Poisonings: Sixth Edition: 2013: Chapter 5 Organophosphates*

(https://www.epa.gov/sites/default/files/documents/rmpp_6thed_ch5_organophosphates.pdf), organophosphates (OPs) are a class of insecticides and nerve agents, several of which are highly toxic. Organophosphates are used in agriculture, homes, gardens and veterinary practices.

Organophosphates poison insects and other animals, including birds, amphibians and mammals, primarily by phosphorylation of the acetylcholinesterase enzyme at nerve endings. The result is a loss of available acetylcholinesterase so that the effector organ becomes overstimulated by the excess acetylcholine (the impulse-transmitting substance) in the nerve ending. The enzyme is critical to normal control of nerve impulse transmission from nerve fibers to smooth and skeletal muscle cells, secretory cells and autonomic ganglia, and within the central nervous system (CNS). Once a critical proportion of the tissue enzyme mass is inactivated by phosphorylation, symptoms and signs of cholinergic poisoning become manifest. At sufficient dosage, loss of enzyme function allows accumulation of acetylcholine peripherally at cholinergic neuroeffector junctions (muscarinic effects), skeletal nerve-muscle junctions and autonomic ganglia (nicotinic effects), as well as centrally. A life-threatening severity of poisoning is signified by loss of consciousness, incontinence, seizures and respiratory depression.

[0012] Nerve agents (NAs) are among the most lethal agents of chemical warfare. Chang et al., *Ann Intern Med.* 2019 Jan. 1; 170(1):59-61. doi: 10.7326/M18-2428. Epub 2018 Dec. 18. Certain

nerve agents are organophosphorus compounds that, in a manner similar to organophosphate insecticides, inhibit acetylcholinesterase, which normally inactivates acetylcholine at neuronal junctions. See, *Id.* Cholinergic signs and symptoms that are seen after exposure to NAs are salivation, lacrimation, urination, diarrhea, gastrointestinal cramps, and emesis. *Id.* Prominent bradycardia, broncho-spasm, bronchorrhea, weakness, and fasciculations may be evident and depend on such variables as dose, exposure route, and clinical course. *Id.*

[0013] Organophosphate nerve agents have been classified as G-series and V-series. The G-series includes GA (tabun), GB (sarin), GD (soman), GF (cyclosarin), and GE (ethysarin). The V-series includes VX (O-Ethyl-S-[2(diisopropylamino)ethyl]methylphosphonothioate), VE (O-Ethyl-S-[2-(diethylamino)ethyl]ethylphosphonothioate), VG (O,O-Diethyl-S-[2-(diethylamino)ethyl]phosphorothioate), VM (O-Ethyl-S-[2-(diethylamino)ethyl]methylphosphonothioate) and VR (N-diethyl-2-(methyl-(2-methylpropoxy)phosphoryl)sulfanylethanamine). VR is the compound from which Soviet newcomer agents (Novichok) Novichok 5 and Novichok 7, are derived.

[0014] Treatment of organophosphate NA poisoning includes the following medical countermeasures: atropine (blocks acetylcholine receptors from excessive stimulation), pralidoxime (prevents acetylcholinesterase deactivation by NA), and benzodiazepine (controls seizures or severe fasciculations). *Id.*

[0015] Tetramethylenedisulfotetramine (TETS) is a nerve agent considered to be a credible chemical threat agent that poses a serious public health risk to military personnel and civilians. Mundy et al., *Toxicology and Applied Pharmacology*, 426 (1 Sep. 2021) 115643. Despite a worldwide ban on its production, TETS is still available on the black market as a rodenticide, and the weaponization of TETS together with accidental and suicidal poisonings has caused a significant number of human deaths. *Id.* Acute exposure can produce seizures that rapidly progress to status epilepticus in humans. *Id.* TETS induces seizures by antagonism of γ -aminobutyric acid type A receptors (GABA.sub.AR). See, *Id.* Currently, a medical countermeasure specific for TETS-induced seizures is not available. *Id.* Intoxicated individuals are likely to be treated with benzodiazepines, which are the standard of care for organophosphate-induced seizures. *Id.* While benzodiazepines are often effective as an immediate antiseizure treatment for GABA.sub.AR antagonist-induced seizures, latent neuropathology in these types of exposures is often not diminished by treatment. *Id.* Clinical reports describe incidences of recurrent seizures that are unresponsive even after initial successful mitigation with a benzodiazepine. *Id.* Due to the use of nerve agents by terrorists and rogue nations and increasing exposure of the general populous to organophosphate pesticides there is a need for as many therapeutic countermeasures for organophosphate and TETS poisoning as can be created.

[0016] Fused amino pyridine compounds and pharmaceutically acceptable salts thereof are disclosed in WO2021/180952, incorporated herein by reference. As described in WO2021/180952, these compounds and their pharmaceutically acceptable salts selectively modulate KCC2 and are used to treat or prevent KCC2 mediated disease, including neurological disorders. As described in the OMIM database, the SLC12A5 gene encodes the neuronal KCC2 channel that is the major extruder of intracellular chloride in mature neurons. In the presence of low intraneuronal chloride, the binding of GABA and glycine to their ionotropic receptors results in chloride influx with subsequent hyperpolarization contributing to neuronal inhibition. Hyperpolarizing GABA.sub.AR currents are critically dependent upon efficient Cl.sup.- extrusion, which is facilitated by KCC2.

[0017] Consistent with its role in supporting fast synaptic inhibition, humans with mutations in KCC2 have been identified, resulting in severe epilepsy, developmental delay, and premature death. Likewise, deficits in KCC2 expression levels are seen in epileptic foci resected from patients with treatment resistant epilepsy that correlate with depolarizing GABA.sub.AR currents (Palma et al., *Proc. Natl. Acad. Sci. USA* 103, 8465-8468 (2006)).

[0018] Conditional inactivation of KCC2 in the hippocampus leads to neuronal Cl.sup.-

accumulation and depolarizing GABA.sub.AR currents. These deficits in inhibition parallel the development of spontaneous seizures, neuronal apoptosis and reactive astrogliosis (Kelley et al., EBioMedicine 32, 62-71 (2018)). Likewise, studies in rodents have further suggested that deficits in KCC2 activity contribute to onset of Status Epilepticus (SE) and the subsequent development of BDZ resistant/refractory seizures, termed refractory SE (RSE) (Moore et al., Proc. Natl. Acad. Sci. U.S.A. 115, 10166-10171 (2018); Barmashenko et al., Epilepsia 52, 1570-1578 (2011); Deeb et al., Eur. J. Neuroscience 38, 2453-2467 (2013); Lee et al., 2022; Pathak et al., Journal of neuroscience: the official journal of the Society for Neuroscience 27, 14012-14022 (2007); (Silayeva et al., Proc. Natl. Acad. Sci. U.S.A. 112, 3523-3528. (2015)).

[0019] Direct modulation of KCC2 by interaction with small molecules has been reported. Delpire et al (Proc Natl Acad Sci USA. 2009 Mar. 31; 106(13): 5383-5388) describe an assay to identify small molecule inhibitors of KCC2 and Zhang et al (Journal of Biomolecular Screening 15(2): 2010) describe an assay used to identify positive modulators of KCC2.

SUMMARY

[0020] Methods of treating a seizure disorder are provided and, in embodiments, include administering to a subject diagnosed with the seizure disorder an effective amount of a compound according to Formula (I):

##STR00002##

or a pharmaceutically acceptable salt thereof, wherein: [0021] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; [0022] R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0023] A is selected from

##STR00003##

or a N-oxide thereof; [0024] R.sup.3 is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3 alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; [0025] R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; [0026] R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; [0027] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; [0028] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0029] R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3

substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; [0030] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl; [0031] R.sup.10 is —C.sub.1-6alkyl; [0032] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0033] R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0034] m is 0 or 1; and [0035] n is 1, 2 or 3.

[0036] In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject in need thereof to provide improvement in one or more symptoms of the seizure disorder. In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject in need thereof to provide improvement in next day functioning of the subject. In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject in need thereof prior to the onset of clinical seizures after detection of abnormal EEG to reduce or prevent symptoms of the seizure disorder. In embodiments, methods of treating an abnormal EEG signature include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject having the abnormal EEG signature. In embodiments, methods of treating a seizure disorder include prophylactically administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject in danger of being exposed to a nerve agent or an organophosphate pesticide. In embodiments, a method of prophylactically treating exposure of a subject to a nerve agent or an organophosphate pesticide includes administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to subject prior to exposure of the subject to the nerve agent or organophosphate pesticide. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in treating a seizure disorder in a subject. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in providing improvement in next day functioning of a subject with a seizure disorder. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in administration to a subject with a seizure disorder prior to the onset of clinical seizures after detection of abnormal EEG to reduce or prevent symptoms of the seizure disorder. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in treating a subject having an abnormal EEG signature. In embodiments, provided herein is a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in prophylactically treating a subject in danger of being exposed to a nerve agent or an organophosphate pesticide. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for the manufacture of a medicament for the treatment of a seizure disorder. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for the manufacture of a medicament for the prophylactic treatment of a subject in danger of being exposed to a nerve agent or an organophosphate pesticide.

[0037] In embodiments, the seizure disorders include epilepsy, epilepsy with generalized tonic-clonic seizures, epilepsy with myoclonic absences, frontal lobe epilepsy, temporal lobe epilepsy, focal cortical dysplasia, Landau-Kleffner Syndrome, Dravet syndrome, Rasmussen's syndrome, Doose syndrome, CDKL5 disorder, infantile spasms (West syndrome), Lennox-Gastaut syndrome

(LGS), Rett syndrome, Ohtahara syndrome, CDKL5 disorder, tuberous sclerosis complex (TSC), childhood absence epilepsy, essential tremor, and acute repetitive seizures, benign rolandic epilepsy, status epilepticus, refractory status epilepticus, super-refractory status epilepticus (SRSE), PCDH19 pediatric epilepsy, benzodiazepine resistant seizures including diazepam resistant seizures or lorazepam resistant seizures, seizures caused by exposure to nerve agents, seizures caused by exposure to pesticides, drug withdrawal induced seizures, alcohol withdrawal induced seizures, and increased seizure activity or breakthrough seizures (increased seizure activity; also called serial or cluster seizures). In embodiments, the seizure disorder is status epilepticus. In embodiments, the seizure disorder is temporal lobe epilepsy. In embodiments, the seizure disorder is tuberous sclerosis complex. In embodiments, the seizure disorder is focal cortical dysplasia. In embodiments, the seizure disorder is benzodiazepine resistant seizures. In embodiments, the seizure disorder is diazepam resistant seizures. In embodiments, the seizure disorder is lorazepam resistant seizures. In embodiments, the seizure disorder is seizures caused by exposure to nerve agents. In embodiments, the seizure disorder is seizures caused by exposure to pesticides.

[0038] In embodiments, a compound of Formula (I) is Compound A:

##STR00004##

[0039] In embodiments, a compound of Formula (I) is Compound B:

##STR00005##

[0040] In embodiments, a compound of Formula (I) is Compound C:

##STR00006##

[0041] In embodiments, a compound of Formula (I) is Compound D:

##STR00007##

[0042] In embodiments, a compound of Formula (I) is Compound E:

##STR00008##

[0043] In embodiments, a compound of Formula (I) is Compound F:

##STR00009##

[0044] In embodiments, a compound of Formula (I) is Compound G:

##STR00010##

[0045] In embodiments, a compound of Formula (I) is Compound H:

##STR00011##

Description

BRIEF DESCRIPTION OF THE FIGURES

[0046] FIG. 1A is a drug screening workflow chart illustrating the steps taken to identify KCC2 direct activators, a multi-tiered program starting from a compound library of 1.3 million compounds. Identification of a subseries of fused pyrimidine ring compounds, denoted as subseries A (SSA) were identified as lead-hits from the compound library screen.

[0047] FIG. 1B depicts immunoblotting results for HEK-293 cells expressing KCC2 exposed to vehicle (–) or 30 μ M SSA1 (+) for 90 min and then heated to 37-58° C. Soluble fractions were subsequently immunoblotted with KCC2 and GAPDH antibodies.

[0048] FIG. 1C is a graph depicting thermal aggregation curves for KCC2 under control conditions and in the presence of SSA-0126. *=significantly different to control $p < 0.001$; t-test; $n = 4$ transfections. The level of remaining soluble native KCC2 at each temperature was normalized to that seen at 37° C. ($I \times I_{37^\circ \text{C.}}$).

[0049] FIG. 1D is a graph depicting isothermal dose response curves constructed for destabilization of KCC2 by SSA1 at 50° C., $n = 4$ transfections

[0050] FIG. 1E shows the structure of Compound A (also referred to as CmpA).

[0051] FIG. 1F is a graph showing the effects of Compound A on KCC2 activity expressed in

HEK-293 cells as measured using TI flux. Data is normalized to values seen with vehicle (100%), n=3.

[0052] FIG. 1G is a chart showing the EC.sub.50 of Compound A for KCC2, KCC2-S940A, KCC3, KCC4 and NKCC1 as measured in expressing HEK-293 cells, n=4 transfections. In all panels p values were determined using unpaired t-tests with Welch's correction.

[0053] FIG. 2A depicts three graphs showing E.sub.Gly values versus time as measured from HEK-293 cells expressing KCC2 together with GlyR α 1 using perforated patch clamp recordings. Individual shifts in E.sub.Gly are shown for cells incubated with Compound A (0.3 and 3 μ M) or vehicle (V) for 15 min.

[0054] FIG. 2B depicts a first bar graph showing Δ E.sub.Gly (mM) by concentration of vehicle and Compound A and a second bar graph showing reduction in [Cl.sup.-] (mM) by concentration of vehicle and Compound A. Mean shifts in E.sub.Gly were determined for cells treated with vehicle, p=0.1227, Compound A 0.3 μ M (*p=0.0051), and 3 μ M (*p=0.0350), respectively, n=5 transfections. Compound A induced changes in [Cl.sup.-] were calculated from E.sub.Gly using the Nernst equation for vehicle (V), (p=0.1227), 300 nM, (*p=0.0051), or 3 μ M (*p=0.0350) CmpA, n=5 transfections.

[0055] FIG. 2C depicts immunoblotting results and a bar graph showing surface/total KCC2 (% conc.) by concentration of Compound A. HEK-293 cells (UT) or those expressing KCC2 were exposed to V, 0.3 μ M, or 3 μ M 350 for 15 min and biotinylated with NHS-biotin. Surface and total extracts were subsequently immunoblotted with KCC2 and actin antibodies. The ratio of surface/total KCC2 immunoreactivity was determined and normalized to V (100%) (n=3 transfections).

[0056] FIG. 2D depicts immunoblotting results obtained by exposing untransfected HEK-293 cells (U) or those expressing KCC2 to vehicle (V), 0.3, or 3 μ M, Compound A, for 15 min and biotinylated with NHS-Biotin. Surface and total extracts were subsequently immunoblotted with KCC2 and actin antibodies. The ratio of surface/total KCC2 immunoreactivity was determined and normalized to V (100%) for 0.3 (*p=0.659) or 3 μ M Compound A (*p=0.739), n=3 transfections.

[0057] FIG. 2E depicts a first bar graph showing pS940/KCC2 (% concentration) of Compound A (μ M) and a second bar graph showing pT1007/KCC2 (% concentration) of Compound A (μ M). HEK-293 cells were treated with V (-) or 3 μ M Compound A (+) for 15 min. Cell lysates were immunoblotted with KCC2, pS940, KCC2 and actin antibodies. The ratios of pS940/KCC2 (p=0.491) and pT1007/KCC2 (p=0.721) were then compared to V (100%), n=3 transfections. In all panels p values were determined using unpaired t-tests with Welch's correction.

[0058] FIG. 3A depicts a graph showing current (pA) versus holding potential (mV) for vehicle and Compound A. 18-21 Div hippocampal neurons were subjected to gramicidin perforated patch clamp recordings in the presence of bumetanide (10 μ M) and TTX (500 nM). After attaining perforation, cultures were exposed to 300 nM Compound A, or V (1% BCD) for 15 min. E.sub.GABA was then determine using voltage ramps protocols and representative current-voltage (I-V) plots are shown for neurons at 0 (blue) and 15 (red) min treatment for V and Compound A treated neurons.

[0059] FIG. 3B depicts a first bar graph showing neuronal E.sub.GABA values by exposure to vehicle (V) over time and a second bar graph defining neuronal E.sub.GABA values by exposure to Compound A over time. E.sub.GABA values were measured at 0 and 15 min following treatment with V (p=0.2190) or 300 nM Compound A (*p=0.0104), n=5 cultures.

[0060] FIG. 3C depicts a bar graph showing neuronal [Cl.sup.-] values by exposure to Compound A over time. [Cl.sup.-] values were calculated from E.sub.GABA values for V (p=0.1646) or Compound A (*p=0.0153) treated neurons, n=5 cultures.

[0061] FIG. 3D depicts a first bar graph showing neuronal Basal E.sub.GABA (mV) upon exposure to vehicle (V) and Compound A, and a second bar graph showing neuronal 11K A E.sub.GABA upon exposure to vehicle (V) and Compound A. 18-21 Div hippocampal neurons were incubated

with 300 nM Compound A or vehicle for 1 h. Neurons were subject to whole cell recording using an intracellular solution containing 30 mM Cl_{sup}-. 5 min later basal E_{sub}.GABA values were determined and compared between treatments, *p=0.0253, n=4 cultures. Neurons were subsequently exposed to 10 μ M VU0463271 and the magnitude of the shift in E_{sub}.GABA(VU0463271 Δ E_{sub}.GABA) was then determined and compared between groups, *p=0.040 n=4 cultures.

[0062] FIG. 3E depicts a first electrograph showing the effect of vehicle on neuronal excitability in C57Bl/6 brain slices and a second electrograph showing the effect of Compound A on neuronal excitability in C57Bl/6 brain slices. C57Bl/6 brain slices were incubated in ACSF supplemented with vehicle or Compound A (1 μ M) for 1 h at 34° C., and continuously perfused to ACSF deficient in Mg_{sup}.+2 (0-Mg) (arrow). Field recordings were then performed within the entorhinal cortex as a means of monitoring neuronal excitability. The green line indicates the onset of the first SLE, while the red indicates the development of LRDs.

[0063] FIG. 3F depicts a first bar graph showing the effect of Compound A and Vehicle (V) on latency to first seizure-like events (SLE) (minutes) and a second bar graph showing the effect of Compound A and Vehicle (V) on latency to late recurrent discharges (LRD) (minutes). The time to the 1_{sup}.st SLE, p=0.1582, and to LRDs *p=0.0028, was compared between treatments, n=7 mice. In all panels p values were determined using unpaired t-tests with Welch's correction.

[0064] FIG. 4A depicts a bar graph showing accumulation of Compound A in the mouse brain within 30 minutes after intravenous injection of 25 mg/kg Compound A in 5% β -cyclodextrin (BCD). 15 and 60 minutes later drug levels were measured via LC-MS/MS, n=3 mice.

[0065] FIG. 4B depicts a bar graph showing accumulation of Compound A in the mouse brain after intravenous injection of 50 mg/kg Compound A in 5% β -cyclodextrin (BCD). 1 and 2 h later drug levels were measured via LC-MS/MS, n=3 mice.

[0066] FIG. 4C is a graph showing accumulation of Compound A in the mouse brain within 30 minutes after subcutaneous injection of 50 mg/Kg Compound A in 5% β -cyclodextrin (BCD). Drug accumulation was then measured over a time course of 8h via LC-MS/MS, n=3-4 mice.

[0067] FIG. 4D depicts a first bar graph showing the effect of Compound A and vehicle (V) on distance traveled after mice were dosed with vehicle (V) or 50 mg/kg Compound A and a second bar graph showing the effect of Compound A and vehicle (V) on time spent in a center zone after mice were dosed with vehicle or 50 mg/kg Compound A. 2-3h after injection, the mice were placed in the center of a 60 cm $\times$ 60 cm open field and allowed to explore for 10 min. The total distance traveled (p=0.365) and time in the center of the area (p=0.425) were then quantified and compared between treatments using t-tests; n=9 mice.

[0068] FIG. 4E depicts a first bar graph showing the effect of Compound A on forelimb clonus induced by pentylenetetrazol (PTZ) after mice were injected intravenously with 25 mg/kg Compound A or vehicle, followed 30 min later with ascending concentrations of PTZ, and a second bar graph showing the effect of Compound A on hindlimb clonus induced by pentylenetetrazol (PTZ) after mice were injected intravenously with 25 mg/kg Compound A or vehicle, followed 30 min later with ascending concentrations of PTZ. Mice were subject to video recording and the doses required to induced fore and hindlimb doses were determined. The respective doses to induced fore (p=0.120) and hindlimb clonus (*p=0.001) were then compared using an unpaired t-tests with Welch's correction n=14 mice.

[0069] FIG. 5A depicts an electroencephalogram (EEG) demonstrating anticonvulsant properties of Compound A and its effects on the development of kainic acid (KA) induced SE in mice. Example EEG traces are shown for mice injected with 50 mg/kg Compound A or vehicle (SC) 2h prior to dosing IP with 20 mg/kg KA (black arrow). 2h following KA injection mice were dosed with IP with 10 mg/kg diazepam (DZ) (light gray arrow) and EEG recordings were extended for a further 1 h.

[0070] FIG. 5B depicts a first bar graph showing the effect of Vehicle (V) and Compound A on

time to first seizure, a second bar graph showing the effect of Vehicle (V) and Compound A on time to SE, and a third bar graph showing the effect of Vehicle (V) and Compound A % total time in epileptiform activity. The time to the 1.^{sup}.st seizure (*p=0.0235), the onset of SE (*p=0.0329) and the % total time in epileptiform activity (*p=0.023;) were compared between treatment groups, n=9 mice.

[0071] FIG. 5C depicts a first graph showing power ($\mu\text{V}.\text{sup.2}$) versus frequency (Hz) and reduction of EEG power by Compound A as compared to Vehicle (V) and a second graph (bar graph) showing reduction of total power ($\mu\text{V}.\text{sup.2}$) by Compound A as compared to Vehicle (V). EEG recordings were subjected to Fast Fourier transform (FFT) and spectral plots are shown for frequencies between 0-100 Hz for mice treated with vehicle or Compound A 2h after kainic acid (KA) injection. Total EEG power was then compared between treatments, *p=0.0024, t-test; n=9 mice.

[0072] FIG. 5D depicts a first graph showing power ($\mu\text{V}.\text{sup.2}$) versus frequency (Hz) and the effect of Vehicle (V) before and after treatment with diazepam (DZ) and a second graph (bar graph) showing the effect of Vehicle (V) on total power ($\mu\text{V}.\text{sup.2}$) before and after treatment with diazepam (DZ). EEG recordings from mice pretreated with Vehicle (V) 2h following kainic acid (KA) injection and 10 min after dosing with diazepam (DZ) were subjected to Fast Fourier transformation (FFT) and spectral plots are shown for frequencies between 0-100 Hz. Total EEG power was then compared *p=0.004, n=9 mice.

[0073] FIG. 5E depicts a first graph showing power ($\mu\text{V}.\text{sup.2}$) versus frequency (Hz) and the effect of Compound A before and after treatment with diazepam (DZ) and a second graph (bar graph) showing the effect of Compound A on total power ($\mu\text{V}.\text{sup.2}$) before and after treatment with diazepam (DZ). EEG recordings from mice pretreated with vehicle 2h following KA injection and 10 min after dosing with DZ were subject to Fast Fourier transformation (FFT) and spectral plots are shown for frequencies between 0-100 Hz. Total EEG power was then compared p=.sup.0.104, n=9 mice. In all panels p values were determined using unpaired t-tests with Welch's correction.

[0074] FIG. 6A depicts an electroencephalogram (EEG) for mice injected IP with 20 mg/kg KA (black arrow). 2h following KA injection mice were dosed IP with 10 mg/kg DZ (light gray arrow) and EEG recordings were extended for a further 1 h.

[0075] FIG. 6B depicts a first graph showing power ($\mu\text{V}.\text{sup.2}$) versus frequency (Hz) and the effect of diazepam (DZ) following KA injection and a second graph showing power ($\mu\text{V}.\text{sup.2}$) versus frequency (Hz) and the effect of DZ and Compound A following KA injection. EEG recordings 2h after KA injection and 30 min following DZ or DZ/Compound A treatment were subject to Fast Fourier transformation (FFT) and spectral plots are shown for frequencies between 0-100 Hz.

[0076] FIG. 6C depicts a bar graph showing predrug power ($\mu\text{V}.\text{sup.2}$) $\times 10.\text{sup.}-6$ following KA treatment, diazepam (DZ) treatment and diazepam (DZ)/Compound A treatment. Total EEG power was compared for treatment groups 2h following KA injection, p=0.3732, n=12 mice.

[0077] FIG. 6D depicts a bar graph showing postdrug power ($\mu\text{V}.\text{sup.2}$) $\times 10.\text{sup.}-6$ following KA treatment, diazepam (DZ) treatment and diazepam (DZ)/Compound A treatment. Total EEG power 2h after KA injection was compared to that 30 min following DZ, or DZ/Compound A treatment, *p=0.0072, n=12 mice.

[0078] FIG. 6E depicts a bar graph showing % predrug power following KA treatment, diazepam (DZ) treatment and diazepam (DZ)/Compound A treatment.

[0079] FIG. 6F depicts a bar graph showing minutes to suppression following KA treatment, diazepam (DZ) treatment and diazepam (DZ)/Compound A treatment. The latency to seizure suppression was measured following injection of DZ or DZ/CmpA, p=0.0008, n=12 mice. In all panels p values were determined using unpaired t-tests with Welch's correction.

[0080] FIG. 6G depicts a bar graph showing % insensitive to diazepam (DZ) treatment alone and to diazepam (DZ)/Compound A treatment together.

[0081] FIG. 7A depicts images of mice brain hippocampal sections subjected to TUNEL/DAPI staining after treatment with diazepam (DZ) and diazepam (DZ)/Compound A.

[0082] FIG. 7B depicts a bar graph showing number of dead cells/in hippocampal CA1 cells after treatment with Vehicle and Compound A. The number of TUNEL positive neurons in CA1 was compared between treatment groups, *p=0.0029, unpaired t-test with Welch's correction, n=14 sections.

DETAILED DESCRIPTION

[0083] Compositions and methods of treating a seizure disorder are provided and, in embodiments, include administering to a subject diagnosed with the seizure disorder an effective amount A compound according to Formula (I):

##STR00012##

or a pharmaceutically acceptable salt thereof, wherein: [0084] R_{sup.1} is selected from C_{sub.2-6}alkyl; C_{sub.2-6}alkenyl; C_{sub.2-6}alkynyl; C_{sub.2-6}alkoxy; C_{sub.2-6}alkenyloxy; C_{sub.2-6}alkynyloxy; C_{sub.3-7}cycloalkyl; —O—C_{sub.3-7}cycloalkyl; C_{sub.6-10}aryl; —O—(CH_{sub.2})_{sub.m}—C_{sub.6-10}aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3} and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy wherein —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF_{sub.3}, —NHC(O)O—C_{sub.1-6}alkyl or two substituents together with the carbon to which they are attached form diazirinyl; [0085] R_{sup.2} is selected from —H; -halo; and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; [0086] A is selected from

##STR00013##

or a N-oxide thereof; [0087] R_{sup.3} is selected from —H; —C_{sub.1-6}alkyl; —C_{sub.2-6}alkenyl; —C_{sub.2-6}alkynyl; C_{sub.3-7}cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF_{sub.3}, —C_{sub.1-3}alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF_{sub.3}, —C(O)NR_{sup.8}R_{sup.9} and —NR_{sup.8}R_{sup.9}; [0088] R_{sup.4a} and R_{sup.4b} are each independently selected from —H and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; [0089] R_{sup.4c} and R_{sup.4d} are each independently selected from —H and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}, or R_{sup.4c} and R_{sup.4d} together with the carbon to which they are attached represent carbonyl; [0090] R_{sup.5a}, R_{sup.5b}, R_{sup.5c} and R_{sup.5d} are each independently selected from —H and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; [0091] R_{sup.6} is selected from —H; -halo; —NH_{sub.2}; —CN; —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; —C_{sub.1-3}alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; —C(O)O—C_{sub.1-3}alkyl; —C(O)NR_{sup.8}R_{sup.9}; —C(O)OH; and —NHC(O)—C_{sub.1-3}alkyl; [0092] R_{sup.7} is selected from NR_{sup.10}R_{sup.11}; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C_{sub.1-6}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF_{sub.3} and —OH; —C_{sub.1-3}alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; —C(O)OH; —C_{sub.1-3}alkylene-NHC(O)C_{sub.1-6}alkyl; —C_{sub.1-3}alkylene-NHC(O)OC_{sub.1-6}alkyl; C_{sub.3-5}cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R_{sup.7} is morpholinyl and R_{sup.1} is unsubstituted phenyl, R_{sup.2} is not —H; [0093] R_{sup.8} and R_{sup.9} are each independently selected from —H and —

C.sub.1-6 alkyl; [0094] R.sup.10 is —C.sub.1-6alkyl; [0095] R.sup.11 is selected from —C.sub.1-6-alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0096] R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0097] m is 0 or 1; and [0098] n is 1, 2 or 3.

[0099] In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I) or a pharmaceutically acceptable salt thereof to a subject diagnosed with the seizure disorder to provide improvement in one or more symptoms of the seizure disorder. In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I) or a pharmaceutically acceptable salt thereof to a diagnosed with the seizure disorder to provide improvement in next day functioning of the subject. In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I) or a pharmaceutically acceptable salt thereof to a subject diagnosed with the seizure disorder prior to the onset of clinical seizures after detection of abnormal EEG to reduce or prevent symptoms of the seizure disorder. In embodiments, methods of treating an abnormal EEG signature include administering a compound according to Formula (I) or a pharmaceutically acceptable salt thereof to a subject having the abnormal EEG signature. In embodiments, a method of prophylactically treating exposure of a subject to a nerve agent or an organophosphate pesticide includes administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to subject prior to exposure of the subject to the nerve agent or organophosphate pesticide. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in treating a seizure disorder in a subject. In embodiments, provided herein is a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in providing improvement in next day functioning of a subject having a seizure disorder. In embodiments, provided herein is a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in administration to a subject in danger of being exposed to a nerve agent or an organophosphate pesticide prior to the onset of exposure to the nerve agent or organophosphate pesticide in order to prevent or reduce the toxic effects of exposure to the nerve agent or organophosphate pesticide. In embodiments, provided herein is a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in administration to a subject having a seizure disorder prior to the onset of clinical seizures after detection of abnormal EEG to reduce or prevent symptoms of the seizure disorder. In embodiments, provided herein is a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in treating a subject having an abnormal EEG signature. In embodiments, provided herein is a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for the manufacture of a medicament for the treatment of a seizure disorder.

[0100] Without wishing to be bound by any particular theory seizure disorders can result from impaired Cl.sup.– transport. See, e.g., Watanabe et al., Sci. Signal. 12, eaaw9315 (2019). KCC2 is a K.sup.+—Cl.sup.– cotransporter and responsible for maintaining low Cl.sup.– concentration in neurons of the central nervous system (CNS), essential for postsynaptic inhibition through GABA.sub.A and glycine receptors. As a result of their KCC2 activation activity, the compounds according to Formula (I), or pharmaceutically acceptable salts thereof, restore GABAergic function in patients experiencing seizure disorders involving impaired Cl.sup.– transport, thereby preventing, reducing, relieving, alleviating or eliminating symptoms.

[0101] Examples of seizure disorders include epilepsy, epilepsy with generalized tonic-clonic seizures, epilepsy with myoclonic absences, frontal lobe epilepsy, temporal lobe epilepsy, focal cortical dysplasia, Landau-Kleffner Syndrome, Rasmussen's syndrome, Dravet syndrome, Doose syndrome, CDKL5 disorder, tuberous sclerosis complex (TSC), infantile spasms (West syndrome), juvenile myoclonic epilepsy (JME), vaccine-related encephalopathy, intractable childhood epilepsy (ICE), Lennox-Gastaut syndrome (LGS), Rett syndrome, Ohtahara syndrome, CDKL5 disorder,

childhood absence epilepsy, essential tremor, acute repetitive seizures, benign rolandic epilepsy, status epilepticus, refractory status epilepticus, super-refractory status epilepticus (SRSE), PCDH19 pediatric epilepsy, benzodiazepine resistant seizures including diazepam resistant seizures or lorazepam resistant seizures, seizures caused by exposure to nerve agents, seizures caused by exposure to pesticides, drug withdrawal induced seizures, alcohol withdrawal induced seizures, increased seizure activity or breakthrough seizures (also called serial or cluster seizures). In embodiments, the seizure disorder is associated with a sodium channel protein type 1 subunit alpha (Scn1a)-related disorder.

[0102] In embodiments, the seizure disorder is status epilepticus (SE). SE is characterized by an epileptic seizure of greater than five minutes or more than one seizure within a five-minute period without the person returning to normal between them. SE can be a dangerous condition that can lead to mortality if treatment is delayed. SE can be convulsive, with a regular pattern of contraction and extension of the arms and legs, or non-convulsive, with a change in a person's level of consciousness of relatively long duration but without large scale bending and extension of the limbs due to seizure activity. Convulsive SE (CSE) may be further classified into (a) tonic-clonic SE, (b) tonic SE, (c) clonic SE and (d) myoclonic SE. Non-convulsive SE (NCSE) is characterized by abnormal mental status, unresponsiveness, ocular motor abnormalities, persistent electrographic seizures, and possible response to anticonvulsants.

[0103] In embodiments, the seizure disorder is caused by exposure to or suspected exposure to nerve agents or organophosphate pesticides. G-series nerve agents include GA (tabun), GB (sarin), GD (soman), GF (cyclosarin), and GE (ethysarin). The V-series includes VX (O-Ethyl-S-[2-(diisopropylamino)ethyl]methylphosphonothioate), VE (O-Ethyl-S-[2-(diethylamino)ethyl]ethylphosphonothioate), VG (O,O-Diethyl-S-[2-(diethylamino)ethyl]phosphorothioate), VM (O-Ethyl-S-[2-(diethylamino)ethyl]methylphosphonothioate) and VR (N-diethyl-2-(methyl-(2-methylpropoxy)phosphoryl)sulfanylethanamine). VR is the compound from which Soviet newcomer agents (Novichok) Novichok 5 and Novichok 7, are derived.

[0104] In embodiments, a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, may be used to treat or reduce the toxic effects of exposure to one or more nerve agents or pesticides, including organophosphate nerve agents and organophosphate pesticides. In embodiments, a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, may be used to treat, reduce the risks of or prevent the effects of seizure disorders resulting from exposure to nerve agents and organophosphate pesticides. Nerve agents antagonize gamma-aminobutyric acid (GABA) neurotransmission, which in part mediates seizures and CNS neuropathology. See, e.g., Medscape, Drugs and Diseases >Emergency Medicine, Huebner et al., CBRNE—Nerve Agents, G-series—Tabun, Sarin, Soman, Pathophysiology (<https://emedicine.medscape.com/article/831648-overview?form=fpf#a5>). The GABAergic activity of compounds according to Formula (I), or pharmaceutically acceptable salts thereof, counteract the GABA antagonism of the nerve agents and organophosphate pesticides, thus restoring tonic inhibition, reducing neuronal hyperexcitability thereby reducing or eliminating seizures. In embodiments, a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered to a subject in advance of an exposure to a nerve agent, thereby preventing or reducing toxic effects of prospective exposure to one or more nerve agents or pesticides.

[0105] In embodiments, methods of treating a seizure disorder include administering a compound according to Formula (I) or a pharmaceutically acceptable salt thereof to a subject diagnosed with the seizure disorder, wherein the subject exhibits improvement in one or more symptoms of the disorder.

[0106] Symptoms of a seizure disorder may include, but are not limited to, episodes involving ataxia, gait impairment, speech impairment, vocalization, impaired cognition, abnormal motor activity, clinical seizure, subclinical seizure, hypotonia, hypertonia, drooling, and mouthing

behavior, aura, convulsions, repetitive movements, and unusual sensations. In embodiments, the methods and compositions provided may reduce or prevent one or more different types of seizures. Generally, a seizure can include convulsions, repetitive movements, unusual sensations, and combinations thereof. Seizures can be categorized as focal seizures (also referred to as partial seizures) and generalized seizures. Focal seizures affect only one side of the brain, while generalized seizures affect both sides of the brain. Specific types of focal seizures include simple focal seizures, complex focal seizures, and secondarily generalized seizures. Simple focal seizures can be restricted or focused on a particular lobe (e.g., temporal lobe, frontal lobe, parietal lobe, or occipital lobe). Complex focal seizures generally affect a larger part of one hemisphere than simple focal seizures, but commonly originate in the temporal lobe or the frontal lobe. When a focal seizure spreads from one side (hemisphere) to both sides of the brain, the seizure is referred to as a secondarily generalized seizure. Specific types of generalized seizures include absences (also referred to as petit mal seizures), tonic seizures, atonic seizures, myoclonic seizures, tonic clonic seizures (also referred to as grand mal seizures), and clonic seizures. Symptoms of a seizure disorder can include those associated with TSC such as seizures, cognitive impairment, autism, gelastic seizures—a disorder characterized by spells of involuntary laughter with interval irritability and depressed mood. Symptoms of a seizure disorder can include those associated with infantile spasms such as seizures, cognitive impairment, developmental regression, and hypsarrhythmia. Symptoms of a seizure disorder can include those associated with Lennox Gastaut syndrome such as seizures, developmental delays, cognitive impairment and behavioral disturbances. Symptoms of a seizure disorder can include those associated with CDKL5 disorder such as seizures, scoliosis, visual impairment, sensory issues, gastrointestinal difficulties, low or poor muscle tone, hand wringing movements or mouthing of the hands, marked developmental delay, limited or absent speech, lack of eye contact or poor eye contact, gastroesophageal reflux, constipation, small, cold feet, breathing irregularities such as hyperventilation, grinding of the teeth, episodes of laughing or crying for no reason, very limited hand skills, some autistic-like tendencies, cortical visual impairment (CVI), aka “cortical blindness”, apraxia, eating/drinking challenges, sleep difficulties, characteristics such as a sideways glance, and a habit of leg crossing. Symptoms of a seizure disorder can include those associated with focal cortical dysplasia such as seizures, cognitive impairment, autism, gelastic seizures, cortical blindness and behavioral disturbances such as irritability and depressed mood.

[0107] In embodiments, the terms “effective amount” or “therapeutically effective amount” as applied to a seizure disorder refer to an amount of a compound, material, composition, medicament, or other material that is effective to achieve a particular pharmacological and/or physiologic effect in connection with seizure disorder symptoms such as, but not limited to, one or more of the following: reducing or eliminating episodes involving ataxia, reducing or eliminating gait impairment, reducing or eliminating speech impairment, reducing or eliminating vocalization, reducing or eliminating impaired cognition, reducing or eliminating abnormal motor activity, reducing or eliminating clinical seizure, reducing or eliminating subclinical seizure, reducing or eliminating hypotonia, reducing or eliminating hypertonia, reducing or eliminating drooling, and mouthing behavior, reducing or eliminating aura, reducing or eliminating convulsions, reducing or eliminating repetitive movements, reducing or eliminating unusual sensations, reducing or eliminating one or more different types of seizures, reducing or eliminating convulsions, reducing or eliminating repetitive movements, reducing or eliminating unusual sensations, reducing or eliminating simple focal seizures, reducing or eliminating complex focal seizures, reducing or eliminating generalized seizures, reducing or eliminating absences (also referred to as petit mal seizures), reducing or eliminating tonic seizures, reducing or eliminating atonic seizures, reducing or eliminating myoclonic seizures, reducing or eliminating tonic clonic seizures (also referred to as grand mal seizures), reducing or eliminating clonic seizures.

[0108] As examples, reducing or eliminating TSC symptoms such as seizures, cognitive

impairment, autism, gelastic seizures—a disorder characterized by spells of involuntary laughter with interval irritability and depressed mood. As examples, reducing or eliminating symptoms associated with infantile spasms such as seizures, cognitive impairment, developmental regression, and hypsarrhythmia. As examples, reducing or eliminating symptoms associated with Lennox Gastaut syndrome such as seizures, developmental delays, cognitive impairment and behavioral disturbances. As examples, reducing or eliminating symptoms associated with CDKL5 disorder such as seizures, scoliosis, visual impairment, sensory issues, gastrointestinal difficulties, low or poor muscle tone, hand wringing movements or mouthing of the hands, marked developmental delay, limited or absent speech, lack of eye contact or poor eye contact, gastroesophageal reflux, constipation, small, cold feet, breathing irregularities such as hyperventilation, grinding of the teeth, episodes of laughing or crying for no reason, very limited hand skills, some autistic-like tendencies, cortical visual impairment (CVI), aka “cortical blindness”, apraxia, eating/drinking challenges, sleep difficulties, characteristics such as a sideways glance, and a habit of leg crossing. As examples, reducing or eliminating symptoms associated with focal cortical dysplasia such as seizures, cognitive impairment, autism, gelastic seizures, cortical blindness and behavioral disturbances such as irritability and depressed mood. In the case of treatment of benzodiazepine resistant seizures, reducing or eliminating benzodiazepine resistant seizures symptoms also includes reversing resistance to or tolerance to treatment of seizure disorders with benzodiazepines such as diazepam or lorazepam.

[0109] In embodiments, an effective amount results in enhancing cognitive function, increasing daytime activity, improving learning (either the rate or ease of learning), improving attention, improving social behavior, and/or improving cerebrovascular function. In embodiments, effective amount refers to an amount which may be suitable to prevent a decline in any one or more of the above qualities, or, in embodiments, to improve any one or more of the above qualities. In embodiments, an effective amount may be suitable to reduce either the extent or rate of decline in a subject's cognitive skills or functioning, and/or the effective amount may be suitable to delay the onset of such decline. Such effectiveness may be achieved, for example, by administering compositions described herein to an individual or to a population. In embodiments, the reduction, or delay of such a decline, or the improvement in an individual or population can be relative to a cohort, e.g., a control subject or a cohort population that has not received the treatment or been administered the composition or medicament. The terms “effective amount” and “therapeutically effective amount” are used interchangeably herein.

[0110] The dosage amount can vary according to a variety of factors such as subject-dependent variables (e.g., age, immune system, health, etc.), the disease or disorder being treated, as well as the route of administration and the pharmacokinetics of the agent being administered.

[0111] Many pharmaceutical products are administered as a fixed dose, at regular intervals, to achieve therapeutic efficacy. Duration of action is typically reflected by a drug's plasma half-life. Since efficacy is often dependent on sufficient exposure within the central nervous system administration of CNS drugs with a short half-life may require frequent maintenance dosing. CNS drugs with a long half-life may require less frequent maintenance dosing.

[0112] It should be understood that in the context of Formula (I) and other Formulas disclosed herein that unless otherwise indicated, the term “alkyl” includes both linear and branched chain alkyl groups. The prefix C.sub.p-q in C.sub.p-q alkyl and other terms (where p and q are integers) indicates the range of carbon atoms that are present in the group, for example C.sub.1-3alkyl includes C.sub.1alkyl (methyl), C.sub.2alkyl (ethyl) and C.sub.3alkyl (propyl as n-propyl and isopropyl).

[0113] The term “C.sub.p-q alkoxy” comprises —O—C.sub.p-q alkyl groups and —C.sub.p-q alkyl groups where the O atom is within the alkyl chain, for example, —CH.sub.2—O—CH.sub.3.

[0114] The term “C.sub.p-q alkenyl” includes both linear and branched chain alkyl groups containing at least two carbon atoms and at least one double carbon-carbon bond.

[0115] The term “C.sub.p-q alkenyloxy” comprises —O—C.sub.p-q alkenyl groups and —C.sub.p-q alkenyl groups where the O atom is within the alkenyl chain.

[0116] The term “C.sub.p-q alkynyl” includes both linear and branched chain alkyl groups containing at least two carbon atoms and at least one triple carbon-carbon bond.

[0117] The term “C.sub.p-q alkynyloxy” comprises —O—C.sub.p-q alkynyl groups and —C.sub.p-q alkynyl groups where the O atoms is within the alkynyl chain.

[0118] C.sub.p-q cycloalkyl refers to a cyclic non-aromatic group of p-q carbon atoms and no heteroatoms. For example, a 3 to 7 membered cycloalkyl refers to a ring containing 3 to 7 carbon atoms. Suitable C.sub.3-7cycloalkyl rings include cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl.

[0119] Aryl is a 6 to 10 membered monocyclic or bicyclic aromatic ring containing no heteroatoms. Aryl includes phenyl.

[0120] Heterocycloalkyl is a monocyclic saturated or partially unsaturated, non-aromatic ring having, for example, 3 to 7 members, such as 3 to 6 members, 5 to 7 members such as 5 or 6 members, where at least one member and up to 4 members, particularly 1, 2 or 3 members of the ring are heteroatoms selected from N, O and S, and the remaining ring atoms are carbon atoms, in stable combinations known to those of skill in the art. Heterocycloalkyl ring nitrogen and sulphur atoms are optionally oxidised. Suitable heterocycloalkyl rings include morpholinyl, thiazolidinyl, homomorpholine, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and tetrahydrofuranyl. In embodiments, when R.sup.7 is heterocycloalkyl, optionally two substituents on the same ring carbon together with the carbon to which they are attached form a 5 to 7 membered heterocycloalkyl ring, thereby creating a spirocyclic ring system. For example, in embodiments, R.sup.7 is morpholinyl and two substituents on the same ring carbon together form a tetrahydropyran.

[0121] Heteroaryl is a polyunsaturated, monocyclic 5 or 6 membered aromatic ring containing at least one and up to 3 heteroatoms, particularly, 1 or 2 heteroatoms selected from N, O and S, and the remaining ring atoms are carbon atoms. Heteroaryl ring nitrogen and sulphur atoms are optionally oxidised. Suitable heteroaryl rings include pyridinyl, isoxazolyl, oxadiazolyl, imidazolyl, pyrazinyl, oxazolyl, thiophenyl and thiazolyl.

[0122] The term “halo” is fluorine, chlorine or bromine.

[0123] The use of the dashed bond “custom-character” in rings A of Formula (I) represents the fusion of the pyrimidine ring.

[0124] Where the term “optionally” is used, it is intended that the subsequent feature may or may not occur. As such, use of the term “optionally” includes instances where the feature is present, and also instances where the feature is not present. For example, a group “optionally substituted with 1, 2 or 3 —F substituents” includes group with and without an —F substituent.

[0125] The term “substituted” means that one or more hydrogens (for example 1 or 2 hydrogens, or alternatively 1 hydrogen) on the designated group is replaced by the indicated substituent(s) (for example 1, 2 or 3 substituents, or alternatively 1 or 2 substituents, or alternatively 1 substituent), provided that any atom(s) bearing a substituent maintains a permitted valency. Substituent combinations encompass only stable compounds and stable synthetic intermediates. “Stable” means that the relevant compound or intermediate is sufficiently robust to be isolated and have utility either as a synthetic intermediate or as an agent having potential therapeutic utility. If a group is not described as “substituted”, or “optionally substituted”, it is to be regarded as unsubstituted (i.e. that none of the hydrogens on the designated group have been replaced).

[0126] The term “pharmaceutically acceptable” is used to specify that an object (for example a salt, dosage form or excipient) is suitable for use in patients. An example list of pharmaceutically acceptable salts can be found in the *Handbook of Pharmaceutical Salts: Properties, Selection and Use*, P. H. Stahl and C. G. Wermuth, editors, Weinheim/Zürich: Wiley-VCH/VHCA, 2002. In embodiments, a suitable pharmaceutically acceptable salt of a compound according to Formula (I)

is, for example, a salt formed within the human or animal body after administration of a compound according to Formula (I), to said human or animal body.

[0127] In embodiments, a suitable pharmaceutically acceptable salt of a compound according to Formula (I) is, for example, an acid addition salt. An acid addition salt of a compound according to Formula (I) may be formed by bringing the compound into contact with a suitable inorganic or organic acid under conditions known to the skilled person. In embodiments, compounds described herein may form base addition salts. A base-addition salt of a compound according to Formula (I) may be formed by bringing the compound into contact with a suitable inorganic or organic base under conditions known to the skilled person.

[0128] In embodiments, a compound according to Formula (I) may be provided as an acid addition salt, a zwitter ion hydrate, zwitter ion anhydrate, hydrochloride or hydrobromide salt, or in the form of the zwitter ion monohydrate. Acid addition salts, include but are not limited to, maleic, fumaric, benzoic, ascorbic, succinic, oxalic, bis-methylenesalicylic, methanesulfonic, ethane-disulfonic, acetic, propionic, tartaric, salicylic, citric, gluconic, lactic, malic, mandelic, cinnamic, citraconic, aspartic, stearic, palmitic, itaconic, glycolic, pantothenic, p-amino-benzoic, glutamic, benzene sulfonic or theophylline acetic acid addition salts, as well as the 8-halotheophyllines, for example 8-bromo-theophylline. In embodiments, inorganic acid addition salts, including but not limited to, hydrochloric, hydrobromic, hydroiodic, sulfuric, sulfamic, phosphoric or nitric acid addition salts may be used.

[0129] In embodiments, there is provided a compound according to Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, there is provided a compound according to Formula (I). In embodiments there is provided a pharmaceutically acceptable salt of a compound according to Formula (I).

[0130] Compounds and salts described herein may exist in solvated forms and unsolvated forms. For example, a solvated form may be a hydrated form, such as a hemi-hydrate, a monohydrate, a dihydrate, a trihydrate or an alternative quantity thereof. The description herein encompasses all such solvated and unsolvated forms of compounds according to Formula (I), particularly to the extent that such forms possess KCC2 modulating activity, as for example measured using the tests described herein.

[0131] The following embodiments of moiety A may be applied to the description of the compounds according to Formula (I), provided herein:

[0132] Moiety A is selected from:

##STR00014##

or a N-oxide thereof.

[0133] In embodiments, moiety A is

##STR00015##

[0134] In embodiments, moiety A is

##STR00016##

[0135] In embodiments, A is

##STR00017##

[0136] In embodiments, moiety A is

##STR00018##

[0137] In embodiments, moiety A is

##STR00019##

[0138] In embodiments, there is provided a compound according to Formula (II):

##STR00020##

or a pharmaceutically acceptable salt thereof, wherein R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4a}, R^{sup.4b} and R^{sup.7} are as defined for Formula (I).

[0139] In embodiments, there is provided a compound according to Formula (II) or a pharmaceutically acceptable salt thereof, wherein R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4a}, R^{sup.4b}

and R.sup.7 are as defined for Formula (I) and when R.sup.7 is morpholinyl, either: [0140] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; heteroaryl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy, wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and aryl is substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy, wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; or [0141] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; and heteroaryl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy, wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and R.sup.2 is selected from -halo and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3.

[0142] In embodiments, there is provided a compound according to Formula (II) or a pharmaceutically acceptable salt thereof, wherein: [0143] R.sup.1 is selected from —C.sub.2-6alkyl; —C.sub.2-6alkoxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; phenyl optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; —O— phenyl optionally substituted with 1 or 2 -halo substituents; —O—CH.sub.2-phenyl; and thiophenyl; wherein —C.sub.2-6alkyl and —C.sub.2-6alkoxy are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0144] R.sup.2 is selected from —H, —F and —CH.sub.3; [0145] R.sup.3 is selected from —C.sub.2-4alkynyl and —C.sub.1-3alkyl optionally substituted with —NR.sup.8R.sup.9; [0146] R.sup.4a and R.sup.4b are both —H; [0147] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1 or 2 substituents selected from —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy; cyclopropyl; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl; [0148] R.sup.8 and R.sup.9 are each independently selected from —C.sub.1-6alkyl; [0149] R.sup.10 is selected from —C.sub.1-3alkyl; [0150] R.sup.11 is selected from —C.sub.1-3alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0151] R.sup.12 is

selected from a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0152] n is 1 or 2.

[0153] In embodiments, there is provided a compound according to Formula (II) or a pharmaceutically acceptable salt thereof, wherein: [0154] R^{sup.1} is selected from —C_{sub.2-6}alkyl; —C_{sub.2-6}alkoxy; C_{sub.3-7}cycloalkyl; —O—C_{sub.3-7}cycloalkyl; phenyl substituted with 1 or 2 substituents selected from -halo, —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy wherein —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF_{sub.3}, —NHC(O)O—C_{sub.1-6}alkyl or two substituents together with the carbon to which they are attached form diazirinyl; —O-phenyl optionally substituted with 1 or 2 -halo substituents; —O—CH_{sub.2}-phenyl; and thiophenyl; wherein —C_{sub.2-6}alkyl and —C_{sub.2-6}alkoxy are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; and R^{sup.2} is selected from —H, —F and —CH_{sub.3}; or [0155] R^{sup.1} is selected from —C_{sub.2-6}alkyl; —C_{sub.2-6}alkoxy; C_{sub.3-7}cycloalkyl; —O—C_{sub.3-7}cycloalkyl; unsubstituted phenyl; —O-phenyl optionally substituted with 1 or 2 -halo substituents; —O—CH_{sub.2}-phenyl; and thiophenyl; wherein —C_{sub.2-6}alkyl and —C_{sub.2-6}alkoxy are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; and R^{sup.2} is selected from —F and —CH_{sub.3}; [0156] R^{sup.3} is selected from —C_{sub.2-4}alkynyl and —C_{sub.1-3}alkyl optionally substituted with —NR^{sup.8}R^{sup.9}; [0157] R^{sup.4a} and R^{sup.4b} are both —H; [0158] R^{sup.7} is selected from —NR^{sup.10}R^{sup.11}; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1 or 2 substituents selected from —CN; —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF_{sub.3} and —OH; —C_{sub.1-3}alkoxy; cyclopropyl; —C(O)OH; —C_{sub.1-3}alkylene-NHC(O)C_{sub.1-6}alkyl; —C_{sub.1-3}alkylene-NHC(O)OC_{sub.1-6}alkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl; [0159] R^{sup.8} and R^{sup.9} are each independently selected from —C_{sub.1-6}alkyl; [0160] R^{sup.10} is selected from —C_{sub.1-3}alkyl; [0161] R^{sup.11} is selected from —C_{sub.1-3}alkyl optionally substituted with 1 or 2 substituents selected from —F and —C_{sub.1-3}alkoxy; and —(CH_{sub.2})_{sub.n}R^{sup.12}; [0162] R^{sup.12} is selected from a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; and [0163] n is 1 or 2.

[0164] In embodiments, there is provided a compound according to Formula (III):

##STR00021##

or a pharmaceutically acceptable salt thereof, wherein R^{sup.1}, R^{sup.2}, R^{sup.5a}, R^{sup.5b}, R^{sup.5c}, R^{sup.5d} and R^{sup.7} are as defined for Formula (I).

[0165] In embodiments, there is provided a compound according to Formula (III) or a pharmaceutically acceptable salt thereof, wherein:

[0166] R^{sup.1} is selected from C_{sub.3-7}cycloalkyl and C_{sub.6-10}aryl, wherein the aryl is optionally substituted with a —C_{sub.2-8}alkoxy substituent wherein the alkoxy is optionally substituted with 1 or 2 —CF_{sub.3} substituents; [0167] R^{sup.2} is —H; [0168] R^{sup.5a}, R^{sup.5b}, R^{sup.5c} and R^{sup.5d} are each —H; [0169] R^{sup.7} is selected from —NR^{sup.10}R^{sup.11}; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C_{sub.1-6}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF_{sub.3} and —OH; —C_{sub.1-3}alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; —C(O)OH; —C_{sub.1-3}alkylene-NHC(O)C_{sub.1-6}alkyl; —C_{sub.1-3}alkylene-NHC(O)OC_{sub.1-4}alkyl; and C_{sub.3-5}cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when

R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H.
[0170] In embodiments, there is provided a compound according to Formula (III) or a pharmaceutically acceptable salt thereof, wherein: [0171] R.sup.1 is selected from C.sub.3-7cycloalkyl and C.sub.6-10aryl, wherein the aryl is optionally substituted with a —C.sub.2-8alkoxy substituent wherein the alkoxy is optionally substituted with 1 or 2 —CF.sub.3 substituents; [0172] R.sup.2 is —H; [0173] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each —H; [0174] R.sup.7 is selected from a 5 to 7 membered monocyclic heterocycloalkyl and a 5 or 6 membered monocyclic heteroaryl, wherein the heterocycloalkyl and heteroaryl are optionally substituted with a substituent selected from —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —OH; and cyclopropyl.

[0175] In embodiments, there is provided a compound according to Formula (IV):

##STR00022##

or a N-oxide or pharmaceutically acceptable salt thereof, wherein R.sup.1, R.sup.2, R.sup.6 and R.sup.7 are as defined for Formula (I).

[0176] In embodiments, there is provided a compound according to Formula (IV), or a N-oxide or pharmaceutically acceptable salt thereof, wherein: [0177] R.sup.1 selected from C.sub.3-7cycloalkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0178] R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0179] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0180] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; [0181] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl; [0182] R.sup.10 is —C.sub.1-6alkyl; [0183] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0184] R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0185] n is 1, 2 or 3.

[0186] In embodiments, there is provided a compound according to Formula (IV), or a N-oxide or pharmaceutically acceptable salt thereof, wherein: [0187] R.sup.1 selected from C.sub.3-7cycloalkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0188] R.sup.2 is —H; [0189] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0190] R.sup.7 is selected from a 5 to 7 membered monocyclic heterocycloalkyl optionally substituted with 1, 2 or 3 groups selected from —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; and C.sub.3-5cycloalkyl; [0191] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl.

[0192] The following embodiments of moieties R.sup.1, R.sup.2, R.sup.3, R.sup.4a, R.sup.4b, R.sup.4c, R.sup.4d, R.sup.5, R.sup.6a, R.sup.6b, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the description of the compounds

according to Formula (I) provided herein. The following embodiments of moieties R.sup.1, R.sup.2, R.sup.3, R.sup.4a, R.sup.4b, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the description of the compounds according to Formula (II) provided herein. The following embodiments of moieties R.sup.1, R.sup.2, R.sup.5a, R.sup.5b, R.sup.5c, R.sup.5d, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the description of the compounds according to Formula (III) provided herein. The following embodiments of moieties R.sup.1, R.sup.2, R.sup.6, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the descriptions of the compounds according to Formula (IV) provided herein.

[0193] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0194] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein —O—(CH.sub.2).sub.m—C.sub.6-10aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and C.sub.6-10aryl is substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl

[0195] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkoxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl and thiophenyl; wherein alkyl, alkoxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0196] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; C.sub.2-4alkoxy; C.sub.4-6cycloalkyl; —O—C.sub.4-6cycloalkyl; phenyl; —O—(CH.sub.2).sub.m-phenyl; and thiophenyl; wherein phenyl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are

attached form diazirinyl.

[0197] In embodiments, R^{sup.1} is selected from C_{sub.2-6}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; C_{sub.2-4}alkoxy; C_{sub.4-6}cycloalkyl; —O—C_{sub.4-6}cycloalkyl; phenyl; —O—(CH_{sub.2})_{sub.m}-phenyl; and thiophenyl; wherein —O—(CH_{sub.2})_{sub.m}-phenyl is optionally substituted with 1 or 2 substituents selected from -halo, —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy wherein —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF_{sub.3} and —NHC(O)O—C_{sub.1-6}alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and phenyl is substituted with 1 or 2 substituents selected from -halo, —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy wherein —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF_{sub.3} and —NHC(O)O—C_{sub.1-6}alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0198] In embodiments, R^{sup.1} is selected from —CF_{sub.2}CF_{sub.3}; propyl; butyl; pentyl; propoxy; cyclobutyl; cyclohexyl; —O-cyclopentyl; thiophenyl; phenyl; —O-phenyl; —O—CH_{sub.2}-phenyl; wherein phenyl is optionally substituted with 1 or 2 substituents selected from —F, —Cl, —CH_{sub.3}, —O—(CH_{sub.2})_{sub.5}C≡CH, —O—(CH_{sub.2})_{sub.7}, —O—(CH_{sub.2})_{sub.2}C(N=N)(CH_{sub.2})_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.2}NHC(O)OC(CH_{sub.3})_{sub.3}, —O—CH_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.5}CF_{sub.3} and —O—(CH_{sub.2})_{sub.7}.

[0199] In embodiments, R^{sup.1} is selected from —CF_{sub.2}CF_{sub.3}; propyl; butyl; pentyl; propoxy; cyclobutyl; cyclohexyl; —O-cyclopentyl; thiophenyl; phenyl substituted with 1 or 2 substituents selected from —F, —Cl, —CH_{sub.3}, —O—(CH_{sub.2})_{sub.5}C≡CH, —O—(CH_{sub.2})_{sub.7}, —O—(CH_{sub.2})_{sub.2}C(N=N)(CH_{sub.2})_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.2}NHC(O)OC(CH_{sub.3})_{sub.3}, —O—CH_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.5}CF_{sub.3} and —O—(CH_{sub.2})_{sub.7}; —O-phenyl; —O—CH_{sub.2}-phenyl; wherein —O-phenyl and —O—CH_{sub.2}-phenyl is optionally substituted with 1 or 2 substituents selected from —F, —Cl, —CH_{sub.3}, —O—(CH_{sub.2})_{sub.5}C≡CH, —O—(CH_{sub.2})_{sub.7}, —O—(CH_{sub.2})_{sub.2}C(N=N)(CH_{sub.2})_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.2}NHC(O)OC(CH_{sub.3})_{sub.3}, —O—CH_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.5}CF_{sub.3} and —O—(CH_{sub.2})_{sub.7}.

[0200] In embodiments, R^{sup.1} is cyclohexyl. In another embodiment, R^{sup.1} is phenyl substituted with —F, —Cl, —CH_{sub.3}, —O—(CH_{sub.2})_{sub.5}C≡CH, —O—(CH_{sub.2})_{sub.7}, —O—(CH_{sub.2})_{sub.2}C(N=N)(CH_{sub.2})_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.2}NHC(O)OC(CH_{sub.3})_{sub.3}, —O—CH_{sub.2}C≡CH, —O—(CH_{sub.2})_{sub.5}CF_{sub.3} and —O—(CH_{sub.2})_{sub.7}. In embodiments, R^{sup.1} is phenyl.

[0201] R^{sup.2} is selected from —H, -halo and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}. In embodiments, R^{sup.2} is —H. In embodiments, R^{sup.2} is -halo. In embodiments, R^{sup.2} is —F. In embodiments, R^{sup.2} is —C_{sub.1-3}alkyl. In embodiments, R^{sup.2} is methyl.

[0202] R^{sup.3} is selected from —H; —C_{sub.1-6}alkyl; —C_{sub.2-6}alkenyl; —C_{sub.2-6}alkynyl; —C_{sub.3-7}cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups, for example 1 or 2 groups, selected from —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF_{sub.3}, —C(O)NR^{sup.8}R^{sup.9} and —NR^{sup.8}R^{sup.9}.

[0203] In embodiments, R^{sup.3} is selected from —H; —C_{sub.2-4}alkynyl; —C_{sub.1-3}alkyl optionally substituted with —C(O)NR^{sup.8}R^{sup.9} or —NR^{sup.8}R^{sup.9}; and a 5 or 6 membered heterocycloalkyl optionally substituted with C_{sub.1-3}alkyl.

[0204] In embodiments, R^{sup.3} is selected from —H; —C_{sub.2-4}alkynyl; —C_{sub.1-3}alkyl optionally substituted with —C(O)NR^{sup.8}R^{sup.9} or —NR^{sup.8}R^{sup.9}; and a 5 or 6 membered

nitrogen containing heterocycloalkyl optionally substituted with C.sub.1-3alkyl.

[0205] In embodiments, R.sup.3 is selected from —H; —C.sub.2-4alkynyl; —C.sub.1-3alkyl optionally substituted with —C(O)NR.sup.8R.sup.9 or —NR.sup.8R.sup.9; and piperidinyl optionally substituted with C.sub.1-3alkyl.

[0206] In embodiments, R.sup.3 is selected from methyl, ethyl, i-propyl, —(CH.sub.2).sub.2N(CH.sub.3).sub.2, —(CH.sub.2).sub.3N(CH.sub.3).sub.2, —CH.sub.2C≡CH, —CH.sub.2C(O)N(CH.sub.3).sub.2 and N-methylpiperidine. In embodiments, R.sup.3 is selected from ethyl, i-propyl, —(CH.sub.2).sub.2N(CH.sub.3).sub.2, —(CH.sub.2).sub.3N(CH.sub.3).sub.2, —CH.sub.2C≡CH, —CH.sub.2C(O)N(CH.sub.3).sub.2 and N-methylpiperidine.

[0207] In embodiments, R.sup.3 is selected from —C.sub.2-4alkynyl and —C.sub.1-3alkyl optionally substituted with —NR.sup.8R.sup.9.

[0208] In embodiments, R.sup.3 is selected from ethyl, i-propyl, —(CH.sub.2).sub.2N(CH.sub.3).sub.2, —(CH.sub.2).sub.3N(CH.sub.3).sub.2 and —CH.sub.2C≡CH.

[0209] In embodiments, R.sup.3 is i-propyl.

[0210] R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.4a is methyl and R.sup.4b is —H. In embodiments, R.sup.4a and R.sup.4b are both —H.

[0211] R.sup.4c and R.sup.4d are each independently selected from —H and C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl. In embodiments, R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl. In embodiments, R.sup.4c and R.sup.4d are each independently selected from —H and C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.4c and R.sup.4d are both —H or together with the carbon to which they are attached represent carbonyl. In embodiments, R.sup.4c and R.sup.4d are both —H.

[0212] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl. In embodiments, R.sup.5a is methyl and R.sup.5b, R.sup.5c and R.sup.5d are each —H. In embodiments, R.sup.5a, R.sup.5b and R.sup.5c are each —H and R.sup.5d is methyl. In embodiments, R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d each represent —H.

[0213] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl. In embodiments, R.sup.6 is selected from —H; —Br; —NH.sub.2; —CN; methoxy; ethyl; —C(O)OCH.sub.3; —C(O)NH.sub.2; —C(O)OH; and —NHC(O)CH.sub.3.

[0214] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN; —C.sub.1-6 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3 alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to

which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl.

[0215] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl; a 5 or 6 membered monocyclic heteroaryl selected from pyridinyl, dihydropyranyl, oxazolyl, imidazolyl and thiazolyl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl.

[0216] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl; a 5 or 6 membered monocyclic heteroaryl selected from pyridinyl, dihydropyranyl, oxazolyl, imidazolyl and thiazolyl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN, methyl, ethyl, propyl, cyclopropyl, methoxy, —CH.sub.2CF.sub.3, —CH.sub.2OH, —CH.sub.2CH.sub.2OH, —C(O)OH, —(CH.sub.2).sub.2NHC(O)CH.sub.3 and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl.

[0217] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11 wherein R.sup.10 is selected from methyl, ethyl or propyl and R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.pR.sup.12; a 5 to 7 membered monocyclic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl; a 5 or 6 membered monocyclic heteroaryl selected from pyridinyl, dihydropyranyl, oxazolyl, imidazolyl and thiazolyl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1 or 2 groups selected from —CN, methyl, ethyl, propyl, cyclopropyl, methoxy, —CH.sub.2CF.sub.3, —CH.sub.2OH, —CH.sub.2CH.sub.2OH, —C(O)OH, —(CH.sub.2).sub.2NHC(O)CH.sub.3 and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form tetrahydropyranyl.

[0218] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11 wherein R.sup.10 is selected from methyl, ethyl or propyl and R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.nR.sup.12.

[0219] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11 wherein R.sup.10 is selected from methyl, ethyl or propyl; R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.nR.sup.12; n is 1 or 2; and R.sup.12 is selected from isoxazolyl, oxadiazolyl, cyclopropyl, pyrazinyl, tetrahydrofuranlyl and pyridinyl.

[0220] In embodiments, R.sup.7 is selected from a 5 to 7 membered monocyclic heterocycloalkyl optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl.

[0221] In embodiments, R.sup.7 is a 5 to 7 membered monocyclic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-

pyranyl wherein the heterocycloalkyl is optionally substituted with 1 or 2 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —CH.sub.2NHC(O)CH.sub.3; —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl.

[0222] In embodiments, R.sup.7 is a 5 to 7 membered monocyclic heterocycloalkyl optionally substituted with 1 or 2 substituents selected from methyl, ethyl, propyl, cyclopropyl, —CH.sub.2CH.sub.2OH, —CH.sub.2OH, —C(O)OH, —CH.sub.2CF.sub.3, and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form tetrahydropyran.

[0223] In embodiments, R.sup.7 is morpholinyl optionally substituted with 1 or 2 substituents selected from methyl, ethyl, propyl, cyclopropyl, —CH.sub.2CH.sub.2OH, —CH.sub.2OH, —C(O)OH, —CH.sub.2CF.sub.3, and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form tetrahydropyran (i.e. R.sup.7 becomes a spirocyclic group).

[0224] In embodiments, R.sup.7 is 2-methylmorpholin-4-yl.

[0225] R.sup.8 is selected from —H and —C.sub.1-6alkyl. In embodiments, R.sup.8 is selected from —H and —C.sub.1-3alkyl. In embodiments, R.sup.8 is —H. In embodiments, R.sup.8 is —C.sub.1-3alkyl. In embodiments, R.sup.8 is methyl.

[0226] R.sup.9 is selected from —H and —C.sub.1-6alkyl. In embodiments, R.sup.9 is selected from —H and —C.sub.1-3alkyl. In embodiments, R.sup.9 is —H. In embodiments, R.sup.9 is —C.sub.1-3alkyl. In embodiments, R.sup.9 is methyl.

[0227] R.sup.10 is —C.sub.1-6alkyl. In embodiments, R.sup.10 is —C.sub.1-3alkyl. In embodiments, R.sup.10 is methyl. In another embodiment, R.sup.10 is ethyl. In embodiments, R.sup.10 is propyl.

[0228] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; or —(CH.sub.2).sub.nR.sup.12. In embodiments, R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and ethoxy. In embodiments, R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.nR.sup.12. In embodiments, R.sup.11 is selected from —(CH.sub.2).sub.nR.sup.12.

[0229] R.sup.12 is selected from a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl. In embodiments, R.sup.12 is selected from isoxazolyl, oxadiazolyl, cyclopropyl, pyrazinyl, tetrahydrofuranyl and pyridinyl.

[0230] m is 0 or 1. In embodiments, m is 0. In embodiments, m is 1.

[0231] n is 1, 2 or 3. In embodiments, n is 1 or 2. In embodiments, n is 1. In embodiments, n is 2. In embodiments, n is 3.

[0232] In embodiments, the compound according to Formula (I) is selected from: [0233] 2-(diethylamino)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0234] 4-[(4-cyclohexylphenyl)amino]-2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0235] 6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-2-(1,3-thiazolidin-3-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0236] 2-[(2R,6S)-2,6-dimethylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0237] 6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-2-(thiomorpholin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0238] 2-[(2S)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0239] 2-[(2R)-2-

methylmorpholin-4-yl]-6-(propan-2-yl)-4-[[4-(propan-2-yl)phenyl]amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0240] 2-[(2S,6S)-2,6-dimethylmorpholin-4-yl]-6-(propan-2-yl)-4-[[4-(propan-2-yl)phenyl]amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0241] 2-(3-methylmorpholin-4-yl)-6-(propan-2-yl)-4-[[4-(propan-2-yl)phenyl]amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0242] 2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-4-[[4-(propan-2-yl)phenyl]amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0243] 4-[(4-cyclohexylphenyl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0244] 4-[(4-cyclohexylphenyl)amino]-2-(2-methylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0245] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0246] 4-[(4-cyclohexylphenyl)amino]-2-((2R)-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0247] 4-[(4-cyclohexylphenyl)amino]-2-((2S)-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0248] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-[2-(2,2,2-trifluoroethyl)morpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0249] tert-butyl {[2-(2R)-4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}morpholin-2-yl]methyl}carbamate; [0250] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-[2-(propan-2-yl)morpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0251] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-(1,3-thiazolidin-3-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0252] 4-[(4-cyclohexylphenyl)amino]-2-[(2-ethoxyethyl)(methyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0253] 4-[(4-cyclohexylphenyl)amino]-2-(2-ethylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0254] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[(1,2-oxazol-3-yl)methyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0255] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[2-(1,2,4-oxadiazol-3-yl)ethyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0256] 4-[(4-cyclohexylphenyl)amino]-2-(1,4-oxazepan-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0257] 4-[(4-cyclohexylphenyl)amino]-2-(1,9-dioxo-4-azaspiro[5.5]undecan-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0258] 4-[(4-cyclohexylphenyl)amino]-2-(3-methoxypyrrolidin-1-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0259] 4-[(4-cyclohexylphenyl)amino]-2-[2-(2-hydroxyethyl)morpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0260] 4-[(4-cyclohexylphenyl)amino]-2-(dipropylamino)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0261] 4-[(4-cyclohexylphenyl)amino]-2-[(cyclopropylmethyl)(methyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0262] 4-[(4-cyclohexylphenyl)amino]-2-[2-(hydroxymethyl)morpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0263] 4-[(4-cyclohexylphenyl)amino]-2-[3-(hydroxymethyl)morpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0264] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[(pyrazin-2-yl)methyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0265] 4-[(4-cyclohexylphenyl)amino]-2-(diethylamino)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0266] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[(oxolan-2-yl)methyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0267] 4-[(4-cyclohexylphenyl)amino]-2-[(2,2-difluoroethyl)(methyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0268] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[2-(pyridin-2-yl)ethyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0269] (3S)-4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}morpholine-3-carboxylic acid; [0270] N-[2-(4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}morpholin-2-yl)ethyl]acetamide; [0271] 6-(propan-2-yl)-4-[[4-(propan-2-yl)phenyl]amino]-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0272] 6-isopropyl-4-((4-isopropylphenyl)amino)-2-(pyridin-4-

yl)-5,6-dihydro-7H-pyrrolo [3,4-d]pyrimidin-7-one (Compound B); [0273] 4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}pyridine-2-carbonitrile; [0274] 4-[(4-cyclohexylphenyl)amino]-2-(2-cyclopropylpyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0275] 4-[(4-cyclohexylphenyl)amino]-2-(2-methoxypyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0276] 4-[(4-cyclohexylphenyl)amino]-2-(2-methylpyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one (Compound C); [0277] 4-[(4-cyclohexylphenyl)amino]-2-(3,6-dihydro-2H-pyran-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0278] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0279] 4-[(4-cyclohexylphenyl)amino]-2-(1-methyl-1H-pyrazol-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0280] 4-[(4-cyclohexylphenyl)amino]-2-(1,3-oxazol-5-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0281] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-(1,3-thiazol-5-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0282] 2-(3,6-dihydro-2H-pyran-4-yl)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0283] 4-{[4-(4-fluorophenoxy)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0284] 2-(2-cyclopropylmorpholin-4-yl)-4-({4'-[(hept-6-yn-1-yl)oxy][1,1'-biphenyl]-4-yl}amino)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0285] 2-(2-cyclopropylmorpholin-4-yl)-4-{{4'-heptyloxy}[1,1'-biphenyl]-4-yl}amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0286] 4-[(4'-{2-[3-(but-3-yn-1-yl)-3H-diaziren-3-yl]ethoxy}[1,1'-biphenyl]-4-yl)amino]-2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0287] 2-[(2R)-2-methylmorpholin-4-yl]-4-[(4-pentylphenyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0288] 4-{[4-(butan-2-yl)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0289] 4-{[4-(benzyloxy)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0290] 2-(2-cyclopropylmorpholin-4-yl)-4-{[4-(pentafluoroethyl)phenyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0291] 2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-4-[(4-propylphenyl)amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0292] 2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-[(4-propylphenyl)amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0293] 2-[(2R)-2-methylmorpholin-4-yl]-4-{[4-(pentafluoroethyl)phenyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0294] 2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-4-({4-[(propan-2-yl)oxy]phenyl}amino)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0295] 4-[(4-cyclobutylphenyl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0296] 4-{[4-(cyclopentyloxy)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0297] 2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(2,2,2-trifluoroethyl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0298] tert-butyl {2-[(4'-{[2-(2-cyclopropylmorpholin-4-yl)-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-yl]amino}[1,1'-biphenyl]-4-yl)oxy]ethyl}carbamate; [0299] 6-ethyl-2-[(2R)-2-methylmorpholin-4-yl]-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0300] 4-[(4-cyclohexylphenyl)amino]-6-ethyl-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0301] tert-butyl {2-[(4'-{[2-(morpholin-4-yl)-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-yl]amino}[1,1'-biphenyl]-4-yl)oxy]ethyl}carbamate; [0302] 4-[(4'-{2-[3-(but-3-yn-1-yl)-3H-diaziren-3-yl]ethoxy}[1,1'-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0303] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-({4'-[(prop-2-yn-1-yl)oxy][1,1'-biphenyl]-4-yl}amino)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0304] 4-[(4-cyclohexylphenyl)amino]-6-[3-(dimethylamino)propyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-

d]pyrimidin-7-one; [0305] 4-[(4-cyclohexylphenyl)amino]-6-[2-(dimethylamino)ethyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0306] 4-[(4-cyclobutylphenyl)amino]-6-[3-(dimethylamino)propyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0307] 4-[(4-cyclobutylphenyl)amino]-6-[2-(dimethylamino)ethyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0308] 2-(morpholin-4-yl)-4-[[4-(propan-2-yl)phenyl]amino]-6-(prop-2-yn-1-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0309] 4-[(4-cyclohexylphenyl)amino]-2-(oxan-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0310] 4-[(4-cyclohexylphenyl)amino]-2-(1H-imidazol-1-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0311] 2-(3,6-dihydro-2H-pyran-4-yl)-4-[(2'-methyl[1,1'-biphenyl]-4-yl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0312] 4-[(4'-{2-[3-(but-3-yn-1-yl)-3H-diaziren-3-yl]ethoxy}[1,1'-biphenyl]-4-yl)amino]-2-(3,6-dihydro-2H-pyran-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0313] 2-(3,6-dihydro-2H-pyran-4-yl)-4-[(2-fluoro[1,1'-biphenyl]-4-yl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0314] 2-(morpholin-4-yl)-4-[[4-(pentafluoroethyl)phenyl]amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0315] 4-[(2-fluoro[1,1'-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0316] 4-[(3',4'-dichloro[1,1'-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0317] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-[[4-(propan-2-yl)phenyl]amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0318] 4-[(4-tert-butylphenyl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0319] 4-[(2-methyl[1,1'-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0320] 4-[(4'-chloro[1,1'-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0321] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0322] N-(4-cyclobutylphenyl)-2-(3,6-dihydro-2H-pyran-4-yl)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0323] N-(4-cyclohexylphenyl)-2-(2-cyclopropylmorpholin-4-yl)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0324] 2-(2-cyclopropylmorpholin-4-yl)-N-[4'-(heptyloxy)[1,1'-biphenyl]-4-yl]-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0325] 2-[(2R)-2-methylmorpholin-4-yl]-N-{4'-[(6,6,6-trifluorohexyl)oxy][1,1'-biphenyl]-4-yl}-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0326] N-(4-cyclohexylphenyl)-2-(2-methylpyridin-4-yl)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0327] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine (Compound D); [0328] 6-bromo-N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine; [0329] N-(4-cyclohexylphenyl)-2-(3,6-dihydro-2H-pyran-4-yl)pyrido[2,3-d]pyrimidin-4-amine; [0330] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-8-oxo-8 λ 5~pyrido[2,3-d]pyrimidin-4-amine; [0331] N-(4-cyclohexylphenyl)-6-ethyl-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine; [0332] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carbonitrile (Compound H); [0333] methyl 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carboxylate; [0334] methyl (R)-4-[(4-cyclohexylphenyl)amino]-2-(2-methylmorpholino)pyrido[2,3-d]pyrimidine-6-carboxylate (Compound E); [0335] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carboxylic acid; [0336] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carboxamide; [0337] 4-[(4-cyclohexylphenyl)amino]-2-(2-cyclopropylmorpholin-4-yl)pyrido[2,3-d]pyrimidine-6-carboxamide; [0338] N-(4-cyclohexylphenyl)-6-methoxy-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine; [0339] (R)—N-(4-cyclohexylphenyl)-6-methoxy-2-(2-methylmorpholino)pyrido[2,3-d]pyrimidin-4-amine (Compound F); [0340] N-{4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-6-yl}acetamide; [0341] N-4-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-4,6-diamine; [0342] N-(4-cyclohexylphenyl)-2-

(morpholin-4-yl)-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-amine; [0343] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-amine; [0344] 2-{4-[(4-cyclohexylphenyl)amino]-2-(3,6-dihydro-2H-pyran-4-yl)-5,7-dihydro-6H-pyrrolo[3,4-d]pyrimidin-6-yl}-N,N-dimethylacetamide (Compound G); [0345] N-(4-cyclohexylphenyl)-2-(2-cyclopropylmorpholin-4-yl)-6-(1-methylpiperidin-4-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-amine; [0346] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-{[4-(thiophen-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0347] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-{[4-(thiophen-3-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0348] and pharmaceutically acceptable salts thereof.

[0349] In embodiments, a compound according to Formula (I) is

##STR00023##

known as (R)—N-(4-cyclohexylphenyl)-2-(2-methylmorpholino)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine (also referred herein to as N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine, or also referred to herein as Compound A or CmpA), or a pharmaceutically acceptable salt thereof.

[0350] In embodiments, the compound of Formula (I) is Compound B:

##STR00024##

6-isopropyl-4-((4-isopropylphenyl)amino)-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo [3,4-d]pyrimidin-7-one

Compound B

[0351] In embodiments, the compound of Formula (I) is Compound C:

##STR00025##

4-[(4-cyclohexylphenyl)amino]-2-(2-methylpyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one

Compound C

[0352] In embodiments, the compound of Formula (I) is Compound D:

##STR00026##

N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine

Compound D

[0353] In embodiments, the compound of Formula (I) is Compound E:

##STR00027##

methyl (R)-4-((4-cyclohexylphenyl)amino)-2-(2-methylmorpholino) pyrido[2,3-d]pyrimidine-6-carboxylate

Compound E

[0354] In embodiments, the compound of Formula (I) is Compound F:

##STR00028##

(R)—N-(4-cyclohexylphenyl)-6-methoxy-2-(2-methylmorpholino) pyrido[2,3-d]pyrimidin-4-amine

Compound F

[0355] In embodiments, the compound of Formula (I) is Compound G:

##STR00029##

2-(4-((4-cyclohexylphenyl)amino)-2-(3,6-dihydro-2H-pyran-4-yl)-5,7-dihydro-6H-pyrrolo[3,4-d]pyrimidin-6-yl)-N,N-dimethylacetamide

Compound G

[0356] In embodiments, the compound according to Formula (I) is:

##STR00030##

known as (R)-4-((4-cyclohexylphenyl)amino)-2-(2-methylmorpholino)pyrido [2,3-d]pyrimidine-6-carbonitrile (also referred to as 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carbonitrile or Compound H), or a pharmaceutically acceptable salt thereof.

[0357] Atoms of the compounds and salts described herein may exist as their isotopes. All

compounds of Formula (I) where an atom is replaced by one or more of its isotopes (for example a compound of Formula (I) where one or more carbon atoms is an ^{11}C or ^{13}C carbon isotope, or where one or more hydrogen atoms is a ^2H or ^3H isotope, or where one or more nitrogen atoms is a ^{15}N isotope or where one or more oxygen atoms is an ^{17}O or ^{18}O isotope) are encompassed herein.

[0358] Compounds herein may exist in one or more geometrical, optical, enantiomeric, and diastereomeric forms, including, but not limited to, cis- and trans-forms, E- and Z-forms, and R-, S- and meso-forms. Unless otherwise stated a reference to a particular compound includes all such isomeric forms, including racemic and other mixtures thereof. Where appropriate such isomers can be separated from their mixtures by the application or adaptation of known methods (e.g., chromatographic techniques and recrystallisation techniques). Where appropriate such isomers can be prepared by the application or adaptation of known methods. In embodiments, a single stereoisomer is obtained by isolating it from a mixture of isomers (e.g., a racemate) using, for example, chiral chromatographic separation.

[0359] In embodiments, a single stereoisomer is obtained through direct synthesis from, for example, a chiral starting material.

[0360] In embodiments, there is provided a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which is a single optical isomer being in an enantiomeric excess (% e.e.) of $\geq 95\%$, $\geq 98\%$ or $\geq 99\%$. In embodiments, the single optical isomer is present in an enantiomeric excess (% e.e.) of $\geq 99\%$.

[0361] In embodiments, there is provided an N-oxide of a compound according to Formula (I) as herein defined, or a pharmaceutically acceptable salt thereof.

[0362] Compounds according to Formula (I), where R^7 is $-\text{NR}^{10}\text{R}^{11}$ (i.e. R^7 is linked by an aliphatic N atom), may for example be prepared as described in WO2021/180952 by the reaction of a compound according to Formula (V):

##STR00031##

or a salt thereof, where R^1 , R^2 and A are as defined herein, with an amine. The reaction is conveniently performed in a suitable solvent and at a suitable temperature, for example, di-isopropylethylamine in dimethylsulfoxide at a temperature of 20-100° C., or TsOH in butanol at 80° C.

[0363] When R^7 is attached via a carbon atom, the compound according to Formula (I) can be made by the reaction of a compound according to Formula (V) with a boronic acid or ester of the a compound according to Formula (VI), where R^7 is as defined herein and each R is the same or different and represents $-\text{H}$, an aliphatic chain, or where together the two R groups form a ring with the boron and two oxygen atoms. The reaction is conveniently performed with a suitable base in the presence of a palladium catalyst and a solvent at a suitable temperature. For example, cesium carbonate or sodium carbonate and a palladium catalyst such as $\text{Pd}(\text{PPh}_3)_4$, in aqueous dioxane at a temperature in the range of 80-100° C.

##STR00032##

[0364] When R^7 is linked via an aromatic N atom, the compound according to Formula (I) can be made by reaction of a compound of Formula (V) with the anion of R^7 . For example, by reaction of the anion of imidazole, generated by treatment with a suitable base (for example, sodium hydride), in a suitable solvent (for example dimethylformamide), with a compound of Formula (V).

[0365] A compound according to the Formula (V) may be prepared from a compound according to Formula (VII), or a salt thereof, where A is as defined herein, and a compound according to Formula (VIII), or a salt thereof, where R^1 and R^2 are as defined herein, in the presence of a base in a suitable solvent (for example, di-isopropylethylamine in tert butanol or dimethylsulfoxide) and at a suitable temperature (for example 20-100° C.).

##STR00033##

[0366] A compound according to Formula (I) may also be made in one pot from the reaction between a compound according to Formula (VII) with the stepwise addition of a compound according to Formula (VIII) and an amine R^{sup.7}. The reaction is conveniently performed in the presence of a base (for example, di-isopropylethylamine) in a suitable solvent (dimethylsulfoxide) at a suitable temperature (for example, a temperature of 20-100° C.).

[0367] A compound according to Formula (VII) may be made, for example, from a compound according to Formula (IX). Suitable conditions for this transformation are heating at a temperature of about 80° C. in POCl₃ in the presence of an amine base such as diethylphenylamine.

##STR00034##

[0368] A compound according to the Formula (IX) may, for example, be prepared from a compound according to the Formula (X) by reaction with propan-2-amine and formaldehyde in a suitable solvent (for example, ethanol) at a suitable temperature (for example, a temperature of 0-80° C.).

##STR00035##

[0369] A compound according to the Formula (I), when R^{sup.5a} and R^{sup.5b} are both H, may also be made from reaction of a compound according to the Formula (XI), or a salt thereof, where R^{sup.1}, R^{sup.2} and R^{sup.7} are as defined herein, with a suitable amine, for example N, N-dimethylpropane-1,3-diamine. Suitable conditions for this reaction are HCl in ethanol at a temperature of about 190° C. in a sealed tube.

##STR00036##

[0370] It will be appreciated that certain of the various ring substituents in the compounds herein may be introduced by standard aromatic substitution reactions or generated by conventional functional group modifications either prior to or immediately following the processes mentioned above. For example, compounds according to Formula (I) may be converted into further compounds according to Formula (I) by conventional functional group modifications. Such reactions and modifications include, for example, introduction of a substituent by means of an aromatic substitution reaction, C—H activation reaction, reduction of substituents, alkylation of substituents and oxidation of substituents. The reagents and reaction conditions for such procedures are well known in the chemical art. Particular examples of aromatic substitution reactions include the introduction of a halogen group.

[0371] It will also be appreciated that in some of the reactions mentioned herein it may be necessary/desirable to protect any sensitive groups in the compounds. The instances where protection is necessary or desirable and suitable methods for protection are known to those skilled in the art. Conventional protecting groups may be used in accordance with standard practice (for illustration see T. W. Green, Protective Groups in Organic Synthesis, John Wiley and Sons, 1991). Thus, if reactants include groups such as amino, carboxy or hydroxy it may be desirable to protect the group in some of the reactions mentioned herein.

[0372] As a result of their KCC2 activation activity, the compounds according to Formula (I), and pharmaceutically acceptable salts thereof are useful in therapy, for example, in the treatment of seizure disorders mediated at least in part by KCC2. In embodiments, the compounds of Formula (I), or pharmaceutically acceptable salts thereof, increase KCC2 activity, diminish neuronal hyperexcitability, and provide a GABAergic effect, resulting in beneficial therapeutic effects on seizure disorders whose symptoms are related to insufficient GABA levels and/or neuronal hyperexcitability.

[0373] The term “therapy” is intended to have its normal meaning of dealing with a disease in order to entirely or partially relieve one, some or all of its symptoms, or to correct or compensate for the underlying pathology. In embodiments, the term “therapy” may also include “prophylaxis” if “prophylaxis” is specifically referred to. The term “prophylaxis” is intended to have its normal meaning and includes primary prophylaxis to prevent the development of the disease and secondary prophylaxis whereby the disease has already developed and the patient is temporarily or

permanently protected against exacerbation or worsening of the disease or the development of new symptoms associated with the disease. Nonetheless, prophylactic (preventive) and therapeutic (curative) treatment are two separate embodiments of the disclosure herein. In embodiments, the term “treatment” can be used synonymously with “therapy”. Similarly, the term “treat” can be regarded as “applying therapy” where “therapy” is as defined herein. As used herein, the terms “treat”, “treatment” or “treating” as applied to seizure disorders encompass any manner in which the symptoms or pathology of a condition, disorder or disease associated with seizure disorders are ameliorated or otherwise beneficially altered. In embodiments, “treat”, “treatment” or “treating” can refer to inhibiting a disease or condition, e.g., arresting or reducing its development or at least one clinical or subclinical symptom thereof. In embodiments, “treat”, “treatment” or “treating” can refer to relieving the disease or condition, e.g., causing regression of the disease or condition or at least one of its clinical or subclinical symptoms. In embodiments, “treating cognitive impairment” means ameliorating, beneficially altering and/or providing relief from one or more of the symptoms of cognitive impairment. The benefit to a subject being treated may be statistically significant, mathematically significant, or at least perceptible to the subject and/or the physician.

[0374] In embodiments, there is provided a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in therapy for seizure disorders. For example, in embodiments, Compound A, Compound B, Compound C, Compound D, Compound E, Compound F, Compound G, or Compound H, or a pharmaceutically acceptable salt of any of the foregoing, is for use in therapy for seizure disorders. In embodiments, there is provided the use of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, e.g., Compound A, Compound B, Compound C, Compound D, Compound E, Compound F, Compound G, or Compound H, or a pharmaceutically acceptable salt of any of the foregoing, for the manufacture of a medicament for treating a seizure disorder.

[0375] The term “therapeutically effective amount” refers to an amount of a compound according to Formula (I) as described herein which is effective to provide “therapy” in a subject, or to “treat” a disease or disorder in a subject. In the case of seizure disorders, the therapeutically effective amount may cause any of the changes observable or measurable in a subject as described in the definition of “therapy”, “treatment” and “prophylaxis” above. As recognized by those skilled in the art, effective amounts may vary depending on route of administration, excipient usage, and co-usage with other agents. For example, where a combination therapy is used, the amount of the compound according to Formula (I) or pharmaceutically acceptable salt described in this specification and the amount of the other pharmaceutically active agent(s) are, when combined, jointly effective to treat a targeted disorder in the animal patient. In this context, the combined amounts are in a “therapeutically effective amount” if they are, when combined, sufficient to decrease the symptoms of a seizure disorder responsive to activation of KCC2 as described herein. Typically, such amounts may be determined by one skilled in the art by, for example, starting with the dosage range described in this specification for the compound according to Formula (I) or pharmaceutically acceptable salt thereof and an approved or otherwise published dosage range(s) of the other pharmaceutically active compound(s). “Subjects” include mammals, for example, humans.

[0376] The compounds according to Formula (I), and pharmaceutically acceptable salts thereof, may be administered as pharmaceutical compositions, including one or more pharmaceutically acceptable excipients. Therefore, in embodiments, there is provided a pharmaceutical composition including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable excipient.

[0377] The excipient(s) selected for inclusion in a particular composition will depend on factors such as the mode of administration and the form of the composition provided. Suitable pharmaceutically acceptable excipients are well known to persons skilled in the art and are described, for example, in the *Handbook of Pharmaceutical Excipients*, Sixth edition,

Pharmaceutical Press, edited by Rowe, Ray C; Sheskey, Paul J; Quinn, Marian. Pharmaceutically acceptable excipients may function as, for example, adjuvants, diluents, carriers, stabilizers, flavorings, colorants, fillers, binders, disintegrants, lubricants, glidants, thickening agents and coating agents. As persons skilled in the art will appreciate, certain pharmaceutically acceptable excipients may serve more than one function and may serve alternative functions depending on how much of the excipient is present in the composition and what other excipients are present in the composition.

[0378] The pharmaceutical compositions may be in a form suitable for oral use (for example as tablets, lozenges, hard or soft capsules, films, dragees, aqueous or oily suspensions, emulsions, dispersible powders or granules, syrups or elixirs), for topical use (for example as creams, ointments, gels, lotions, transdermal patches, or aqueous or oily solutions or suspensions), for administration by inhalation (for example as a finely divided powder or a liquid aerosol), for administration by insufflation (for example as a finely divided powder) or for parenteral administration (for example as a sterile aqueous or oily solution or suspension for intravenous, subcutaneous or intramuscular dosing), or as a suppository for rectal dosing. The compositions may be obtained by conventional procedures using conventional pharmaceutical excipients well known in the art. Thus, compositions intended for oral use may contain, for example, one or more coloring, sweetening, flavoring and/or preservative agents.

[0379] In embodiments, a pharmaceutical composition herein may contain a compound according to Formula (I) in any of the amounts set forth herein and a diluent in an amount from 30% to 90% by weight. Examples of diluents include lactose, microcrystalline cellulose, starch, calcium phosphate, calcium carbonate, sucrose, mannitol, maltodextrin and sorbitol. In embodiments, a pharmaceutical composition herein may contain a compound according to Formula (I) and a lubricant in an amount, e.g., from 0.25% to 5.0% by weight. Examples of lubricants include magnesium stearate, stearic acid, sodium stearyl fumarate, talc, polyethylene glycols and silicon dioxide. Examples of water-soluble lubricants include sodium benzoate, polyethylene glycol, and adipic acid. In embodiments, a pharmaceutical composition herein may contain a compound according to Formula (I) and a disintegrant in an amount, e.g., from 1.0% to 10.0% by weight. There are two classes of disintegrants: traditional disintegrants, such as starch, and super disintegrants, which include croscarmellose sodium, crospovidone, and sodium starch glycolate.

[0380] The compound according to Formula (I) will normally be administered to a subject, e.g., a warm-blooded animal at a unit dose within the range 2.5-5000 mg/m² body area of the animal, or approximately 0.05-100 mg/kg, and this normally provides a therapeutically-effective dose. A unit dose form such as a tablet, capsule, film, patch, vial will can contain, for example 0.1-500 mg of active ingredient. The daily dose will necessarily be varied depending upon the host treated, the particular route of administration, any therapies being co-administered, and the severity of the illness being treated.

[0381] The pharmaceutical compositions described herein include compounds according to Formula (I), or a pharmaceutically acceptable salt thereof, for use in therapy for a seizure disorder.

[0382] As such, in embodiments, there is provided a pharmaceutical composition for use in therapy for a seizure disorder, including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable excipient.

[0383] In embodiments, there is provided a pharmaceutical composition for use in the treatment of a seizure disorder or condition in which activation of KCC2 is beneficial, including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable excipient.

[0384] In embodiments, treatment of a seizure disorder is effected by administering to a subject in need thereof about 0.01 mg to about 1500 mg according to Formula (I), or a pharmaceutically acceptable salt thereof, such as Compound A, Compound B, Compound C, Compound D, Compound E, Compound F, Compound G, or Compound H, or a pharmaceutically acceptable salt

of any of the foregoing. In embodiments, methods include treating a seizure disorder by administering to a subject in need thereof about 0.01 mg to about 1500 mg according to Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, the amount Formula (I), or a pharmaceutically acceptable salt thereof, can be, e.g., between 0.1 and 1500 mg/day, or 0.01 mg/kg/day to 15 mg/kg/day, for treatment of a seizure disorder. For example, the daily dosage can be, e.g., in the range of about 0.01 to 1500 mg, 0.1 to 1250 mg, 0.1 to 1000 mg, 0.1 to 750 mg, 0.1 to 500 mg, 0.1 to 450 mg, 0.1 to 300 mg, 0.1 to 250 mg, 0.1 to 200 mg, 0.1 to 175 mg, 0.1 to 150 mg, 0.1 to 125 mg, 0.1 to 100 mg, 0.1 to 75 mg, 0.1 to 50 mg, 0.1 to 30 mg, 0.1 to 25 mg, 0.1 to 20 mg, 0.1 to 15 mg, 0.1 to 10 mg, 0.1 to 5 mg, 0.1 to 1 mg, 1 to 1500 mg, 1 to 1000 mg, 1 to 500 mg, 1 to 300 mg, 1 to 250 mg, 1 to 200 mg, 1 to 175 mg, 1 to 150 mg, 1 to 125 mg, 1 to 100 mg, 1 to 75 mg, 1 to 50 mg, 1 to 30 mg, 1 to 25 mg, 1 to 20 mg, 1 to 15 mg, 1 to 10 mg, 1 to 5 mg, 5 to 1500 mg, 5 to 1000 mg, 5 to 500 mg, 5 to 300 mg, 5 to 250 mg, 5 to 200 mg, 5 to 175 mg, 5 to 150 mg, 5 to 125 mg, 5 to 100 mg, 5 to 75 mg, 5 to 50 mg, 5 to 30 mg, 5 to 25 mg, 5 to 20 mg, 5 to 15 mg, 5 to 10 mg, 10 to 1500 mg, 10 to 1000 mg, 10 to 500 mg, 10 to 300 mg, 10 to 250 mg, 10 to 200 mg, 10 to 175 mg, 10 to 150 mg, 10 to 125 mg, 10 to 100 mg, 10 to 75 mg, 10 to 50 mg, 10 to 30 mg, 10 to 25 mg, 10 to 20 mg, 10 to 15 mg, 15 to 1500 mg, 15 to 1000 mg, 15 to 500 mg, 15 to 300 mg, 15 to 250 mg, 15 to 200 mg, 15 to 175 mg, 15 to 150 mg, 15 to 125 mg, 15 to 100 mg, 15 to 75 mg, 15 to 50 mg, 15 to 30 mg, 15 to 25 mg, 15 to 20 mg, 20 to 1500 mg, 20 to 1000 mg, 20 to 500 mg, 20 to 300 mg, 20 to 250 mg, 20 to 200 mg, 20 to 175 mg, 20 to 150 mg, 20 to 125 mg, 20 to 100 mg, 20 to 75 mg, 20 to 50 mg, 20 to 30 mg, 20 to 25 mg, 25 to 1500 mg, 25 to 1000 mg, 25 to 500 mg, 25 to 300 mg, 25 to 250 mg, 25 to 200 mg, 25 to 175 mg, 25 to 150 mg, 25 to 125 mg, 25 to 100 mg, 25 to 75 mg, 25 to 50 mg, 25 to 30 mg, 30 to 1500 mg, 30 to 1000 mg, 30 to 500 mg, 30 to 300 mg, 30 to 250 mg, 30 to 200 mg, 30 to 175 mg, 30 to 150 mg, 30 to 125 mg, 30 to 100 mg, 30 to 75 mg, 30 to 50 mg, 35 to 1500 mg, 35 to 1000 mg, 35 to 500 mg, 35 to 300 mg, 35 to 250 mg, 35 to 200 mg, 35 to 175 mg, 35 to 150 mg, 35 to 125 mg, 35 to 100 mg, 35 to 75 mg, 35 to 50 mg, 40 to 1500 mg, 40 to 1000 mg, 40 to 500 mg, 40 to 300 mg, 40 to 250 mg, 40 to 200 mg, 40 to 175 mg, 40 to 150 mg, 40 to 125 mg, 40 to 100 mg, 40 to 75 mg, 40 to 50 mg, 50 to 1500 mg, 50 to 1000 mg, 50 to 500 mg, 50 to 300 mg, 50 to 250 mg, 50 to 200 mg, 50 to 175 mg, 50 to 150 mg, 50 to 125 mg, 50 to 100 mg, 50 to 75 mg, 75 to 1500 mg, 75 to 1000 mg, 75 to 500 mg, 75 to 300 mg, 75 to 250 mg, 75 to 200 mg, 75 to 175 mg, 75 to 150 mg, 75 to 125 mg, 75 to 100 mg, 100 to 1500 mg, 100 to 1000 mg, 100 to 500 mg, 100 to 300 mg, 100 to 250 mg, 100 to 200 mg, 100 to 175 mg, 100 to 150 mg, 100 to 125 mg, 125 to 1500 mg, 125 to 1000 mg, 125 to 500 mg, 125 to 300 mg, 125 to 250 mg, 125 to 200 mg, 125 to 175 mg, 125 to 150 mg, 150 to 1500 mg, 150 to 1000 mg, 150 to 500 mg, 150 to 300 mg, 150 to 250 mg, 150 to 200 mg, 150 to 175 mg, 175 to 1500 mg, 175 to 1000 mg, 175 to 500 mg, 175 to 300 mg, 175 to 250 mg, 175 to 200 mg, 200 to 1500 mg, 200 to 1000 mg, 200 to 500 mg, 200 to 300 mg, 200 to 250 mg, 250 to 1500 mg, 250 to 1000 mg, 250 to 500 mg, 250 to 300 mg, 7.5 to 15 mg, 2.5 to 5 mg, 1 to 5 mg, with doses of, e.g., about 0.25 mg, 0.5 mg, 0.75 mg, 1 mg, 1.2 mg, 1.5 mg, 1.75 mg, 2.0 mg, 2.5 mg, 3.0 mg, 3.5 mg, 4.0 mg, 4.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg, 15 mg, 17.5 mg, 20 mg, 22.5 mg, 25 mg, 27.5 mg, 30 mg, 35 mg, 40 mg, 45 mg, 50 mg, 75 mg, 100 mg, 125 mg, 150 mg, 175 mg, 200 mg, 225 mg, 250 mg, 275 mg, 300 mg, 400 mg and 500 mg being examples.

[0385] In embodiments, pharmaceutical compositions for treating a seizure disorder may include Formula (I), or a pharmaceutically acceptable salt thereof, in an amount of, e.g., about 0.01 to 1500 mg, 0.01 to 1250 mg, 0.01 to 1000 mg, 0.01 to 750 mg, 0.01 to 500 mg, 0.01 to 250 mg, 0.01 to 100 mg, 0.01 to 50 mg, 0.01 to 25 mg, 0.01 to 10 mg, 0.01 to 5 mg, 0.01 to 1 mg, 0.1 to 500 mg, 0.1 to 450 mg, 0.1 to 300 mg, 0.1 to 250 mg, 0.1 to 200 mg, 0.1 to 175 mg, 0.1 to 150 mg, 0.1 to 125 mg, 0.1 to 100 mg, 0.1 to 75 mg, 0.1 to 50 mg, 0.1 to 30 mg, 0.1 to 25 mg, 0.1 to 20 mg, 0.1 to 15 mg, 0.1 to 10 mg, 0.1 to 5 mg, 0.1 to 1 mg, 0.5 to 500 mg, 0.5 to 450 mg, 0.5 to 300 mg, 0.5 to 250 mg, 0.5 to 200 mg, 0.5 to 175 mg, 0.5 to 150 mg, 0.5 to 125 mg, 0.5 to 100 mg, 0.5 to 75 mg, 0.5 to 50 mg, 0.5 to 30 mg, 0.5 to 25 mg, 0.5 to 20 mg, 0.5 to 15 mg, 0.5 to 10 mg, 0.5 to 5 mg, 0.5

to 1 mg, 1 to 500 mg, 1 to 450 mg, 1 to 300 mg, 1 to 250 mg, 1 to 200 mg, 1 to 175 mg, 1 to 150 mg, 1 to 125 mg, 1 to 100 mg, 1 to 75 mg, 1 to 50 mg, 1 to 30 mg, 1 to 25 mg, 1 to 20 mg, 1 to 15 mg, 1 to 10 mg, 1 to 5 mg, 5 to 500 mg, 5 to 450 mg, 5 to 300 mg, 5 to 250 mg, 5 to 200 mg, 5 to 175 mg, 5 to 150 mg, 5 to 125 mg, 5 to 100 mg, 5 to 75 mg, 5 to 50 mg, 5 to 30 mg, 5 to 25 mg, 5 to 20 mg, 5 to 15 mg, 5 to 10 mg, 10 to 500 mg, 10 to 450 mg, 10 to 300 mg, 10 to 250 mg, 10 to 200 mg, 10 to 175 mg, 10 to 150 mg, 10 to 125 mg, 10 to 100 mg, 10 to 75 mg, 10 to 50 mg, 10 to 30 mg, 10 to 25 mg, 10 to 20 mg, 10 to 15 mg, 15 to 500 mg, 15 to 450 mg, 15 to 300 mg, 15 to 250 mg, 15 to 200 mg, 15 to 175 mg, 15 to 150 mg, 15 to 125 mg, 15 to 100 mg, 15 to 75 mg, 15 to 50 mg, 15 to 30 mg, 15 to 25 mg, 15 to 20 mg, 20 to 500 mg, 20 to 450 mg, 20 to 300 mg, 20 to 250 mg, 20 to 200 mg, 20 to 175 mg, 20 to 150 mg, 20 to 125 mg, 20 to 100 mg, 20 to 75 mg, 20 to 50 mg, 20 to 30 mg, 20 to 25 mg, 25 to 500 mg, 25 to 450 mg, 25 to 300 mg, 25 to 250 mg, 25 to 200 mg, 25 to 175 mg, 25 to 150 mg, 25 to 125 mg, 25 to 100 mg, 25 to 80 mg, 25 to 75 mg, 25 to 50 mg, 25 to 30 mg, 30 to 500 mg, 30 to 450 mg, 30 to 300 mg, 30 to 250 mg, 30 to 200 mg, 30 to 175 mg, 30 to 150 mg, 30 to 125 mg, 30 to 100 mg, 30 to 75 mg, 30 to 50 mg, 40 to 500 mg, 40 to 450 mg, 40 to 400 mg, 40 to 250 mg, 40 to 200 mg, 40 to 175 mg, 40 to 150 mg, 40 to 125 mg, 40 to 100 mg, 40 to 75 mg, 40 to 50 mg, 50 to 500 mg, 50 to 450 mg, 50 to 300 mg, 50 to 250 mg, 50 to 200 mg, 50 to 175 mg, 50 to 150 mg, 50 to 125 mg, 50 to 100 mg, 50 to 75 mg, 75 to 500 mg, 75 to 450 mg, 75 to 300 mg, 75 to 250 mg, 75 to 200 mg, 75 to 175 mg, 75 to 150 mg, 75 to 125 mg, 75 to 100 mg, 100 to 500 mg, 100 to 450 mg, 100 to 300 mg, 100 to 250 mg, 100 to 200 mg, 100 to 175 mg, 100 to 150 mg, 100 to 125 mg, 125 to 500 mg, 125 to 450 mg, 125 to 300 mg, 125 to 250 mg, 125 to 200 mg, 125 to 175 mg, 125 to 150 mg, 150 to 500 mg, 150 to 450 mg, 150 to 300 mg, 150 to 250 mg, 150 to 200 mg, 200 to 500 mg, 200 to 450 mg, 200 to 300 mg, 200 to 250 mg, 250 to 500 mg, 250 to 450 mg, 250 to 300 mg, 300 to 500 mg, 300 to 450 mg, 300 to 400 mg, 300 to 350 mg, 350 to 500 mg, 350 to 450 mg, 350 to 400 mg, 400 to 500 mg, 400 to 450 mg, with 0.1 mg, 0.25 mg, 0.5 mg, 0.75 mg, 1 mg, 2.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg, 15 mg, 17.5 mg, 20 mg, 22.5 mg, 25 mg, 30 mg, 35 mg, 40 mg, 45 mg, 50 mg, 55 mg, 60 mg, 65 mg, 70 mg, 75 mg, 80 mg, 85 mg, 90 mg, 95 mg, 100 mg, 125 mg, 150 mg, 175 mg, 200 mg, 225 mg, 250 mg, 275 mg, 300 mg, 325 mg, 350 mg, 375 mg, 400 mg, 425 mg, 450 mg, 475 mg, and 500 mg being examples.

[0386] Typically, dosages may be administered to a subject having a seizure disorder once, twice, three or four times daily, every other day, once weekly, or once a month. In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is administered to a subject having a seizure disorder twice a day, (e.g., morning and evening), three times a day (e.g., at breakfast, lunch, and dinner), or four times a day (e.g., breakfast, lunch, dinner and at bedtime) at a dose of 0.01-1000 mg/administration. In embodiments, the pharmaceutical compositions described herein may be administered by continuous infusion.

[0387] In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is administered to a subject having a seizure disorder 1500 mg/per day, 1400 mg/per day, 1300 mg/per day, 1200 mg/per day, 1000 mg/per day, 900 mg/per day, 800 mg/per day, 700 mg/day, 600 mg/per day, 500 mg/per day, 400 mg/per day, 300 mg/per day, 200 mg/per day, 100 mg/per day, 95 mg/per day, 90 mg/per day, 85 mg/per day, 80 mg/per day, 75 mg/per day, 70 mg/per day, 65 mg/per day, 60 mg/per day, 55 mg/per day, 50 mg/per day, 45 mg/per day, 40 mg/per day, 35 mg/per day, 30 mg/per day, 25 mg/per day, 20 mg/per day, 15 mg/per day, 10 mg/per day, 5 mg/per day, 4 mg/per day, 3 mg/per day, 3 mg/per day, 2 mg/per day, 1 mg/per day, in one or more doses. Dosages can be lower for infants and children than for adults. In embodiments, an infant or pediatric dose can be about 0.1 to 1500 mg per day once or in 2, 3 or 4 divided doses. In embodiments, a pediatric dose can be 0.05 mg/kg/day to 1500 mg/kg/day. In embodiments, the subject may be started at a low dose and the dosage is escalated over time. It should be understood that the above amounts are exemplary and doses according to Formula (I), or a pharmaceutically acceptable salt thereof, can include different amounts and varying ranges within a continuum between the minimum or maximum amounts described above in connection with Formula (I), or a pharmaceutically

acceptable salt thereof.

[0388] In embodiments, methods of treating a seizure disorder are provided which include administering to a subject in need thereof a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the seizure disorder for more than 1 hour after administration to the subject. In embodiments, methods of treating a seizure disorder are provided which include administering to a subject in need thereof a pharmaceutical composition including (Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the seizure disorder for more than 2 hours after administration to the subject. In embodiments, methods of treating a seizure disorder are provided which include administering to a subject in need thereof a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the seizure disorder for more than 3 hours after administration to the subject. In embodiments, methods of treating a seizure disorder are provided which include administering to a subject in need thereof a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the seizure disorder for more than 4 hours after administration to the subject. In embodiments, methods of treating a seizure disorder are provided which include administering to a subject in need thereof a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the seizure disorder for more than 6 hours after administration to the subject. In embodiments, methods of treating a seizure disorder are provided which include administering to a subject in need thereof a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the seizure disorder for more than 8, 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration to the subject. In embodiments, improvement in at least one symptom for 12 hours after administration of the pharmaceutical composition to the subject is provided in accordance with the present disclosure. In embodiments, the pharmaceutical compositions provide improvement in next day functioning of the subject. For example, the pharmaceutical compositions may provide improvement in one or more symptoms of the seizure disorder for more than about, e.g., 2 hours, 4 hours, 6 hours, 8 hours, 10 hours, 12 hours, 14 hours, 16 hours, 18 hours, 20 hours, 22 hours or 24 hours after administration and waking from a night of sleep.

[0389] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, in an effective amount, after a warning sign of an impending seizure is detected to reduce or prevent seizure activity. In embodiments, a continuing regimen of administration of active agents herein according to Formula (I), or a pharmaceutically acceptable salt thereof, is effective to reduce or prevent occurrence of seizure activity.

[0390] In embodiments, the methods described herein are effective to reduce, delay, or prevent one or more other clinical symptoms of a seizure disorder. For example, the effect of a composition including Formula (I), or a pharmaceutically acceptable salt thereof, on a particular symptom, pharmacologic, or physiologic indicator can be compared to an untreated subject, or the condition of the subject prior to treatment. In embodiments, the symptom, pharmacologic, and/or physiologic indicator is measured in a subject prior to treatment, and again one or more times after treatment is initiated. In embodiments, the control is a reference level, or average determined based on measuring the symptom, pharmacologic, or physiologic indicator in one or more subjects that do not have the disease or condition to be treated (e.g., healthy subjects). In embodiments, the effect of the treatment is compared to a conventional treatment that is known the art.

[0391] Effective treatment of a seizure disorder (e.g., acute repetitive seizure, status epilepticus, TSC, focal cortical dysplasia, temporal lobe epilepsy, diazepam resistant seizures, seizures from

exposure to nerve agents, etc.) herein may be established by showing reduction in the frequency or severity of symptoms (e.g., more than, e.g., 10%, 20%, 30% 40%, 50% or more) after a period of time compared with baseline. For example, after a baseline period of 1 month, the subjects may be randomly allocated Formula (I), or a pharmaceutically acceptable salt thereof, or placebo as add-on therapy to standard therapies, during a double-blind period of, e.g., 2 months.

[0392] In embodiments, primary outcome measurements may include the percentage of responders on Formula (I), or a pharmaceutically acceptable salt thereof, and on placebo, defined as having experienced at least a 10% to 50% or more reduction of symptoms during the second month of the double-blind period compared with baseline.

[0393] In embodiments, administration of (Formula (I), or a pharmaceutically acceptable salt thereof, may be used to prevent epilepsy when administration occurs before seizures manifest themselves in a subject. Detecting abnormal EEG signature in a subject provides a modality for predicting developing epilepsy in the subject. For example, patients diagnosed with KCC2 mutation and/or deficiency, e.g., in TSC, are at high risk of developing epilepsy, including infantile spasms and neurological deficits such as cognitive impairment. An abnormal EEG signature such as multifocal spikes in infants diagnosed with TSC serves to identify candidates for antiepileptogenic therapy with Formula (I), or a pharmaceutically acceptable salt thereof, which is implemented before seizures occur or shortly thereafter. In other words, antiepileptogenic treatment at an early stage of epileptogenesis is instituted. In this manner, the neurological outcome of the subject is improved. Improved neurological outcome refers to a decrease in or elimination of one or more symptoms associated with a seizure disorder. Accordingly, Formula (I), or a pharmaceutically acceptable salt thereof, is used to treat abnormal EEG signature.

[0394] Cognitive impairment (aka impairment of cognition) may be associated with subjects having a seizure disorders. Cognitive impairment may be measured against normal cognitive function, which refers to the normal physiologic activity of the brain, including, but not limited to, one or more of the following: mental stability, memory/recall abilities, problem solving abilities, reasoning abilities, thinking abilities, judging abilities, ability to discriminate or make choices, capacity for learning, ease of learning, perception, intuition, attention, and awareness, as measured by any criteria suitable in the art.

[0395] Cognitive impairment also includes deficits in mental activities that are mild or that otherwise do not significantly interfere with daily life. Mild cognitive impairment (MCI) is an example of such a condition. A subject with mild cognitive impairment may display symptoms of dementia (e.g., difficulties with language or memory) but the severity of these symptoms is such that a diagnosis of dementia may not be appropriate.

[0396] One skilled in the art will appreciate that there are numerous human and animal models that may be used to evaluate and compare the relative safety and efficacy of compounds described herein for the treatment of cognitive impairment associated with seizure disorders. In humans, cognitive function may be measured, for example and without limitation, by the clinical global impression of change scale (CGI); the Mini Mental State Exam (MMSE) (aka the Folstein Test); the Neuropsychiatric Inventory (NPI); the Clinical Dementia Rating Scale (CDR); the Cambridge Neuropsychological Test Automated Battery (CANTAB), the Sandoz Clinical Assessment-Geriatric (SCAG) scale, the Benton Visual Retention Test (BVRT), Montreal Cognitive Assessment (MoCA) or Digit Symbol Substitution Test (DSST).

[0397] In animal model systems, cognitive function may be measured in various conventional ways known in the art, including using a Morris Water Navigation Task, Barnes maze, radial arm maze task, T maze and the like. Other tests known in the art may also be used to assess cognitive function, such as novel object recognition and odor recognition tasks.

[0398] Cognitive function may also be measured using imaging techniques such as Positron Emission Tomography (PET), functional magnetic resonance imaging (fMRI), Single Photon Emission Computed Tomography (SPECT), or any other imaging technique that allows one to

measure brain function. In animals, cognitive function may also be measured with electrophysiological techniques.

[0399] In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is administered via a pharmaceutical composition for treatment of seizure disorders. Pharmaceutical compositions herein encompass dosage forms. Dosage forms herein encompass unit doses. In embodiments, as discussed below, various dosage forms including conventional formulations and modified release formulations can be administered one or more times daily. Any suitable route of administration may be utilized, e.g., oral, rectal, nasal, pulmonary, vaginal, sublingual, transdermal, intravenous, intraarterial, intramuscular, intraperitoneal and subcutaneous routes. As mentioned previously, suitable dosage forms include tablets, capsules, oral liquids, powders, aerosols, transdermal modalities such as topical liquids, patches, creams and ointments, parenteral formulations and suppositories.

[0400] In embodiments, provided herein are methods of a seizure disorder including administering to a subject in need thereof a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile according to Formula (I), or a pharmaceutically acceptable salt thereof, wherein the in vivo plasma profile according to Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration according to Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 50% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0401] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 55% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration. In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 55% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0402] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 60% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration. of a compound according to Formula (I) in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0403] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 65% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration. In embodiments,

provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of (Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 65% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0404] In embodiments, provided herein are methods of treating a seizure disorder wherein the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about 4 hours after administration of the pharmaceutical composition is less than about 75% of the administered dose. In embodiments, provided herein are methods wherein the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about, e.g., 6 hours, 8 hours, 10 hours, 12 hours, 15 hours, or 20 hours after administration of the pharmaceutical composition is less than about 75%.

[0405] In embodiments, provided herein are methods of treating a seizure disorder wherein the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about 4 hours after administration of the pharmaceutical composition is less than about 80% of the administered dose. In embodiments, provided herein are methods wherein the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about, e.g., 6 hours, 8 hours, 10 hours, 12 hours, 15 hours, or 20 hours after administration of the pharmaceutical composition is less than about 80% of the administered dose.

[0406] In embodiments, provided herein are methods of treating a seizure disorder wherein the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about 4 hours after administration of the pharmaceutical composition is between about 65% to about 85% of the administered dose. In embodiments, the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, within the subject after about, e.g., 6 hours, 8 hours, 10 hours, 12 hours, 15 hours, or 20 hours after administration of the pharmaceutical composition is between about 65% to about 85% of the administered dose.

[0407] In embodiments, as mentioned previously, pharmaceutical compositions herein may be provided with conventional release or modified release profiles. Pharmaceutical compositions may be prepared using a pharmaceutically acceptable “carrier” or “excipient” composed of materials that are considered safe and effective. The “carrier” includes all components present in the pharmaceutical formulation other than the active ingredient or ingredients. The term “carrier” includes, but is not limited to, diluents, binders, lubricants, disintegrants, fillers, and coating compositions. Those with skill in the art are familiar with such pharmaceutical carriers and methods of compounding pharmaceutical compositions using such carriers.

[0408] In embodiments, pharmaceutical compositions herein are modified release dosage forms which provide modified release profiles. Modified release profiles may exhibit immediate release, delayed release, or extended release profiles. Conventional (or unmodified) release oral dosage forms such as tablets, capsules, suppositories, syrups, solutions and suspensions typically release medications into the mouth, stomach or intestines as the tablet, capsule shell or suppository dissolves, or, in the case of syrups, solutions and suspensions, when they are swallowed. The pattern of drug release from modified release (MR) dosage forms is deliberately changed from that of a conventional dosage form to achieve a desired therapeutic objective and/or better patient compliance. Types of MR drug products include orally disintegrating dosage forms (ODDFs) which provide immediate release, extended release dosage forms, delayed release dosage forms (e.g., enteric coated), and pulsatile release dosage forms.

[0409] An ODDF is a solid dosage form containing a medicinal substance or active ingredient which disintegrates rapidly, usually within a matter of seconds when placed upon the tongue. The disintegration time for ODDFs generally range from one or two seconds to about a minute. ODDFs

are designed to disintegrate or dissolve rapidly on contact with saliva. This mode of administration can be beneficial to people who may have problems swallowing tablets whether it be from physical infirmity or psychiatric in nature. Subjects with seizure disorders may exhibit such behavior. ODDF's can provide rapid delivery of medication to the blood stream through mucosa resulting in a rapid onset of action. Examples of ODDFs include orally disintegrating tablets, capsules and rapidly dissolving films and wafers.

[0410] Extended release dosage forms (ERDFs) have extended release profiles and are those that allow a reduction in dosing frequency as compared to that presented by a conventional dosage form, e.g., a solution or unmodified release dosage form. ERDFs provide a sustained duration of action of a drug. Suitable formulations which provide extended release profiles are well-known in the art. For example, coated slow release beads or granules ("beads" and "granules" are used interchangeably herein) in which a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is applied to beads, e.g., confectioners nonpareil beads, and then coated with conventional release retarding materials such as waxes, enteric coatings and the like. In embodiments, beads can be formed in which a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is mixed with a material to provide a mass from which the drug leaches out. In embodiments, the beads may be engineered to provide different rates of release by varying characteristics of the coating or mass, e.g., thickness, porosity, using different materials, etc. Beads having different rates of release may be combined into a single dosage form to provide variable or continuous release. The beads can be contained in capsules or compressed into tablets.

[0411] In embodiments, modified dosage forms herein incorporate delayed release dosage forms having delayed release profiles. Delayed release dosage forms can include delayed release tablets or delayed release capsules. A delayed release tablet is a solid dosage form which releases a drug (or drugs) such as a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, at a time other than promptly after administration. A delayed release capsule is a solid dosage form in which the drug is enclosed within either a hard or soft soluble container made from a suitable form of gelatin, and which releases a drug (or drugs) at a time other than promptly after administration. For example, enteric-coated tablets, capsules, particles and beads are well-known examples of delayed release dosage forms. Enteric coated tablets, capsules and particles and beads pass through the stomach and release the drug in the intestine. In embodiments, a delayed release tablet is a solid dosage form containing a conglomerate of medicinal particles that releases a drug (or drugs) at a time other than promptly after administration. In embodiments, the conglomerate of medicinal particles are covered with a coating which delays release of the drug. In embodiments, a delayed release capsule is a solid dosage form containing a conglomerate of medicinal particles that releases a drug (or drugs) at a time other than promptly after administration. In embodiments, the conglomerate of medicinal particles is covered with a coating which delays release of the drug.

[0412] Delayed release dosage forms are known to those skilled in the art. For example, coated delayed release beads or granules in which a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is applied to beads, e.g., confectioners nonpareil beads, and then coated with conventional release delaying materials such as waxes, enteric coatings and the like. In embodiments, beads can be formed in which a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is mixed with a material to provide a mass from which the drug leaches out. In embodiments, the beads may be engineered to provide different rates of release by varying characteristics of the coating or mass, e.g., thickness, porosity, using different materials, etc. In embodiments, enteric coated granules of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, can be contained in an enterically coated capsule or tablet which releases the granules in the small intestine. In embodiments, the granules have a coating which remains intact until the coated granules reach at least the ileum and thereafter provide a delayed release of the drug in the colon. Suitable enteric coating materials are well known in the art, e.g., Eudragit® coatings such methacrylic acid and methyl methacrylate polymers and others.

The granules can be contained in capsules or compressed into tablets.

[0413] In embodiments, a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is incorporated into porous inert carriers that provide delayed release profiles. In embodiments, the porous inert carriers incorporate channels or passages from which the drug diffuses into surrounding fluids. In embodiments, a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is incorporated into an ion-exchange resin to provide a delayed release profile. Delayed action may result from a predetermined rate of release of the drug from the resin when the drug-resin complex contacts gastrointestinal fluids and the ionic constituents dissolved therein. In embodiments, membranes are utilized to control rate of release from drug containing reservoirs. In embodiments, liquid preparations may also be utilized to provide a delayed release profile. For example, a liquid preparation consisting of solid particles dispersed throughout a liquid phase in which the particles are not soluble. The suspension is formulated to allow at least a reduction in dosing frequency as compared to that drug presented as a conventional dosage form (e.g., as a solution or a prompt drug-releasing, conventional solid dosage form). For example, a suspension of ion-exchange resin constituents or microbeads.

[0414] In embodiments, pharmaceutical compositions described herein are suitable for parenteral administration, including, e.g., intramuscular (i.m.), intravenous (i.v.), subcutaneous (s.c.), intraperitoneal (i.p.), or intrathecal (i.t.). Parenteral compositions must be sterile for administration by injection, infusion or implantation into the body and may be packaged in either single-dose or multi-dose containers. In embodiments, liquid pharmaceutical compositions for parenteral administration to a subject include a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in any of the respective amounts described above. In embodiments, the pharmaceutical compositions for parenteral administration are formulated as a total volume of about, e.g., 10 ml, 20 ml, 25 ml, 50 ml, 100 ml, 200 ml, 250 ml, or 500 ml. In embodiments, the compositions are contained in a bag, a glass vial, a plastic vial, or a bottle.

[0415] In embodiments, pharmaceutical compositions for parenteral administration include respective amounts described above for a compound according to Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, pharmaceutical compositions for parenteral administration include about 0.05 mg to about 500 mg a compound according to Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, pharmaceutical compositions for parenteral administration to a subject include a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, at a respective concentration of about 0.005 mg/ml to about 500 mg/ml. In embodiments, the pharmaceutical composition for parenteral administration includes a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, at a respective concentration of, e.g., about 0.05 mg/ml to about 50 mg/ml, about 0.1 mg/ml to about 50 mg/ml, about 0.1 mg/ml to about 10 mg/ml, about 0.05 mg/ml to about 25 mg/ml, about 0.05 mg/ml to about 10 mg/ml, about 0.05 mg/ml to about 5 mg/ml, or about 0.05 mg/ml to about 1 mg/ml. In embodiments, the pharmaceutical composition for parenteral administration includes a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, at a respective concentration of, e.g., about 0.05 mg/ml to about 15 mg/ml, about 0.5 mg/ml to about 10 mg/ml, about 0.25 mg/ml to about 5 mg/ml, about 0.5 mg/ml to about 7 mg/ml, about 1 mg/ml to about 10 mg/ml, about 5 mg/ml to about 10 mg/ml, or about 5 mg/ml to about 15 mg/ml.

[0416] In embodiments, a pharmaceutical composition for parenteral administration is provided wherein the pharmaceutical composition is stable for at least six months. In embodiments, the pharmaceutical compositions for parenteral administration exhibit no more than about 5% decrease in a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, e.g., 3 months or 6 months. In embodiments, the amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, degrades at no more than about, e.g., 2.5%, 1%, 0.5% or 0.1%. In embodiments, the degradation is less than about, e.g., 5%, 2.5%, 1%, 0.5%, 0.25%, 0.1%,

for at least six months.

[0417] In embodiments, pharmaceutical compositions for parenteral administration are provided wherein the pharmaceutical composition remains soluble. In embodiments, pharmaceutical compositions for parenteral administration are provided that are stable, soluble, local site compatible and/or ready-to-use. In embodiments, the pharmaceutical compositions herein are ready-to-use for direct administration to a subject in need thereof.

[0418] The pharmaceutical compositions for parenteral administration provided herein may include one or more excipients, e.g., solvents, solubility enhancers, suspending agents, buffering agents, isotonicity agents, stabilizers or antimicrobial preservatives. When used, the excipients of the parenteral compositions will not adversely affect the stability, bioavailability, safety, and/or efficacy of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, used in the composition. Thus, parenteral compositions are provided wherein there is no incompatibility between any of the components of the dosage form.

[0419] In embodiments, parenteral compositions including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, include a stabilizing amount of at least one excipient. For example, excipients may be selected from the group consisting of buffering agents, solubilizing agents, tonicity agents, antioxidants, chelating agents, antimicrobial agents, and preservative. One skilled in the art will appreciate that an excipient may have more than one function and be classified in one or more defined group.

[0420] In embodiments, parenteral compositions including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, and an excipient wherein the excipient is present at a weight percent (w/v) of less than about, e.g., 10%, 5%, 2.5%, 1%, or 0.5%. In embodiments, the excipient is present at a weight percent between about, e.g., 1.0% to 10%, 10% to 25%, 15% to 35%, 0.5% to 5%, 0.001% to 1%, 0.01% to 1%, 0.1% to 1%, or 0.5% to 1%. In embodiments, the excipient is present at a weight percent between about, e.g., 0.001% to 1%, 0.01% to 1%, 1.0% to 5%, 10% to 15%, or 1% to 15%.

[0421] In embodiments, parenteral compositions may be administered as needed, e.g., once, twice, thrice or four or more times daily, or continuously depending on the subject's needs.

[0422] In embodiments, parenteral compositions of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, are provided, wherein the pH of the composition is between about 4.0 to about 8.0. In embodiments, the pH of the compositions is between, e.g., about 5.0 to about 8.0, about 6.0 to about 8.0, about 6.5 to about 8.0. In embodiments, the pH of the compositions is between, e.g., about 6.5 to about 7.5, about 7.0 to about 7.8, about 7.2 to about 7.8, or about 7.3 to about 7.6. In embodiments, the pH of the aqueous solution is, e.g., about 6.8, about 7.0, about 7.2, about 7.4, about 7.6, about 7.7, about 7.8, about 8.0, about 8.2, about 8.4, or about 8.6.

[0423] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a pharmaceutical composition including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in a respective amount described herein, wherein the composition provides an in vivo plasma profile having a C.sub.max less than about 800 ng/ml. In embodiments, the composition provides improvement for more than 6 hours after administration to the subject.

[0424] In embodiments, pharmaceutical compositions including a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, provide an in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, having a C.sub.max less than about, e.g., 2000 ng/ml, 1000 ng/ml, 850 ng/ml, 800 ng/ml, 750 ng/ml, 700 ng/ml, 650 ng/ml, 600 ng/ml, 550 ng/ml, 450 ng/ml, 400 ng/ml 350 ng/ml, or 300 ng/ml and wherein the composition provides improvement of next day functioning of the subject. In embodiments, the pharmaceutical composition provides an in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, having a C.sub.max less than about, e.g., 250 ng/ml, 200

ng/ml 150 ng/ml, or 100 ng/ml and wherein the composition provides improvement of next day functioning of the subject. In embodiments, the pharmaceutical composition provides improvement in one or more symptoms of a seizure disorder herein for more than 6 hours after administration. [0425] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides a consistent in vivo plasma profile having a AUC.sub.0-∞ of less than about 900 ng.Math.hr/ml. In embodiments, the pharmaceutical composition provides improvement in next day functioning of the subject. In embodiments, the compositions provide an in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 850 ng.Math.hr/ml, 800 ng.Math.hr/ml, 750 ng.Math.hr/ml, or 700 ng.Math.hr/ml and wherein the pharmaceutical composition provides improvement of next day functioning of the subject. In embodiments, the composition provides improvement in one or more symptoms of a seizure disorder herein for more than 6 hours after administration.

[0426] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a pharmaceutical composition comprising a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides an in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 650 ng.Math.hr/ml, 600 ng.Math.hr/ml, 550 ng.Math.hr/ml, 500 ng.Math.hr/ml, or 450 ng.Math.hr/ml. In embodiments, the composition provides an in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 400 ng.Math.hr/ml, 350 ng.Math.hr/ml, 300 ng.Math.hr/ml, 250 ng.Math.hr/ml, or 200 ng.Math.hr/ml. In embodiments, the pharmaceutical composition provides an in vivo plasma profile of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 150 ng.Math.hr/ml, 100 ng.Math.hr/ml, 75 ng.Math.hr/ml, or 50 ng.Math.hr/ml. In embodiments, the pharmaceutical composition provides improvement of next day functioning of the subject after administration for more than, e.g., 4 hours, 6 hours, 8 hours, 10 hours, or 12 hours, after administration of the composition to the subject.

[0427] In embodiments, the T.sub.max provided by a pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is less than 3 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is less than 2.5 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is less than 2 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is less than 1.5 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is less than 1 hour. In embodiments, the T.sub.max provided by the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is less than 0.5 hour. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 0.25 hour.

[0428] In embodiments, the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, provides a dissolution of at least about 80% within the first 20 minutes of administration to a subject in need thereof. In embodiments, the pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, provides a dissolution of at least about, e.g., 85%, 90% or 95% within the first 20 minutes of administration to a subject in need thereof. In embodiments, the

pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, provides a dissolution of at least 80% within the first 10 minutes of administration to a subject in need thereof.

[0429] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a first pharmaceutical dosage including a sub-therapeutic amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, treating a seizure disorder includes administering to a subject in need thereof a pharmaceutical composition containing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, in a sub-therapeutic amount, wherein the composition provides improvement in one or more symptoms of the disorder for more than 6 hours after administration.

[0430] A sub-therapeutic dosage is an amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, that is less than the amount typically required for a therapeutic effect. In embodiments, a sub-therapeutic dosage is an amount of Formula (I), or a pharmaceutically acceptable salt thereof, that alone may not provide improvement in at least one symptom of a seizure disorder but is sufficient to maintain such improvement. In embodiments, the methods provide administering a first pharmaceutical composition containing an effective amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, that provides improvement in at least one symptom of a seizure disorder and a second composition containing a subtherapeutic amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, that maintains the improvement. In embodiments, after administration of the first pharmaceutical composition, the second pharmaceutical composition may provide a synergistic effect to improve at least one symptom of a seizure disorder.

[0431] In embodiments, provided herein are methods of treating a seizure disorder including administering to a subject in need thereof a first pharmaceutical composition including a first pharmaceutical dosage of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, wherein the first pharmaceutical dosage provides improvement for more than 6 hours after administration, and a second pharmaceutical composition including a sub-therapeutic dosage of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof.

[0432] In embodiments, the first or the second pharmaceutical composition are provided to the subject once in the evening and once in the morning. In embodiments, the total amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, administered to a subject in a 24-hour period is any of the respective amounts described herein.

[0433] In embodiments, the first and/or the second pharmaceutical compositions may be provided with conventional release or modified release profiles. The first and second pharmaceutical compositions may be provided at the same time or separated by an interval of time, e.g., 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 12 hours, etc. In embodiments, the first and the second pharmaceutical compositions may be provided with different drug release profiles to create a two-phase release profile. For example, the first pharmaceutical composition may be provided with an immediate release profile, e.g., ODDF, parenteral, etc., and the second pharmaceutical composition may provide an extended release profile. In embodiments, one or both of the first and second pharmaceutical compositions may be provided with an extended release or delayed release profile. Such compositions may be provided as pulsatile formulations, multilayer tablets or capsules containing tablets, beads, granules, etc. In embodiments, the first pharmaceutical composition is an immediate release composition. In embodiments, the second pharmaceutical composition is an immediate release composition. In embodiments, the first and second pharmaceutical compositions are provided as separate immediate release compositions, e.g., film, tablets or capsules. In embodiments the first and second pharmaceutical compositions are provided 12 hours apart.

[0434] It should be understood that respective dosage amounts of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, that are provided herein are applicable to

all the dosage forms described herein including conventional dosage forms, modified dosage forms, the first and second pharmaceutical compositions, as well as the parenteral formulations described herein. Those skilled in the art will determine appropriate amounts depending on criteria such as dosage form, route of administration, subject tolerance, efficacy, therapeutic goal and therapeutic benefit, among other pharmaceutically acceptable criteria.

[0435] Combination therapies utilizing a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, can include administration of the active agents together in the same admixture, or in separate admixtures. In embodiments, the pharmaceutical composition can include two, three, or more active agents. In embodiments, the combinations result in a more than additive effect on the treatment of the disease or disorder. Thus, treatment is provided for a seizure disorder with a combination of agents that combined, may provide a synergistic effect that enhances efficacy.

[0436] Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosure herein belongs.

[0437] The term “about” or “approximately” as used herein means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e., the limitations of the measurement system. For example, “about” can mean within 3 or more than 3 standard deviations, per the practice in the art. Alternatively, “about” can mean a range of up to 20%, up to 10%, up to 5%, and/or up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value.

[0438] “Improvement” refers to the treatment of seizure disorders such as epilepsy, epilepsy with generalized tonic-clonic seizures, epilepsy with myoclonic absences, frontal lobe epilepsy, temporal lobe epilepsy, focal cortical dysplasia, Landau-Kleffner Syndrome, Rasmussen's syndrome, Dravet syndrome, Doose syndrome, CDKL5 disorder, infantile spasms (West syndrome), juvenile myoclonic epilepsy (JME), vaccine-related encephalopathy, intractable childhood epilepsy (ICE), Lennox-Gastaut syndrome (LGS), Rett syndrome, Ohtahara syndrome, childhood absence epilepsy, essential tremor, acute repetitive seizures, benign rolandic epilepsy, status epilepticus, refractory status epilepticus, super-refractory status epilepticus (SRSE), PCDH19 pediatric epilepsy, benzodiazepine resistant seizures including diazepam resistant seizures or lorazepam resistant seizures, seizures induced by exposure to nerve agents, seizures induced by exposure to pesticides, abnormal EEG signature, drug withdrawal induced seizures, alcohol withdrawal induced seizures, increased seizure activity or breakthrough seizures (also called serial or cluster seizures), measured relative to at least one symptom of the foregoing disorders.

[0439] “Improvement in next day functioning” or “wherein there is improvement in next day functioning” refers to improvement after waking from an overnight sleep period wherein the beneficial effect of administration of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, applies to at least one symptom of a syndrome or disorder herein and is discernable, either subjectively by a subject or objectively by an observer, for a period of time, e.g., 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 12 hours, 24 hours, etc. after waking.

[0440] “PK” refers to the pharmacokinetic profile. C.sub.max is defined as the highest plasma drug concentration estimated during an experiment (ng/ml). T.sub.max is defined as the time when C.sub.max is estimated (min). AUC.sub.0-∞ is the total area under the plasma drug concentration-time curve, from drug administration until the drug is eliminated (ng.Math.hr/ml or µg.Math.hr/ml). The area under the curve is governed by clearance. Clearance is defined as the volume of blood or plasma that is totally cleared of its content of drug per unit time (ml/min).

[0441] “Pharmaceutically acceptable” refers to molecular entities and compositions that are “generally regarded as safe”, e.g., that are physiologically tolerable and do not typically produce an allergic or similar untoward reaction, such as gastric upset and the like, when administered to a

human. In embodiments, this term refers to molecular entities and compositions approved by a regulatory agency of the federal or a state government, as the GRAS list under section 204(s) and 409 of the Federal Food, Drug and Cosmetic Act, that is subject to premarket review and approval by the FDA or similar lists, the U.S. Pharmacopeia or another generally recognized pharmacopeia for use in animals, and more particularly in humans.

[0442] “Effective amount” or “therapeutically effective amount”, previously referred to, can also mean a dosage sufficient to alleviate one or more symptoms of a syndrome, disorder, disease, or condition being treated, or to otherwise provide a desired pharmacological and/or physiologic effect. “Effective amount” or “therapeutically effective amount” may be used interchangeably herein.

[0443] “Co-administered with”, “administered in combination with”, “a combination of”, “in combination with” or “administered along with” may be used interchangeably and mean that two or more agents are administered in the course of therapy. The agents may be administered together at the same time or separately in spaced apart intervals. The agents may be administered in a single dosage form or in separate dosage forms.

[0444] “Subject in need thereof” may include individuals that have been diagnosed with a seizure disorder such as epilepsy, epilepsy with generalized tonic-clonic seizures, epilepsy with myoclonic absences, frontal lobe epilepsy, temporal lobe epilepsy, focal cortical dysplasia, Landau-Kleffner Syndrome, Rasmussen's syndrome, Dravet syndrome, Doose syndrome, CDKL5 disorder, infantile spasms (West syndrome), juvenile myoclonic epilepsy (JME), vaccine-related encephalopathy, intractable childhood epilepsy (ICE), Lennox-Gastaut syndrome (LGS), Rett syndrome, Ohtahara syndrome, CDKL5 disorder, childhood absence epilepsy, essential tremor, acute repetitive seizures, benign rolandic epilepsy, status epilepticus, refractory status epilepticus, super-refractory status epilepticus (SRSE), PCDH19 pediatric epilepsy, benzodiazepine resistant seizures including diazepam resistant seizures or lorazepam resistant seizures, seizures induced by exposure to nerve agents, seizures induced by exposure to pesticides, abnormal EEG signature, drug withdrawal induced seizures, alcohol withdrawal induced seizures, increased seizure activity or breakthrough seizures (also called serial or cluster seizures). The methods may be provided to any individual including, e.g., wherein the subject is a neonate, infant, a pediatric subject (6 months to 12 years), an adolescent subject (age 12-18 years) or an adult (over 18 years). Subjects include mammals. “Subject” and “patient” are used interchangeably herein.

[0445] “Prodrug” refers to a pharmacological substance (drug) that is administered to a subject in an inactive (or significantly less active) form. Once administered, the prodrug is metabolized in the body (in vivo) into a compound having the desired pharmacological activity.

[0446] “Analog” and “Derivative” may be used interchangeably and refer to a compound that possesses the same core as the parent compound, but may differ from the parent compound in bond order, the absence or presence of one or more atoms and/or groups of atoms, and combinations thereof. The derivative can differ from the parent compound, for example, in one or more substituents present on the core, which may include one or more atoms, functional groups, or substructures. In general, a derivative can be imagined to be formed, at least theoretically, from the parent compound via chemical and/or physical processes.

[0447] The term “pharmaceutically acceptable salt”, as used herein, refers to derivatives of the compounds defined herein, wherein the parent compound is modified by making acid or base salts thereof. Example of pharmaceutically acceptable salts include but are not limited to mineral or organic acid salts of basic residues such as amines; and alkali or organic salts of acidic residues such as carboxylic acids. The pharmaceutically acceptable salts include the conventional non-toxic salts or the quaternary ammonium salts of the parent compound formed, for example, from non-toxic inorganic or organic acids. Such conventional non-toxic salts include but are not limited to those derived from inorganic acids such as hydrochloric, hydrobromic, sulfuric, sulfamic, phosphoric, and nitric acids; and the salts prepared from organic acids such as acetic, propionic,

succinic, glycolic, stearic, lactic, malic, tartaric, citric, ascorbic, pantoic, maleic, hydroxymaleic, phenylacetic, glutamic, benzoic, salicylic, sulfanilic, 2-acetoxybenzoic, fumaric, toluenesulfonic, naphthalenesulfonic, methanesulfonic, ethane disulfonic, oxalic, and isethionic salts. The pharmaceutically acceptable salts can be synthesized from the parent compound, which contains a basic or acidic moiety, by conventional chemical methods.

EXAMPLES

[0448] The examples provided herein are included solely for augmenting the disclosure herein and should not be considered to be limiting in any respect.

[0449] The following Examples establish: Compound A is capable of directly binding to KCC2, as measured using cellular thermal shift assay (CETSA). Compound A potently increases KCC2 activity without modifying its plasma membrane stability or the phosphorylation of key regulatory sites, as measured by patch-clamp recording and TI flux. This contrasts to other indirect potentiators of KCC2, which have been shown to increase expression levels on the plasma membrane, enhance its expression, increase transcription of the SLC12A5 gene, or modify its phosphorylation (Gagnon et al., *Nat. Med.* 19, 1524-1528 (2013); Lee et al., *Brain: a journal of neurology* 145, 950-963 (2022); Prael Iii et al., *Front Cell Dev Biol* 10, 912812 (2022); Tang et al., *Sci. Transl. Med.* 11, 10.1126 (2019)). Consistent with its role in limiting neuronal excitability, Compound A reduced neuronal Cl_{sup}- accumulation and the development of late recurrent discharges (LRDs) in acute brain slices exposed to 0-Mg. Compound A rapidly accumulated in the brain when dosed IV, IP, or SC to concentrations ranging from 40-0.6 μ M, dependent upon the route of administration. Importantly, as measured in the open field, Compound A appeared to have no gross effects on mouse behavior.

[0450] The Examples provide an assessment of the impact of Compound A on the development of KA-induced SE in mice. In a refractory setting, the onset of SE was slowed in mice pre-dosed with Compound A compared to vehicle treated controls. Likewise, Compound A decreased KA-induced elevations in total EEG power. Taken together with the Examples using pentylentetrazol (PTZ), these findings suggest that Compound A exhibits anticonvulsant efficacy. KA induced SE in rodents, like that seen in patients, develops resistance to termination by benzodiazepines (BDZs) (Niquet et al., *Ann. N. Y. Acad. Sci.* 1378, 166-173 (2016)). Pretreatment of mice with Compound A prevented the development of refractory status epilepticus (RSE) 4h later. Critically, at this time point, Compound A was present in the brain at >600 nM, sufficient to robustly potentiate KCC2 activity based on an EC₅₀ of 260 nM. In addition to pretreatment, the Examples establish that Compound A was efficacious in restoring the efficacy of BDZs to terminate ongoing RSE when delivered IP. Evidence accrued from patients and animal models suggest that multiple factors, ranging from modifications in the membrane trafficking and subunit composition of GABA_A receptors to deficits in GABA release, are all believed to contribute to RSE (Burman et al., *Nat Rev Neurol* 18, 428-441 (2022)). However, increased intracellular accumulation of Cl_{sup}- leading to depolarizing GABA_A receptor currents are also widely believed to be of significance (Burman et al., *Brain: a journal of neurology* 142, 3482-3501 (2019)). The ability of Compound A to prevent and arrest RSE suggest that deficits in KCC2 activity are central to the pathophysiology of this trauma.

[0451] Prolonged seizure activity in humans and animal models leads to neuronal cell death. In mice that survive KA-induced SE cell death is evident within 48-72h. The Examples establish that in mice treated with Compound A and diazepam, neuronal cell death in the hippocampus was significantly reduced compared to controls treated with diazepam alone. This effect may be due to the ability of Compound A to limit hyperexcitability, but previous in vitro and in vivo studies have shown that reducing KCC2 expression levels, or transiently inhibiting it, are sufficient to induce neuronal apoptosis. Likewise, modifying KCC2 expression levels during development also impacts neuronal viability. Regardless of the precise mechanism, the Examples provide evidence that KCC2 activation by a compound according to Formula (I), or a pharmaceutically acceptable salt thereof,

arrests RSE and limits the extent of the subsequent neuronal injury.

Example 1

Identification of Small Molecules that Activate and Directly Bind to KCC2

[0452] A multi-tiered high-throughput screening assay was used, leveraging a proprietary library of 1.3 million small molecule compounds from AstraZeneca (Cambridge, UK) to identify chemical entities capable of activating KCC2. A single point concentration of 30 μM was used to identify potassium flux using an established thallium (^{111}Tl) influx FLIPR assay as the primary read out in HEK-293 cells overexpressing KCC2 (Cardarelli et al., Nat. Med. 23, 1394-1396 (2017); Conway et al., J Biol Chem 292, 21253-21263 (2017); Lee et al., 2022, supra). Compounds that exhibited $>20\%$ potentiation of KCC2 activity with EC₅₀ of $<30\text{ }\mu\text{M}$ were selected for further optimization. This approach resulted in the identified subseries of fused pyrimidine compounds as KCC2 activators, which are denoted as subseries A (SSA; FIG. 1A).

[0453] Neuronal Cultures. Hippocampal neurons were prepared from Sprague Dawley E18 embryos and plated at 450,000 per dish and maintained at 37°C . in a humidified $5\%\text{ CO}_2$ incubator for 10-25 days before experimentation (Lee et al., Nat. Neurosci. 14, 736-743 (2011)).

[0454] Cell Culture and biotinylation. HEK-293 cells were purchased from ATCC (CRL-1573) and transfected with human KCC2 and GFP expression plasmids using Lipofectamine 2000. Cells were utilized for experimentation 48-72h following transfection. Cells were biotinylated at 4°C . using NHS-Biotin. Following lysis and purification on immobilized avidin beads, cell surface and total fractions were subject to immunoblotting as outlined previously (Conway et al., 2017, supra).

[0455] FLIPR TI Flux assay. HEK-293 cells were washed with HBSS, and then incubated with loading buffer containing TI sensitive fluorescent dye (<https://www.moleculardevices.com>), 20 μM bumetanide 20 μM ouabain and 2.5 mM probenecid for 1 h at room temperature (RT) under control conditions or with drugs in the dark using a FlexStation 3 microplate reader (Molecular Devices) and imaged 525 nm for 30 min. Stimulus buffer was added containing 5 mM K⁺ and 0.5 mM Tl⁺ and cells were imaged every 1.5 s over a 190 second time course. The initial rate Tl⁺ uptake (Relative Fluorescence Units (RFU) per second) was determined following addition of the stimulus buffer as outlined previously (Conway et al., 2017, supra) (Lee et al., 2022, supra).

Example 2

Identification of Compound A

[0456] Determining if compounds in SSA directly interact with KCC2. This was tested using a cellular thermal shift assay (CETSA), which measures the ability of a drug to modify the thermal stability of its target in whole cells (Kawatkar et al., ACS Chem. Biol. 14, 1913-1920 (2019); Martinez et al., Sci Rep 8, 9472 (2018)).

[0457] 2×10^6 transfected HEK-293 cells were incubated with 30 μM drug or vehicle for 90 min at 37°C . and then heated from $37-58^\circ\text{C}$. for 5 min. Cells were rapidly chilled to 25°C . and NP40 was added to a final concentration of prior to snap freezing in liquid nitrogen. Following 5 freeze/thaw cycles lysates were centrifuged at $12,000\times g$ to remove denatured proteins and associated cell debris. The supernatant cell and tissue lysate samples, or immunoprecipitated KCC2 samples were separated by SDS-PAGE and subsequently immunoblotted with KCC2 antibodies (1:2000, Millipore #07-432), KCC2-S940, KCC2-pT1007(1:2000, PhosphoSolutions, #1568 actin and GAPDH (1:3000, SantaCruz #sc-32233) as detailed in previous studies (Lee et al., 2011, supra) (Conway et al., 2017, supra), and the level of soluble KCC2 at each temperature was normalized to that seen at 37°C . This data was then used to determine the temperature at which 50% of KCC2 was denatured.

[0458] Aliquots of HEK-293 cells expressing KCC2 were exposed to 30 μM SSA, heated from $37-55^\circ\text{C}$. The remaining level of soluble native KCC2 at each temperature was measured by immunoblotting and normalized to the level seen at 37°C . which was given a value of 1. SSA-1 reduced the thermal stability of KCC2 compared to vehicle, which demonstrated target engagement. To gain insights into affinity, isothermal dose response analysis was performed at 50°

C. revealing a K_{sub}.D of 395.6±23.5 nM of SSA for KCC2 (FIG. 1B-D). SSA derivatives were further optimized for affinity, optimal drug like properties and predicted brain penetration by in vitro cellular uptake assays, which resulted in the identification of Compound A (FIG. 1E). Dose response measurements with Compound A were then made at 50° C. and this data was used to estimate its relative affinity for KCC2 (Kawatkar et al., 2019, supra) (Martinez et al., 2018, supra). To control for compounds that may indirectly modulate KCC2 activity, via modifying its phosphorylation (Kahle et al., Trends Neurosci. 36, 726-737 (2013); Moore et al., Proc. Natl. Acad. Sci. U.S.A. 115, 10166-10171 (2018); Silayeva et al., Proc. Natl. Acad. Sci. U.S.A. 112, 3523-3528 (2015); Smalley et al., Front. Mol. Neurosci. 13, 563091 (2020)), the effect of Compound A to modify the activity of a panel of 144 protein kinases (SelectScreen; Thermofisher) was studied. This panel included protein kinase C isoforms, OSR1, SPAK (STK39) and with-no-lysine kinases (WNK), all of which phosphorylate KCC2 on multiple residues to indirectly modify its activity (Kahle et al., 2013, supra; Smalley et al., 2020, supra). Compound A did not modify the activity of this kinase panel.

[0459] As measured by TI⁺ flux, Compound A exhibited an EC_{sub}.50 for KCC2 of 261.4±22.2 nM and a maximum potentiation of 90% at a saturating concentration of 3 μM using a 1% β-cyclodextrin (BCD) vehicle (FIG. 1F). Several studies have shown that the activity and plasma membrane accumulation of KCC2 is potentiated via phosphorylation of S940, a process mediated by the activity of protein kinase C and protein phosphatase 1 (Lee et al., 2011, supra; Lee et al., J Biol Chem 282, 29777-29784 (2007); Moore et al., Proc. Natl. Acad. Sci. U.S.A. 115, 10166-10171 (2018); Moore et al., Trends Neurosci. 40, 555-571 (2017); Silayeva et al., Proc. Natl. Acad. Sci. U.S.A. 112, 3523-3528 (2015)). The ability of Compound A to modulate the activity of a KCC2 construct was tested in which this key residue for phospho-dependent modulation of KCC2 was mutated to an alanine (S940A). Significantly, Compound A increased the activity of KCC2-S940A construct (FIG. 1G). Finally, the specificity of Compound A for other solute transporters was examined using the TI⁺ assay using HEK-293 cells expressing KCC3, KCC4, or NKCC1. Compared to KCC2, effects on KCC3, KCC4, and NKCC1 were only seen at concentrations more than 35-fold that of KCC2 (>100 μM) (FIG. 1G).

[0460] These results suggest that Compound A binds directly to KCC2 and acts to increase its activity with minimal effects on other solute transporters that regulate neuronal Cl_{sup}.– accumulation.

Example 3

Compound A Potentiates KCC2 Activity without Modifying its Plasma Membrane Accumulation or Phosphorylation

[0461] To confirm the validity of our measurements using TI⁺ flux, KCC2 was expressed in HEK-293 cells with the α1 subunit of the glycine receptor (GlyRα1) which forms homomeric glycine-activated ion channels when expressed in this system. The gramicidin perforated patch clamp technique was used to measure the reversal potential for GlyRα1-mediated currents (E_{sub}.GLY) in cells exposed to Compound A. The use of this technique, while limited in throughput, facilitates high resolution single-cell recordings without disturbing endogenous intracellular Cl_{sup}.– levels (Cardarelli et al., 2017, supra; Conway et al., 2017, supra). To calculate E_{sub}.GLY, cells were exposed to 50 μM Gly and the polarity of Gly-induced currents was determined at different holding potentials. Brief (15 min) incubation with either 0.3 μM (approx. EC_{sub}.50) or 3 μM Compound A induced shifts of -15.6±4.5 (p=0.022), and -32.2±9.5 (p=0.001) mV in E_{sub}.GLY respectively, while vehicle was without effect (-0.8±0.5 mv; p=0.512) (FIG. 2A-B). As determined using the Nernst equation, these negative shifts in E_{sub}.GLY by Compound A corresponded to reduced intracellular Cl_{sup}.– levels of FIG. 2B; Compound A; 5.2±1.2; vehicle=7.9±1.2 mM; p=0.0104).

[0462] The effects of Compound A on the plasma membrane stability of KCC2 expression was assessed using biotinylation, as described previously (Cardarelli et al., 2017, supra; Conway et al., 2017, supra). Compared to vehicle, 0.3 μM or 3 μM Compound A did not modify the level of

KCC2 on the plasma membrane (FIG. 2C; $94.5 \pm 8.4\%$, $p=0.745$ and $104.7 \pm 8.4\%$ of control; $p=0.736$ respectively). The reliability of the biotinylation procedure was assessed via immunoblotting with antibodies against the cytosolic protein actin. Actin was present in the total lysates but was absent from the surface fractions (FIG. 2C).

[0463] The activity of KCC2 is subject to dynamic modulation via phosphorylation of critical intracellular residues within its C-terminal cytoplasmic domain. Central to this process are serine 940 (S940) and threonine 1007 (T1007), which activate and inhibit KCC2 activity respectively (Moore et al., 2017, supra). Therefore, the effects of Compound A on the phosphorylation of these residues was examined using immunoblotting with characterized phospho-specific antibodies pS940 and pT1007, respectively. Compound A ($3 \mu\text{M}$) did not significantly modify phosphorylation of S940 ($95.3 \pm 10.2\%$ of control; $p=0.764$), or T1007 ($108 \pm 10.2\%$ of control; $p=0.649$) (FIG. 2D).

[0464] The foregoing suggests that Compound A potentiates KCC2 activity without modifying either its plasma membrane accumulation or the phosphorylation of key regulatory sites.

Example 4

Compound A Reduces Neuronal Cl.SUP.– Accumulation and Slows the Development of Seizure-Like Events In Vitro

[0465] To test if Compound A modifies neuronal Cl.sup.– levels gramicidin-perforated patch-clamp recording was used to measure the reversal potentials of GABA.sub.A receptor mediated currents (E.sub.GABA) in 18-20 Div hippocampal neurons (Lee et al., 2011, supra; Moore et al., 2018, supra). At this developmental stage, GABA.sub.AR receptor activation leads to hyperpolarization, a phenomenon critically dependent upon KCC2 (Kontou et al., J Biol Chem, 100364 (2021)).

[0466] Neuronal Patch clamp assays. Neuronal cell culture recordings were performed in bath saline at 34°C . For gramicidin perforated-patch experiments, pipettes contained (in mM): 140 KCl and 10 HEPES, pH 7.4 KOH. For whole-cell experiments, pipettes contained (in mM): 115 K-meth-SO.sub.4, 30 KCl, 2 Mg-ATP, 4 Na-ATP, 0.4 Na-GTP, and 10 HEPES, pH 7.4 KOH. Bath saline contained the following (in mM): 140 NaCl, 2.5 KCl, 2.5 CaCl.sub.2, 2.5 MgCl.sub.2, 10 HEPES, and 11 glucose, pH 7.4 NaOH (Kontou et al., 2021, supra). All compound solutions were applied to cells using a three-barreled $700 \mu\text{m}$ pipe positioned just above the cell (Warner Instruments). Compound A was applied in 1 mg/mL 2-Hydroxypropyl- β -cyclodextrin, and its effects were compared to this vehicle alone. 10 mV voltage-ramp protocols over 1-s periods were used to determine the reversal potentials of the leak-subtracted muscimol-activated GABA.sub.AR currents. All voltages from whole-cell experiments were corrected offline using a calculated liquid junction potential value (13.6 mV) in Clampex (Molecular Devices). For gramicidin perforated-patch experiments, intracellular Cl.sup.– values were back-calculated using measured E.sub.GABA values and the Nernst equation:

$$[00001] E_{\text{ion}} = \frac{RT}{zF} \ln \frac{[\text{ion}]_o}{[\text{ion}]_i}$$

where E is the ion's reversal potential, R is the universal gas constant, T is temperature in kelvins, F is Faraday's constant, z is the charge of the ion, [chloride].sub.i is the intracellular chloride concentration and [chloride].sub.o is the extracellular ion concentration.

[0467] Measurements were performed in the presence of bumetanide ($10 \mu\text{M}$) and tetrodotoxin (TTX; 300 nM) to limit the contributions of NKCC1 and activity dependent shifts in chloride levels, respectively (Lee et al., 2011, supra). Cultures were exposed to Compound A or vehicle for 15 min and voltage ramps were used to determine changes in the reversal potential of GABA.sub.AR receptor currents over time (FIG. 3A). These data were then used to determine shifts subsequently, using [Cl.sup.–] levels. At 15 min 300 nM Compound A (an approximately EC.sub.50 concentration) significantly reduced E.sub.GABA from -75 ± 3.1 to $-85.1 \pm 4.2 \text{ mV}$ (FIG. 3B; $p=0.0104$), a net $-9.1 \pm 2.2 \text{ mV}$ negative shift, reflecting a $2.2 \pm 0.6 \text{ mM}$ reduction in Cl.sup.–

(FIG. 3C; $p=0.0153$). In contrast, vehicle did not significantly modify E.sub.GABA (FIG. 3B; $0=-77.2\pm3.3$, $15=-78.0\pm4.3$ mV respectively, $p=0.580$) or chloride levels (please add the chloride levels to the text).

[0468] To measure the effects of Compound A on KCC2 activity under more dynamic conditions cultures in Compound A were incubated for 1 h and subjected to whole cell patch-clamp recording to artificially impose a 32 mM Cl_{sup}- load on neurons via the patch pipette (Lee et al., 2022, supra; Moore et al., 2018, supra). 5 min after break-in, an initial E.sub.GABA value was determined and the effects of a subsequent 5 min incubation with VU0463271 was assessed. Compound A-treated cells exhibited lower basal E.sub.GABA values compared to controls (FIG. 3C; FIG. 3D -59.3 ± 2.1 and -47.4 ± 2.3 mV, respectively, $p=0.025$). Likewise, VU0463271 induced shifts in E.sub.GABA were larger for cultures treated with Compound A (FIG. 3C; FIG. 3D 17.5 ± 4.6 and 5.4 ± 4.2 mV, respectively, $p=0.009$).

[0469] The effects of Compound A were assessed on the development of seizure like events (SLE) in brain slices exposed to ACSF deficient in Mg_{sup}.+2 (0-Mg), a widely used method to increase neuronal excitability (Anderson et al., Brain research 398, 215-219 (1986)). In vitro seizure assays. Horizontal/Coronal slices (400 μ M) are prepared from 5-7 week old male mice and placed in a submerged chamber for a 60-min recovery period at 32° C. in normal ACSF as detailed previously (Kelley et al., Neuropharmacology 108, 103-110 (2016); Moore et al., 2018, supra). Slices were then transferred to a Warner Instruments recording chamber and electrodes of 0.5-1 mohm were inserted into layer III of the medial entorhinal cortex, several cell layers deep. Slices were perfused for 10 min with normal ACSF before exposure to media deficient in Mg_{sup}.+2 supplemented containing vehicle (1 mg/mL 2-Hydroxypropyl- β -cyclodextrin), or Compound A. Recordings were made within the entorhinal cortex with a Multiclamp 700B amplifier (Molecular Devices) with Clampex 10 acquisition software (Molecular Devices). The time spent in seizure-like activity was measured using threshold detection software in Clampfit. $3\times$ the standard deviation of a 1 min period of the baseline immediately prior to /0-Mg²⁺ onset was used as the cut off for event detection. The duration between each above-threshold event was calculated, and inter-event durations of <20s were considered as a continuation of the same seizure-like event. The field recordings were then employed to monitor the development of seizure-like events (SLEs) and their transition to late recurrent discharges (LRDs) which are believed to be analogous to the development of status epilepticus (SE) (Kelley et al., 2016, supra; Moore et al., 2018, supra). Relative to control, Compound A modified the appearance of the first SLE relative to control (FIG. 3E, F; Compound A= 11.3 ± 3.3 min, and vehicle= 10.5 ± 2.6 min, respectively, $p=0.256$). In addition, Compound A slowed the development of LRDs Cellular Thermal Shift Assay (FIG. 3E, F; Compound A= 28.4 ± 6.4 min, vehicle= 12.4 ± 4.1 min, $p=0.010$).

[0470] Thus, Compound A reduces neuronal Cl_{sup}- levels and slows the development of hyperexcitability in an ex vivo system.

Example 5

Compound A Rapidly Accumulates in the Brain and Protects Against Pentylenetetrazol (PTZ) Induced Convulsions In Vivo

[0471] PK studies. Mice were injected intraperitoneally (IP), intravenously (IV), or subcutaneously (SC), with Compound A and 5% BCD in a maximum volume of 300 μ l. Plasma and brain samples were rapidly frozen on dry ice. Tissue extracts were treated with Acetonitrile and Compound A levels were quantified using LC-MS/MS (Integrated Analytical Solutions, Inc, Berkeley, CA 94710; www.ianalytical.net).

[0472] To examine the ability of Compound A to cross the blood brain barrier, mice were injected with a single 25 mg/kg IV bolus; accumulation in the brain was evident within 30 minutes min in the brain and plasma to 7.6 ± 4.4 and 41.5 ± 11.5 and 11.2 μ M respectively (FIG. 4A). Likewise, a 50 mg/kg IP injection resulted in the accumulation of Compound A in the brain to 2.67 ± 0.21 and 4.1 ± 0.13 μ M 2 and 4 hr later (FIG. 4B). Drug accumulation in the brain was examined following

SC injection (FIG. 4C; 50 mg/kg). Accumulation was detected at 30 min and reached a maximal concentration of 675.75 ± 80.5 nM at 4 hr, a level that was maintained 8h later (FIG. 4C).

[0473] Given the ability of Compound A to accumulate in the brain, it was examined to see if it also causes any gross effects on animal behavior. Mice were injected with Compound A (SC; 50 mg/kg), or vehicle and their behavior was compared in the open field test (Tretter et al., Proc. Natl. Acad. Sci. U.S.A. 106, 20039-20044 (2009); Vien et al., Frontiers in molecular neuroscience 15, 817996 (2022)). Compound A did not modify the total distance traveled during the test trial (FIG. 4D; Compound A= 6134.2 ± 235.3 cm, and vehicle= 6453.7 ± 215.4 cm; $p=0.253$, $n=11$ mice).

Likewise, time spent in the center zone was comparable (FIG. 4D; Compound A= 18.4 ± 2.1 , and vehicle $16.6 \pm 2.5\%$ of total time; $p=0.658$, $n=11$ mice).

[0474] To gauge if Compound A exerts any anticonvulsant efficacy its ability to protect against motor seizures induced by the convulsant (PTZ), a GABA.sub.AR antagonist (Shimada and Yamagata, J Vis Exp. 10.3791/56573 (2018)) was determined. To do this, mice were dosed IV with 12.5 mg/kg Compound A or vehicle, and then the modified Racine scale was used to measure the dose of PTZ required to induce forelimb and hindlimb clonus, stage 3 and 4/5 seizures, was assessed (Van Erum et al., Epilepsy Behav 95, 51-55 (2019)). While Compound A did not protect against forelimb clonus (FIG. 4E; Control= 51.5 ± 3.1 , Compound A= 56.3 ± 3.3 mg/kg; $p=0.120$, $n=14$ mice), the dose of PTZ required to induce hindlimb clonus was significantly increased by Compound A pretreatment (FIG. 4E; Control= 84.5 ± 5.1 mg/kg, Compound A= 137.4 ± 7.5 mg/kg respectively, $p=0.001$, $n=11-14$ mice).

[0475] Collectively, these results suggest that Compound A accumulates in the brain when dosed IV, IP or SC, and does not lead to any stereotyped effects on behavior but does protect against late-stage seizures induced by PTZ, a pharmacological screening tool widely used to identify anticonvulsants.

Example 6

KCC2 Activation Reduces the Severity of SE and the Development of BDZ Resistance, In Vivo

[0476] To further explore the anticonvulsant properties of Compound A, its effects on the development of kainic acid (KA) induced SE in mice was examined using electroencephalographic (EEG) recording (Silayeva et al., 2015, supra). This model was chosen due to the similarities with patients undergoing SE, as KA-induced seizures become refractory to benzodiazepines (BDZs) within minutes (Betjemann and Lowenstein, Lancet Neurol 14, 615-624 (2015); Crawshaw and Cock, (2020a) [Seizure: Eur. J. Epilep. 75 (2020) 145-152] Seizure 80, 282; Lee et al., 2022, supra; Reddy and Kuruba, Int J Mol Sci 14, 18284-18318 (2013)).

[0477] EEG recordings and analysis. WT C57Bl/6J adult male mice (12 weeks) were anesthetized with isoflurane and implanted with EEG/EMG electrodes and allowed to recover for 7 days prior to experimentation. For pre-dosing basal EEGs were recorded for 30 min prior to injection with 50 mg/kg Compound A (SC), or vehicle followed 2h later with 20 mg/kg KA (IP). 2h following KA administration mice were injected with a saturating concentration of diazepam (DZ) 10 mg/kg (IP) and recordings were extended for a further 60 min. Alternatively basal EEGs were recorded for 30 min prior to dosing with 20 mg/kg KA, followed 2h later with 10 mg/kg DZ alone or together with 50 mg/kg Compound A (IP) and recordings were extended for a further 60 min (FIG. 5A).

LabChart 8 software was used for data analysis. Epileptiform activity was identified by changes in the amplitude of electrographic activity characterized by consistent changes in the power of the Fast Fourier transform (FFT) of the EEG and abnormal activity characterized by periods of rhythmic spiking lasting longer than 30s (Moore et al., 2017, supra; Silayeva et al., 2015, supra; Sivakumaran et al., J. Neurosci. 35, 8291-8296 (2015)).

[0478] First, the effects of Compound A on the development of SE were examined, revealing increased latency to the first seizure (FIG. 5B; vehicle= 10.2 ± 2.5 min, Compound A= 14.4 ± 2.9 min, $p=0.0329$, $n=10$ mice) and to SE, compared to vehicle (FIG. 5B; vehicle= 38.3 ± 3.4 min, and Compound A= 54.5 ± 6.9 min, $p=0.0230$, $n=10$ mice). In addition, the time spent in epileptiform

activity was lower in mice pretreated with Compound A (FIG. 5B; vehicle=85±5.4%, and Compound A=62.5±8.4%, p=0.0230, n=9 mice).

[0479] To provide quantitative insights on the effects of KA, raw EEG recordings were subject to Fast-Fourier transformation (8K FFT size, Hann (cosine-bell), 87.50% window overlap) for frequencies between 0-100 Hz 2h after KA injection (Lee et al., 2022, supra). In mice pretreated with Compound A total EEG power was significantly reduced compared to vehicle (FIG. 5C; $4.2 \pm 2.4 \times 10^{-6}$ and $6.9 \pm 3.2 \times 10^{-7}$ $\mu\text{V} \cdot \text{s}^2$ for vehicle and Compound A, respectively; p=0.0024, n=11 mice).

[0480] Finally, the effects of DZ on EEG power were compared min prior to and following DZ treatment. In mice pretreated with Compound A, DZ significantly reduced EEG power (FIG. 5D; $6.9 \pm 3.2 \times 10^{-7}$ and $2.7 \pm 3.2 \times 10^{-7}$ $\mu\text{V} \cdot \text{s}^2$, pre and post DZ respectively; p=0.004, n=9 mice). In contrast, but consistent with previously published studies on the KA model, DZ did not modify EEG power in mice pretreated with vehicle (FIG. 5D; $4.2 \pm 2.4 \times 10^{-6}$ and $2.9 \pm 2.7 \times 10^{-6}$ $\mu\text{V} \cdot \text{s}^2$ pre and post DZ respectfully; p=0.105).

[0481] To confirm that Compound A acts centrally to mediate its effects on seizure activity, Compound A (10 μM), (20 μM) or vehicle was injected directly into the hippocampus of mice implanted with EEG/EMG electrodes (Lee et al., 2022, supra). 30 or 60 min later mice were injected with KA (20 mg/kg) and 2h later with DZ as detailed above. Relative to vehicle Compound A increased the latency to the 1st seizure, to SE, and reduced the percentage of time in epileptiform activity. 2h after KA injection DZ did not significantly modify EEG power in mice treated with vehicle. However, DZ significantly reduced seizure parameters in mice pretreated with Compound A, thus restoring the efficacy of BDZs in terminating RSE.

[0482] Collectively, these observations demonstrate that pre-treatment of mice with Compound A reduces the severity of KA induced SE and restores BZD efficacy, preventing the development of RSE.

Example 7

Compound A Restores the Ability of DZ to Arrest Ongoing SE and Limits Subsequent Neuronal Cell Death

[0483] To evaluate the potential of KCC2 activators to terminate ongoing SE, the KA model described above was employed. Mice were injected with KA (20 mg/Kg; IP). 2h later mice were subsequently dosed with DZ alone or DZ/Compound A IP (DZ=10 mg/kg; Compound A=50/mg/Kg), and recordings were extended for a further 60 min (FIG. 6A). Spectral data were then subjected to FFT and the total EEG power between 1-100 Hz of the 2h following KA treatment (pre-drug) and 30 min following their injection (post-drug) were compared (FIG. 6B). In mice treated with DZ alone variable effects were observed, from large increases to modest reductions in EEG power, but these effects were not significant (FIG. 6C; pre= $1.1 \pm 0.2 \times 10^{-6}$, post= $4.4 \pm 7.4 \times 10^{-7}$ $\mu\text{V} \cdot \text{s}^2$, p=0.214, n=13-14 mice). In contrast, total EEG power was significantly reduced in mice treated with DZ/Compound A (FIG. 6D; pre= $9.2 \pm 1.0 \times 10^{-7}$, post= $2.0 \pm 0.2 \times 10^{-7}$ $\mu\text{V} \cdot \text{s}^2$, p=0.007, n=13-15 mice) reflecting a decrease in power of approximately 72%. The latency of the treatments to modify EEG power was examined by measuring the time taken for this parameter to be reduced to 50% (FIG. 6E). The mean time to suppression in DZ/Compound A treated mice was 2.3 ± 0.5 , compared to 32.5 ± 6.7 min in those treated with DZ alone (FIG. 6F, G; p=0.0008, n=13-14 mice). Finally in mice treated with DZ alone 62% were insensitive to the effects of this agent on EEG power. In contrast, DZ insensitivity was abolished by Compound A (FIG. 6G).

[0484] KA-induced seizures lead to neuronal cell death in many brain regions 48-72h after treatment (Ben-Ari and Cossart, Trends Neurosci. 23, 580-587 (2000)). To determine if KCC2 activators have an impact on the extent of neuronal injury, brain sections from mice that survived for 48h after KA injection were subjected to Terminal deoxynucleotidyl transferase dUTP nick end (TUNEL) staining to visualize dead cells, and counterstained with DAPI. The number of TUNEL

positive cells in the CA1 region of the hippocampus were then compared between treatment groups. Consistent with published studies, high levels of TUNEL positive cells were found in mice injected with DZ alone. However, DZ/Compound A treatment reduced the number of TUNEL positive CA1 cells (FIG. 7A, 7B; DZ=40.93 $\pm$ 3.35 and DZ/Compound A=31.07 $\pm$ 2.64 dead cells/ROI, p=1.sup.4 stacks, from 4 mice).

[0485] These results demonstrate that Compound A restores the ability of DZ to arrest ongoing SE and limits subsequent neuronal cell death.

Example 8

Prospective Assessment of the Efficacy of Compound A in Pediatric Subjects with Status Epilepticus

[0486] This study is designed to determine whether with Compound A leads to an improvement in one or more symptoms of status epilepticus in pediatric patients. Epilepsy is among the most common serious neurologic disorders in childhood. Medicines with novel actions of mechanisms of action are needed to try to address the unmet clinical need for seizure control in subjects with status epilepticus. In addition, the purpose of this study is to evaluate the safety and tolerability of Compound A as treatment in subjects with status epilepticus.

[0487] Compound A will be administered intravenously to pediatric subjects (3 months to 16 years) in amounts ranging from 0.25, 0.5, 1, 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 9, and 10 mg as 50 mL short-term infusion solution intravenously (IV) within 15 minutes (infusion rate 200 mL/h). Subjects whose seizure does not stop or recurs within 10 minutes after the initial dose may receive the same amount of Compound A injection no earlier than 10 minutes following the initial dose. Subjects whose seizure stops within 10 minutes after the initial dose, but recurs thereafter (within 12 hours) may receive the same amount of Compound A injection; a total of 2 doses will be permitted in this study.

Inclusion Criteria:

[0488] Subjects with status epilepticus or repetitive status epilepticus/cluster seizure who have seizures that can be evaluated by investigator's visual observations based on motor symptoms or who have seizures that can be evaluated by EEG.

[0489] Subjects with status epilepticus accompanied by generalized seizure, partial seizure or secondarily generalized seizure lasting 5 minutes or longer

[0490] Subjects with repetitive status epilepticus/cluster seizure accompanied by not less than 3 consecutive episodes of generalized seizure, partial seizure or secondarily generalized seizure in 1 hour.

[0491] Subjects not younger than 3 months (either gender is eligible for the study)

Primary Outcome Measures:

[0492] Percentage of Participants With Clinical Benefit [Time Frame: 30 minutes]

[0493] Percent of participants whose initial seizure stopped within 10 minutes after initial dose and who continued seizure-free for at least 30 minutes after the completion of initial dose (responder rate)

Secondary Outcome Measures:

[0494] Percent of participants whose initial seizure stopped within 10 minutes after the administration of Compound A (either initial or second dose [in 10 to 30 minutes from the initial dose]) and who continued seizure-free for at least 30 minutes. [Time Frame: 1 hour]

[0495] Percent of participants whose seizures stopped within 10 minutes after the administration of Compound A (only the initial dose) and who continued seizure-free for at least 12 hours post-dose. [Time Frame: 12 hours]

[0496] Percent of participants whose seizures stopped within 10 minutes after the administration of Compound A (either initial or second dose [in 10 to 30 minutes from the initial dose]) and who continued seizure-free for at least 12 hours post-dose. [Time Frame: 12 hours]

[0497] Percent of participants whose seizures stopped within 10 minutes after the administration of

Compound A (only the initial dose) and who continued seizure-free for at least 24 hours post-dose.

[Time Frame: 24 hours]

[0498] Percent of participants whose seizures stopped within 10 minutes after the administration of (Compound A (either initial or second dose [in 10 to 30 minutes from the initial dose])) and who continued seizure-free for at least 24 hours post-dose. [Time Frame: 24 hours]

[0499] Time to resolution of seizures from the administration of Compound A (only the initial dose). [Time Frame: 24 hours]

[0500] Time to resolution of seizures from the administration of Compound A (either initial or second dose). [Time Frame: 24 hours]

[0501] Time to relapse from the resolution of seizures following the administration of Compound A (only the initial dose, within 24 hours). [Time Frame: 24 hours]

[0502] Time to relapse from the resolution of seizures following the administration of Compound A (either initial or second dose, within 24 hours). [Time Frame: 24 hours]

Example 9

Prospective Assessment of the Use of Compound A to Prevent Epilepsy in Infants with Tuberous Sclerosis Complex

[0503] A multi-center, randomized, placebo-controlled, double-blind clinical trial. 80 infants with Tuberous Sclerosis Complex who are less than 6 months of age prior to the onset of their first seizure will be enrolled. Early identification of electroencephalography (EEG) biomarkers and early treatment versus delayed treatment with (Compound A in infants with tuberous sclerosis complex (TSC) will be assessed for positive impact on developmental outcomes at 24 months of age. Early treatment should prevent or lower the risk of developing infantile spasms and refractory seizures. This preventative approach should result in more favorable long-term cognitive, behavioral, developmental and psychiatric outcomes and significantly improve overall quality of life.

[0504] Inclusion criteria are infants less than or equal to 6 months of age, no history of seizures or infantile spasms, or evidence of subclinical electrographic seizures on a previous video EEG, and meet genetic or clinical diagnostic criteria for TSC, the latter based on current recommendations for diagnostic evaluation, such as physical exam, neuroimaging, echocardiogram.

[0505] Compound A or placebo will be dosed according to body weight 1 mg to 50 mg/kg/day divided twice daily. Dosing will follow established recommended guidelines (1 mg/kg/day and increased as needed by 10 mg/kg/day every 3 days up to a maximum dose of 50 mg/kg/day, divided twice daily).

[0506] Subjects randomized to Compound A in Arm A will be treated with Compound A up to 50 mg/kg/day or placebo until 24 months of age or until they show evidence of clinical seizures or electrographic seizures on video EEG. If electrographic or clinical seizures occur while on study drug, they will transition into an Open label phase of the study (Arm B) and continue to be followed until 36 months of age. Compound A open label (Arm B) will be given for administration, dosed according to body weight 1 mg to 50 mg/kg/day divided twice daily. Dosing will follow established recommended guidelines (1 mg/kg/day and increased as needed by 10 mg/kg/day every 3 days up to a maximum dose of 50 mg/kg/day, divided twice daily).

[0507] Primary outcome measures: cognitive assessment scores on the Bayley Scales of Infant and Toddler Development at 24 months and Bayley Scales of Infant and Toddler Development at 24 months will be used for the data analysis and compare the developmental impact of early versus delayed treatment with Compound A.

[0508] Secondary outcome measures: 1. Evaluate the number of subjects that develop seizures when treated with Compound A as a seizure prevention, 2. Time to the subject's first clinical seizure will be measured for both subjects on placebo and Compound A, 3. The prevalence of drug resistant epilepsy, 4. Evaluate Vineland II scores and the impact of early versus late treatment with Compound A at 12, 24, and 36 months, 5. Evaluate ADOS2 scores and the impact of early versus

late treatment at 24 and 36 months, 6. Number of subjects with Compound A related adverse events, severe adverse events as assessed by CTCAE v4.0 and risk evaluation and mitigation strategy (REMS) measures.

Example 10

Prospective Assessment of the Efficacy of Compound B in Pediatric Subjects with Status Epilepticus

[0509] This study is designed to determine whether with Compound B leads to an improvement in one or more symptoms of status epilepticus in pediatric patients. Epilepsy is among the most common serious neurologic disorders in childhood. Medicines with novel actions of mechanisms of action are needed to try to address the unmet clinical need for seizure control in subjects with status epilepticus. In addition, the purpose of this study is to evaluate the safety and tolerability of Compound B as treatment in subjects with status epilepticus.

[0510] Compound B will be administered intravenously to pediatric subjects (3 months to 16 years) in amounts ranging from 0.25, 0.5, 1, 1.5, 2, 2.5, 3, 4, 5, 6, 7, 8, 9, and 10 mg as 50 mL short-term infusion solution intravenously (IV) within 15 minutes (infusion rate 200 mL/h). Subjects whose seizure does not stop or recurs within 10 minutes after the initial dose may receive the same amount of Compound B injection no earlier than 10 minutes following the initial dose. Subjects whose seizure stops within 10 minutes after the initial dose, but recurs thereafter (within 12 hours) may receive the same amount of Compound B injection; a total of 2 doses will be permitted in this study.

Inclusion Criteria:

[0511] Subjects with status epilepticus or repetitive status epilepticus/cluster seizure who have seizures that can be evaluated by investigator's visual observations based on motor symptoms or who have seizures that can be evaluated by EEG.

[0512] Subjects with status epilepticus accompanied by generalized seizure, partial seizure or secondarily generalized seizure lasting 5 minutes or longer

[0513] Subjects with repetitive status epilepticus/cluster seizure accompanied by not less than 3 consecutive episodes of generalized seizure, partial seizure or secondarily generalized seizure in 1 hour.

[0514] Subjects not younger than 3 months (either gender is eligible for the study)

Primary Outcome Measures:

[0515] Percentage of Participants With Clinical Benefit [Time Frame: 30 minutes]

[0516] Percent of participants whose initial seizure stopped within 10 minutes after initial dose and who continued seizure-free for at least 30 minutes after the completion of initial dose (responder rate)

Secondary Outcome Measures:

[0517] Percent of participants whose initial seizure stopped within 10 minutes after the administration of Compound B (either initial or second dose [in 10 to 30 minutes from the initial dose]) and who continued seizure-free for at least 30 minutes. [Time Frame: 1 hour]

[0518] Percent of participants whose seizures stopped within 10 minutes after the administration of Compound B (only the initial dose) and who continued seizure-free for at least 12 hours post-dose. [Time Frame: 12 hours]

[0519] Percent of participants whose seizures stopped within 10 minutes after the administration of Compound B (either initial or second dose [in 10 to 30 minutes from the initial dose]) and who continued seizure-free for at least 12 hours post-dose. [Time Frame: 12 hours]

[0520] Percent of participants whose seizures stopped within 10 minutes after the administration of Compound B (only the initial dose) and who continued seizure-free for at least 24 hours post-dose. [Time Frame: 24 hours]

[0521] Percent of participants whose seizures stopped within 10 minutes after the administration of (Compound B (either initial or second dose [in 10 to 30 minutes from the initial dose]) and who

continued seizure-free for at least 24 hours post-dose. [Time Frame: 24 hours]

[0522] Time to resolution of seizures from the administration of Compound B (only the initial dose). [Time Frame: 24 hours]

[0523] Time to resolution of seizures from the administration of Compound B (either initial or second dose). [Time Frame: 24 hours]

[0524] Time to relapse from the resolution of seizures following the administration of Compound B (only the initial dose, within 24 hours). [Time Frame: 24 hours]

[0525] Time to relapse from the resolution of seizures following the administration of Compound B (either initial or second dose, within 24 hours). [Time Frame: 24 hours]

Example 11

Prospective Assessment of the Use of Compound B to Prevent Epilepsy in Infants with Tuberous Sclerosis Complex

[0526] A multi-center, randomized, placebo-controlled, double-blind clinical trial. 80 infants with Tuberous Sclerosis Complex who are less than 6 months of age prior to the onset of their first seizure will be enrolled. Early identification of electroencephalography (EEG) biomarkers and early treatment versus delayed treatment with (Compound B in infants with tuberous sclerosis complex (TSC) will be assessed for positive impact on developmental outcomes at 24 months of age. Early treatment should prevent or lower the risk of developing infantile spasms and refractory seizures. This preventative approach should result in more favorable long-term cognitive, behavioral, developmental and psychiatric outcomes and significantly improve overall quality of life.

[0527] Inclusion criteria are infants less than or equal to 6 months of age, no history of seizures or infantile spasms, or evidence of subclinical electrographic seizures on a previous video EEG, and meet genetic or clinical diagnostic criteria for TSC, the latter based on current recommendations for diagnostic evaluation, such as physical exam, neuroimaging, echocardiogram.

[0528] Compound B or placebo will be dosed according to body weight 1 mg to 50 mg/kg/day divided twice daily. Dosing will follow established recommended guidelines (1 mg/kg/day and increased as needed by 10 mg/kg/day every 3 days up to a maximum dose of 50 mg/kg/day, divided twice daily).

[0529] Subjects randomized to Compound B in Arm A will be treated with Compound A up to 50 mg/kg/day or placebo until 24 months of age or until they show evidence of clinical seizures or electrographic seizures on video EEG. If electrographic or clinical seizures occur while on study drug, they will transition into an Open label phase of the study (Arm B) and continue to be followed until 36 months of age. Compound B open label (Arm B) will be given for administration, dosed according to body weight 1 mg to 50 mg/kg/day divided twice daily. Dosing will follow established recommended guidelines (1 mg/kg/day and increased as needed by 10 mg/kg/day every 3 days up to a maximum dose of 50 mg/kg/day, divided twice daily).

[0530] Primary outcome measures: cognitive assessment scores on the Bayley Scales of Infant and Toddler Development at 24 months and Bayley Scales of Infant and Toddler Development at 24 months will be used for the data analysis and compare the developmental impact of early versus delayed treatment with Compound B.

[0531] Secondary outcome measures: 1. Evaluate the number of subjects that develop seizures when treated with Compound B as a seizure prevention, 2. Time to the subject's first clinical seizure will be measured for both subjects on placebo and Compound B, 3. The prevalence of drug resistant epilepsy, 4. Evaluate Vineland II scores and the impact of early versus late treatment with Compound B at 12, 24, and 36 months, 5. Evaluate ADOS2 scores and the impact of early versus late treatment at 24 and 36 months, 6. Number of subjects with Compound B related adverse events, severe adverse events as assessed by CTCAE v4.0 and risk evaluation and mitigation strategy (REMS) measures.

[0532] It should be understood that the examples and embodiments provided herein are exemplary

examples and embodiments. Those skilled in the art will envision various modifications of the examples and embodiments that are consistent with the scope of the disclosure herein. Such modifications are intended to be encompassed by the claims.

Claims

1. A method of treating a seizure disorder comprising administering a compound according to Formula (I) ##STR00037## or a pharmaceutically acceptable salt thereof, wherein: R^{sup.1} is selected from C_{sub.2-6}alkyl; C_{sub.2-6}alkenyl; C_{sub.2-6}alkynyl; C_{sub.2-6}alkoxy; C_{sub.2-6}alkenyloxy; C_{sub.2-6}alkynyloxy; C_{sub.3-7}cycloalkyl; —O—C_{sub.3-7}cycloalkyl; C_{sub.6-10}aryl; —O—(CH_{sub.2})_{sub.m}—C_{sub.6-10}aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3} and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy wherein —C_{sub.1-3}alkyl, —C_{sub.1-8}alkoxy and —C_{sub.2-8}alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF_{sub.3}, —NHC(O)O—C_{sub.1-6}alkyl or two substituents together with the carbon to which they are attached form diazirinyl; R^{sup.2} is selected from —H; -halo; and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; A is selected from ##STR00038## or a N-oxide thereof; R^{sup.3} is selected from —H; —C_{sub.1-6}alkyl; —C_{sub.2-6}alkenyl; —C_{sub.2-6}alkynyl; C_{sub.3-7}cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF_{sub.3}, —C_{sub.1-3}alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF_{sub.3}, —C(O)NR^{sup.8}R^{sup.9} and —NR^{sup.8}R^{sup.9}; R^{sup.4a} and R^{sup.4b} are each independently selected from —H and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; R^{sup.4c} and R^{sup.4d} are each independently selected from —H and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}, or R^{sup.4c} and R^{sup.4d} together with the carbon to which they are attached represent carbonyl; R^{sup.5a}, R^{sup.5b}, R^{sup.5c} and R^{sup.5d} are each independently selected from —H and —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; R^{sup.6} is selected from —H; -halo; —NH_{sub.2}; —CN; —C_{sub.1-3}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; —C_{sub.1-3}alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF_{sub.3}; —C(O)O—C_{sub.1-3}alkyl; —C(O)NR^{sup.8}R^{sup.9}; —C(O)OH; and —NHC(O)—C_{sub.1-3}alkyl; R^{sup.7} is selected from NR^{sup.10}R^{sup.11}; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C_{sub.1-6}alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF_{sub.3} and —OH; —C_{sub.1-3}alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF_{sub.3}; —C(O)OH; —C_{sub.1-3}alkylene-NHC(O)C_{sub.1-6}alkyl; —C_{sub.1-3}alkylene-NHC(O)OC_{sub.1-6}alkyl; C_{sub.3-5}cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R^{sup.7} is morpholinyl and R^{sup.1} is unsubstituted phenyl, R^{sup.2} is not —H; R^{sup.8} and R^{sup.9} are each independently selected from —H and —C_{sub.1-6}alkyl; R^{sup.10} is —C_{sub.1-6}alkyl; R^{sup.11} is selected from —C_{sub.1-6}alkyl optionally substituted with 1 or 2 substituents selected from —F and —C_{sub.1-3}alkoxy; and —(CH_{sub.2})_{sub.n}R^{sup.12}; R^{sup.12} is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; m is 0 or 1; and n is 1, 2 or 3, to a subject diagnosed with the seizure disorder in an amount of from about 0.01 mg to about 1500 mg.

2. The method of treating a seizure disorder according to claim 1, wherein the total amount of the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, administered to the subject in a twenty-four hour period is between about 1 mg and about 1000 mg.
3. The method of treating a seizure disorder according to claim 1, wherein the seizure disorder is selected from the group consisting of epilepsy, epilepsy with generalized tonic-clonic seizures, epilepsy with myoclonic absences, frontal lobe epilepsy, temporal lobe epilepsy, focal cortical dysplasia, Landau-Kleffner Syndrome, Rasmussen's syndrome, Dravet syndrome, Doose syndrome, CDKL5 disorder, infantile spasms (West syndrome), tuberous sclerosis complex, juvenile myoclonic epilepsy (JME), vaccine-related encephalopathy, intractable childhood epilepsy (ICE), Lennox-Gastaut syndrome (LGS), Rett syndrome, Ohtahara syndrome, childhood absence epilepsy, essential tremor, acute repetitive seizures, benign rolandic epilepsy, status epilepticus, refractory status epilepticus, super-refractory status epilepticus (SRSE), PCDH19 pediatric epilepsy, benzodiazepine resistant seizures, diazepam resistant seizures, seizures induced by exposure to nerve agents, seizures induced by exposure to pesticides, lorazepam resistant seizures, drug withdrawal induced seizures, alcohol withdrawal induced seizures, increased seizure activity and breakthrough seizures.
4. The method of treating a seizure disorder according to claim 1, wherein the method provides improvement in at least one symptom selected from the group consisting of ataxia, gait impairment, speech impairment, vocalization, impaired cognition, abnormal motor activity, clinical seizure, subclinical seizure, hypotonia, hypertonia, drooling, mouthing behavior, aura, convulsions, repetitive movements, unusual sensations, simple focal seizures, complex focal seizures, generalized seizures, absences, tonic seizures, atonic seizures, myoclonic seizures, tonic clonic seizures, clonic seizures, frequency of seizures and severity of seizures.
5. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with tuberous sclerosis complex and the method provides improvement in at least one symptom selected from the group consisting of seizures, cognitive impairment, autism, gelastic seizures, involuntary laughter, interval irritability and depressed mood.
6. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with infantile spasms and the method provides improvement in at least one symptom selected from the group consisting of seizures, cognitive impairment, developmental regression and hypsarrhythmia.
7. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with Lennox Gastaut syndrome and the method provides improvement in at least one symptom selected from the group consisting of developmental delays, cognitive impairment and behavioral disturbances.
8. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with benzodiazepine resistant seizures.
9. The method of treating a seizure disorder according to claim 8, wherein the benzodiazepine resistant seizures are diazepam resistant seizures.
10. The method of treating a seizure disorder according to claim 8, wherein the benzodiazepine resistant seizures are lorazepam resistant seizures.
11. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with seizures which are suspected to have been induced by exposure to nerve agents or pesticides.
12. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with CDKL5 disorder and the method provides improvement in at least one symptom selected from the group consisting of seizures, scoliosis, visual impairment, sensory issues, gastrointestinal difficulties, low or poor muscle tone, hand wringing movements, mouthing of the hands, marked developmental delay, limited or absent speech, lack of eye contact or poor eye contact, gastroesophageal reflux, constipation, small, cold feet, breathing irregularities, grinding of

the teeth, episodes of laughing or crying without cause, limited hand skills, autistic-like tendencies, cortical visual impairment, apraxia, eating/drinking challenges, sleep difficulties, sideways glance and a habit of leg crossing.

13. The method of treating a seizure disorder according to claim 1, wherein the subject has been diagnosed with focal cortical dysplasia and the method provides improvement in at least one symptom selected from the group consisting of developmental delays, cognitive impairment and behavioral disturbances.

14. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound A:
##STR00039##

15. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound B:
##STR00040##

16. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound C:
##STR00041##

17. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound D:
##STR00042##

18. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound E:
##STR00043##

19. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound F:
##STR00044##

20. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound G:
##STR00045##

21. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound H:
##STR00046##

22. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered enterally.

23. The method of treating a seizure disorder according to claim 22, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered orally, sublingually, buccally, transdermally or rectally.

24. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered parenterally.

25. The method of treating a seizure disorder according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered in the form of a liquid for parenteral use, a tablet, a capsule, a caplet, a pill, an oral liquid, a lozenge, a film, a powder, an aerosol, or a patch.

26. The method of treating a seizure disorder according to claim 1, wherein the subject is human.

27. A method of treating a seizure disorder comprising administering a compound according to Formula (I), ##STR00047## or a pharmaceutically acceptable salt thereof, wherein: R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted

with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; A is selected from ##STR00048## or a N-oxide thereof; R.sup.3 is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3 alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3 alkyl; R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6 alkyl; R.sup.10 is —C.sub.1-6alkyl; R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; m is 0 or 1; and n is 1, 2 or 3, to a subject diagnosed with the seizure disorder prior to the onset of clinical seizures after detection of abnormal EEG to reduce or prevent symptoms of the seizure disorder.

28. A method of treating an abnormal EEG signature comprising administering a compound according to Formula (I), ##STR00049## or a pharmaceutically acceptable salt thereof, wherein: R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they

are attached from diazirinyl; R^{sup.2} is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; A is selected from ##STR00050## or a N-oxide thereof; R^{sup.3} is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3 alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR^{sup.8}R^{sup.9} and —NR^{sup.8}R^{sup.9}; R^{sup.4a} and R^{sup.4b} are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R^{sup.4c} and R^{sup.4d} are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R^{sup.4c} and R^{sup.4d} together with the carbon to which they are attached represent carbonyl; R^{sup.5a}, R^{sup.5b}, R^{sup.5c} and R^{sup.5d} are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R^{sup.6} is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR^{sup.8}R^{sup.9}; —C(O)OH; and —NHC(O)—C.sub.1-3 alkyl; R^{sup.7} is selected from NR^{sup.10}R^{sup.11}; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R^{sup.7} is morpholinyl and R^{sup.1} is unsubstituted phenyl, R^{sup.2} is not —H; R^{sup.8} and R^{sup.9} are each independently selected from —H and —C.sub.1-6 alkyl; R^{sup.10} is —C.sub.1-6 alkyl; R^{sup.22} is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR^{sup.12}; R^{sup.12} is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; m is 0 or 1; and n is 1, 2 or 3, to a subject having the abnormal EEG signature, wherein the abnormal EEG signature is indicative of early stage of epileptogenesis.

29. A method of prophylactically treating exposure of a subject to a nerve agent or an organophosphate pesticide comprising administering a compound according to Formula (I) ##STR00051## or a pharmaceutically acceptable salt thereof, prior to the exposure of the subject to the nerve agent or organophosphate pesticide, wherein: R^{sup.1} is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; R^{sup.2} is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; A is selected from ##STR00052## or a N-oxide thereof; R^{sup.3} is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl;

C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3 alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3 alkyl; R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6 alkyl; R.sup.10 is —C.sub.1-6 alkyl; R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; m is 0 or 1; and n is 1, 2 or 3.
